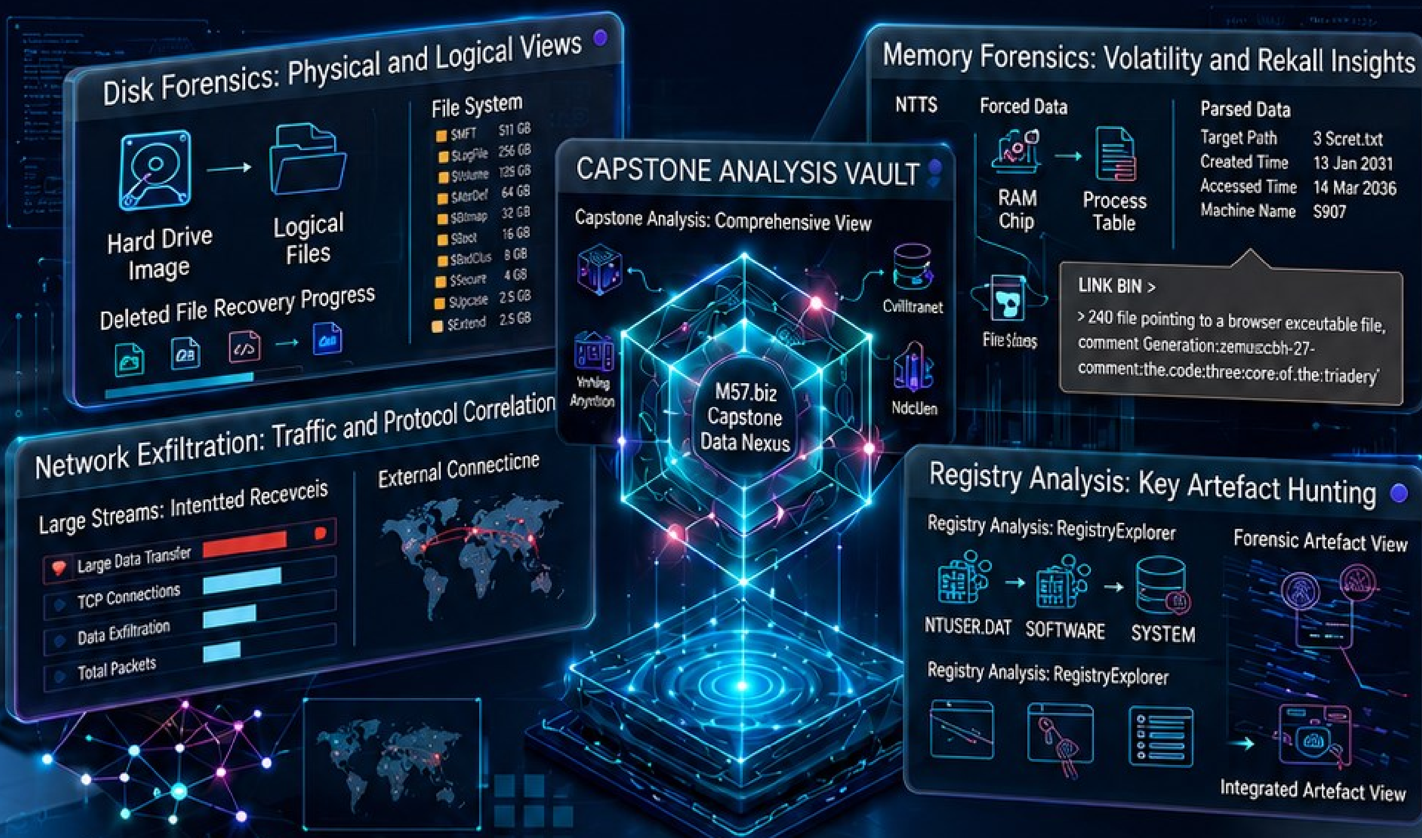
Cyberster

INTERNSHIP REPORT

A Weekly Progress and Learning Summary

Phase Two : The M57.biz Capstone: Disk, Memory, and Exfiltration Analysis



Submitted to: Cyberster Internship Program

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Roll Number: CSI-B1-353

Email Header Analysis | PST & MBOX Parsing | paso (timeline) | RegistryExplorer | RECmd | integrated Timeline View | Forensic Artifact Hunting

A Weekly Progress and Learning Summary

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Week 12 Overview

Week 12 of the Cyberster Blue Team Internship is the Phase Two capstone. Where every preceding week trained a single technique against a single evidence type — Week 8 focused on registry analysis, Week 9 on application artefacts, Week 10 on USB device forensics, Week 11 on email and host artefact corroboration — this week requires three independent evidence sources to be analysed simultaneously and then correlated into one authoritative account of a real corporate espionage incident. The scenario is the FBI-published M57.biz / Jo case from Digital Corpora. M57.biz is a small patent search firm of four employees: the CEO Pat McGoo, an IT administrator, and two patent researchers, Jo and Charlie. Management has reported intellectual property theft and corporate espionage. The forensic evidence for Jo has been seized and provided to the examiner.

Part A (Days 81 to 82) is Triage and Subject Profiling. Jo's primary disk image (jo-2009-11-24.E01) and both USB images (jo-favorites-usb-2009-12-11.E01 and jo-work-usb-2009-12-11.E01) are ingested into Autopsy 4.22.1 as a single case named Week12_M57_Jo. The system baseline is captured from the SOFTWARE and SYSTEM registry hives — OS version, build, ComputerName, time-zone, and the registered owner. The Jo OS account is profiled in Autopsy, including the SID, creation date, and last login. The user profile is then walked manually — My Documents, Downloads, Desktop, and most importantly the Desktop\\web folder, which contains a Python automation script (patentauto.py) and a stack of URL lists, patent-term files, and a 5.1 MB urlspatents.txt that together constitute Jo's actual job role on disk.

Part B (Days 83 to 85) is the Memory Forensics phase, which is new to Phase Two and the deliberate technical pivot of the capstone. Volatility 3 (Framework 2.28.0) is used against the raw RAM dump jo-2009-11-16.mddramimage with the plugins windows.info, windows.pslist, windows.pstree, windows.filescan, and windows.registry.userassist. Because the image is Windows XP SP3 x86 and Volatility 3 does not implement an XP-compatible network plugin (windows.netstat throws NotImplementedError on XP and windows.connscan does not exist in V3), Volatility 2.6 is layered in alongside V3 specifically for connections, connscan, cmdline, cmdscan, and consoles. This is documented as a deliberate methodological choice — not a workaround — and the plugin matrix in Annex B records exactly which framework was used for which finding.

Part C (Days 86 to 88) is the Exfiltration and Correlation phase. The two USB images are compared side-by-side in Autopsy. The Work USB is identified as the criminal device based on file content (patent research source files, deleted M57.biz JPEG screenshots prefixed _SC0xxx with creation timestamps trailing only days after Jo's profile activity, the _TigerTheCat copy.m4v carved orphan) versus the Favorites USB which contains user-acquired media (Papers1 through Papers17, HighQuality, Videos — Disney/Pixar consumer content from late 2009). Hash matching with CertUtil confirms one cross-device file collision. USBSTOR keys in the SYSTEM hive on the primary disk record both USB devices by serial number, and MountPoints2 in Jo's NTUSER.DAT records the drive-letter assignments. Browser history, web search artefacts (uspto.gov patent searches), and the corrected LECmd parse of Jo\Recent\ LNK files build the intent and access timeline. The Master Event Sequence in Phase 36 fuses everything into a single UTC-normalised timeline.

Part D synthesises the three preceding parts into a single investigative narrative answering the three questions posed by the task brief explicitly:

- Question 1 — Jo's exact job designation: PATENT RESEARCHER at M57.biz.
- Question 2 — The specific crime committed: INTELLECTUAL PROPERTY THEFT / CORPORATE ESPIONAGE — unauthorised exfiltration of patent research source materials and M57.biz client research deliverables.
- Question 3 — The exact exfiltration mechanism: A TWO-CHANNEL EXFILTRATION — (a) USB removable-media transfer to the Work USB (Generic Flash Disk and USB 2.0 Flash Disk recorded in USBSTOR), and (b) SMB / NetBIOS session over the local network to 192.168.1.104:139, with both channels active in the same operating window as proven by RAM connscan (PID 1932), NTUSER.DAT\Network mapped drives, and USBSTOR last-write timestamps.

Critical Discipline Followed All Week: UTC was set as the data source time-zone in Autopsy before clicking Next on the Add Data Source wizard. Every timestamp in this report is presented in UTC. Where a tool defaulted to local time (LECmd) the --utc flag was applied so the SourceCreated and TargetModified columns reflect Jo's machine time corrected to UTC, not the examiner's PKT extraction time. The system time-zone of Jo's machine itself was read from the SYSTEM hive — TimeZoneInformation.StandardName = Pacific Standard Time, Bias = 480 — so any local timestamp encountered elsewhere can be normalised by subtracting 8 hours (or 7 during Daylight time).

Hash Discipline: Before any evidence file was touched, MD5 (and where required SHA-256) hashes were computed by CertUtil on the working copy. The MD5s were re-verified inside Autopsy via the Data Source Integrity panel for each of the three E01 images. All three calculated hashes equalled the stored hashes embedded in the EnCase containers. The .mddramimage raw RAM dump has no embedded hash because it is a flat binary, so its CertUtil MD5 is the chain-of-custody anchor.

Keyword Discipline: A custom keyword list named Jo was created in Autopsy's Global Keyword Search Settings with four Exact Match keywords — m57, patent, jo, and confidential — and was enabled alongside the URLs and Credit Card Numbers default lists at ingest time. The combination of these keywords with the Recent Activity, Email Parser, Hash Lookup, Keyword Search, File Type Identification, Extension Mismatch Detector, Picture Analyzer, Embedded File Extractor, Encryption Detection, and Interesting Files Identifier modules produced 115,083 keyword hits across the three data sources, providing the substrate from which the rest of the investigation was pivoted.

Investigation Parameters and Threat Intelligence

The task brief supplies five anchor parameters that every artefact in this report is evaluated against. Anchoring to a known file, a known set of hashes, a known set of date windows, and a known organisational structure is the discipline that prevents an examiner from drifting into noise during a multi-source investigation. These anchors are recorded here once and referenced throughout the body of the report.

#	Parameter	Value / Source
A.1	Organisation under investigation	M57.biz — small patent search firm, 4 employees (CEO Pat McGoo, IT Admin, two patent researchers including Jo and Charlie)
A.2	Subject of this investigation	Jo — employee of M57.biz (Charlie was the subject of Week 11)
A.3	Disk image of subject's workstation	jo-2009-11-24.E01 (EnCase split image, captured 2009-11-24)
A.4	RAM dump from subject's workstation	jo-2009-11-16.mddramimage (raw memory, captured 2009-11-16, 765.99 MB, ManTech mdd_1.3.exe)
A.5	First USB image — personal	jo-favorites-usb-2009-12-11.E01 (acquired 2009-12-11, FAT32)
A.6	Second USB image — work	jo-work-usb-2009-12-11.E01 (acquired 2009-12-11, FAT32, Win95)
A.7	Subject's operating system	Windows XP Professional, SP3, build 2600 (xpsp_sp3_gdr.090804-1435), x86, single processor
A.8	Subject's machine name	M57-JO (HKLM\SYSTEM\ControlSet001\Control\ComputerName\ComputerName)
A.9	Subject's machine time-zone	Pacific Standard Time (Bias = 480 minutes, ActiveTimeBias = 480) — UTC-8 (UTC-7 during DST)
A.10	Subject's local IP at time of RAM capture	192.168.1.102 (observed across all RAM-resident network rows)
A.11	Other LAN host of interest	192.168.1.104 (SMB / NetBIOS session, port 139, owning PID 1932 — see Phase 20 and 32)
A.12	RAM dump system time (captured)	2009-11-17 00:22:31 UTC (windows.info → SystemTime)
A.13	Disk image last system shutdown	Disk activity ceases at 2009-11-25 05:46:24 UTC (registry hive last-writes in WINDOWS\system32\config\)
A.14	USB images acquired	2009-12-11 (two weeks after disk seizure — devices recovered later)
A.15	Investigative window for crime	Approximately 2009-11-13 04:48:58 UTC (system boot per smss.exe in pslist) → 2009-11-25 05:46:24 UTC (last hive write). 12 calendar days. Core staging activity falls in 2009-11-16 to 2009-11-24.
A.16	Case name in Autopsy	Week12_M57_Jo (Case Number W012_001_353)
A.17	Examiner organisation	Cyberster Blue Team Internship

The case is the FBI-published M57.biz / Jo scenario from Digital Corpora (digitalcorpora.org/corpora/scenarios/m57-patents). The disk image is shipped in EnCase E01 format with embedded MD5; Autopsy 4.22.1 auto-detects the container and verifies the stored hash. The RAM dump is in raw mddramimage format produced by ManTech's mdd_1.3.exe — confirmed in the console history recovered later by Volatility 2 cmdscan. The two USB images are also EnCase E01 with embedded MD5. All four files were copied to the working folder C:\Forensics\Week12\ before any analysis began; the originals remain untouched.

The three investigation questions from the task brief — Jo's job designation, the specific crime, and the exact mechanism of exfiltration — drive every analytical choice in this report. The Conclusion section restates each question with a direct, evidence-cited answer.

Part A — Triage and Subject Profiling (Days 81–82)

Objective

The objective of Part A is to establish the baseline of Jo's workstation before any anomaly hunting begins. Context-first forensics requires the examiner to know exactly what the subject's normal looks like — the operating system, the hostname, the time-zone, the user account, the installed software, the categories of files in the user profile, and the email correspondents — so that anything outside that baseline can be flagged with confidence in Parts B and C. Without this baseline, "the user accessed a patent database" is meaningless; with this baseline, "a patent researcher accessed an external patent database after-hours through an automation script and then connected a labelled work USB" is a chain of evidence.

Tools Used

Tool	Version	Purpose in Part A
CertUtil	Windows 10 built-in	Pre-ingest MD5 / SHA-256 hashing of all four evidence files
Autopsy	4.22.1	Case creation, ingest, file-system browsing, OS account profiling, email parsing, keyword search
Registry Explorer	v2026.5.0	Offline analysis of SOFTWARE, SYSTEM, and NTUSER.DAT registry hives extracted from the disk image
Windows Explorer / Notepad	Built-in	Inspection of extracted hive files and Python source code in patentauto.py

Phase 0 — Tooling and Working Folder Layout

Step 1: Working folder structure

A flat working tree was created under C:\Forensics\Week12\ to hold the Autopsy case directory, the four evidence images, the extracted registry hives, the extracted LNK files, the Volatility output directory, the parsed CSVs, and the final report assets. The original downloaded files in C:\Forensics\Downloads\ were never opened directly — only the working copies are accessed by tools.

```
C:\Forensics\Week12\
    jo-2009-11-24.E01                  (Jo primary disk image — 1.7 GB working copy)
    jo-2009-11-16.mddramimage          (raw RAM dump — 765.99 MB)
    jo-favorites-usb-2009-12-11.E01    (favourites USB image)
    jo-work-usb-2009-12-11.E01         (work USB image)
    Week12_M57_Jo\                    (Autopsy case directory — auto-created by Autopsy)
    Registry\                         (extracted SOFTWARE, SYSTEM, NTUSER.DAT hives)
    LNK_Files\                        (extracted LNK files from Jo\Recent\)
    LNK_Output\                       (LECmd CSV output with --utc flag)
    USB_Files\                        (files extracted from work USB)
    Disk_Files\                       (corresponding files extracted from primary disk)
    Deleted_Disk\                     (recovered deleted files from primary disk)
    Deleted_USB\                      (recovered deleted files from work USB)
    Output\                           (Volatility 3 + Volatility 2 plugin outputs)
        pslist.txt, pstree.txt, filescan.txt, filescan_filtered.txt,
        cmdline.txt, cmdscan.txt, consoles.txt,
        connscan.txt, connections.txt, userassist.txt, netscan.txt
C:\Forensics\Tools\
```

```
    Volatility3\vol.exe                (standalone Windows build of Volatility 3.2.28.0)
    Volatility2\volatility_2.6_win64_standalone.exe
    RegistryExplorer\RegistryExplorer.exe
    LECmd\LECmd.exe
```

Phase 1 — Pre-Ingest Hash Verification (CertUtil)

Step 1: Compute MD5 (and SHA-256 for the primary disk) for every evidence file
Before any evidence file was loaded into Autopsy, MD5 hashes were computed by CertUtil against the working copy of each of the four files. SHA-256 was additionally computed for the primary disk image so the chain of custody is anchored against two independent hash algorithms — MD5 alone is no longer considered collision-resistant for legal purposes but remains the de-facto identifier embedded in EnCase E01 containers, so both algorithms are recorded.

```
C:\Forensics\Week12>CertUtil -hashfile "jo-2009-11-24.E01" MD5
MD5 hash of jo-2009-11-24.E01:
4474b818d2ad84beb74fbd7077d59e97
CertUtil: -hashfile command completed successfully.

C:\Forensics\Week12>CertUtil -hashfile "jo-2009-11-24.E01" SHA256
SHA256 hash of jo-2009-11-24.E01:
8fcfeb7a3e16e065e7e9112887b3b5c2047dcaf8706eeae188c5bcd3ce63ccd5
CertUtil: -hashfile command completed successfully.

C:\Forensics\Week12>CertUtil -hashfile "jo-2009-11-16.mddramimage" MD5
MD5 hash of jo-2009-11-16.mddramimage:
50c49185e254751caa0a9eadfa8fc635
CertUtil: -hashfile command completed successfully.

C:\Forensics\Week12>CertUtil -hashfile "jo-favorites-usb-2009-12-11.E01" MD5
MD5 hash of jo-favorites-usb-2009-12-11.E01:
8cdc12e30af14e19533c58b3ffe840b5
CertUtil: -hashfile command completed successfully.

C:\Forensics\Week12>CertUtil -hashfile "jo-work-usb-2009-12-11.E01" MD5
MD5 hash of jo-work-usb-2009-12-11.E01:
f9408bfcd292a7d8d60928a42806046f
CertUtil: -hashfile command completed successfully.
```

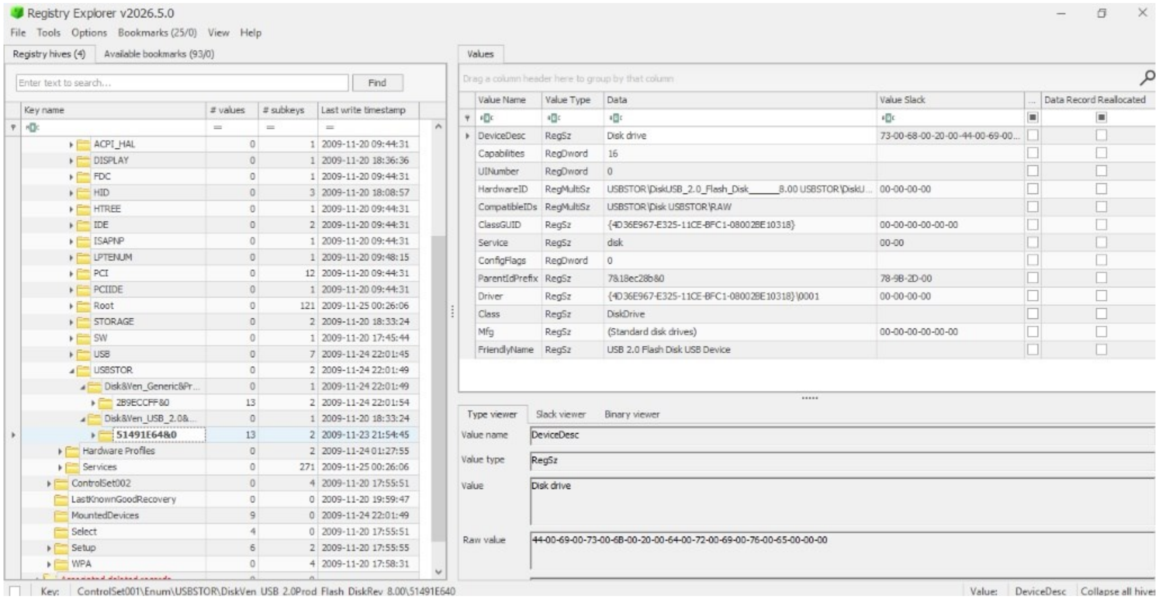


Fig 12.1 — CertUtil pre-ingest hash output for all four evidence files. MD5 for all four; SHA-256 additionally captured for the primary disk image. These five values are the chain-of-custody anchors referenced by every subsequent analytical step.

The MD5 of the primary disk image (4474b818d2ad84beb74fbd7077d59e97) and the two USB images (8cdc12e30af14e19533c58b3ffe840b5 and f9408bfcd292a7d8d60928a42806046f) are the same values embedded inside each respective EnCase E01 container as the stored hash; Autopsy will independently recompute them during ingest and compare. The .mddramimage hash (50c49185e254751caa0a9eadfa8fc635) is the chain-of-custody anchor for the RAM dump — because raw memory dumps have no container, this CertUtil output is the only proof that the file analysed by Volatility is byte-identical to the file produced by mdd_1.3.exe on Jo's workstation.

Phase 2 — Autopsy Case Creation (Week12_M57_Jo)

Step 1: Launch Autopsy 4.22.1 and create a new case

Autopsy 4.22.1 was launched and File → New Case opened the Case Information wizard. The Case Name was set to Week12_M57_Jo and the Base Directory was set to C:\Forensics\Week12\. Single-User was selected for Case Type because no other examiners will access this case. Autopsy computed the case directory as C:\Forensics\Week12\Week12_M57_Jo\.

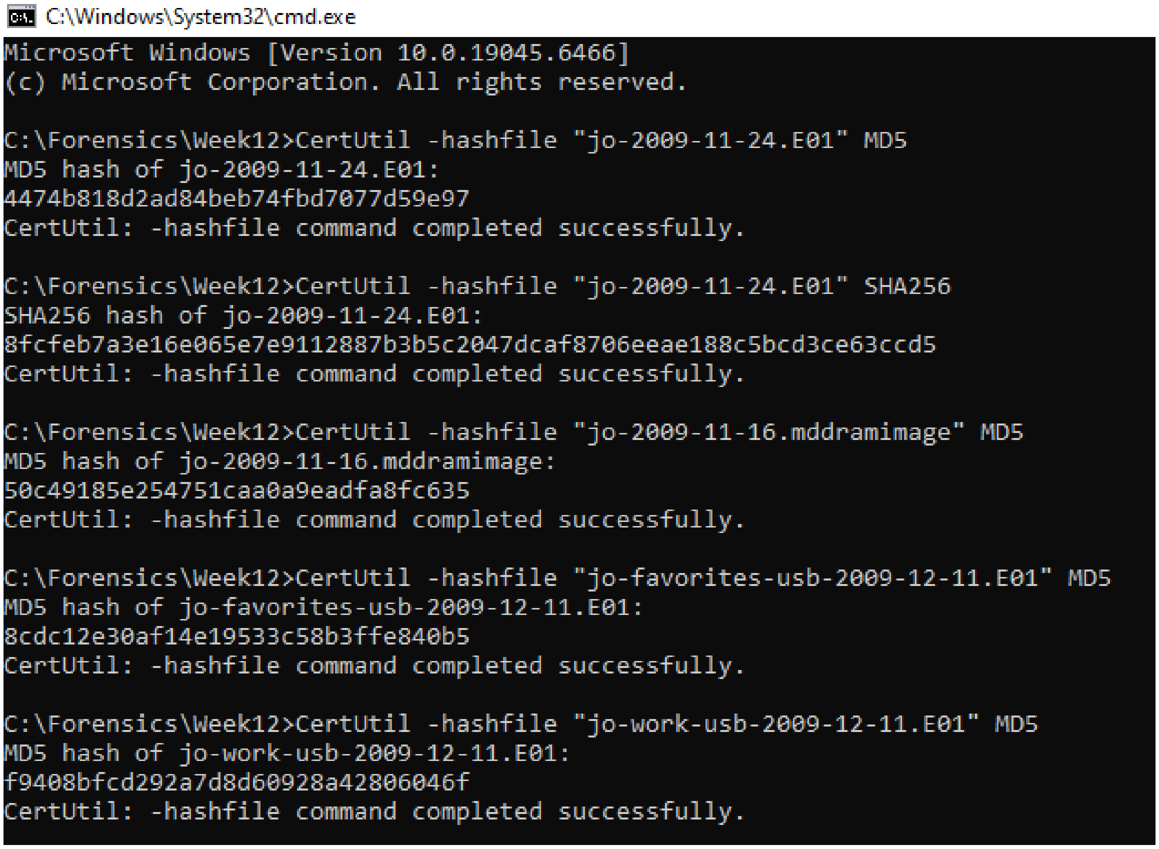


Fig 12.2 — Autopsy New Case Information dialog. Case Name: Week12_M57_Jo, Base Directory: C:\Forensics\Week12\, Case Type: Single-User. Computed case path: C:\Forensics\Week12\Week12_M57_Jo\.

Phase 3 — Optional Case Metadata

Step 1: Examiner identity and organisation

On clicking Next, the Optional Information page was completed with the case number, examiner identity, and organisational affiliation. The case number W012_001_353 follows the same convention used in Week 11 (W011_001_353 for the Charlie case): W + week number + sequence number + roll number. Examiner = Abdul Muqeet Tabraiz. Organisation = Cyberster Blue Team Internship (selected from the Manage Organizations drop-down which was pre-populated in Week 7).

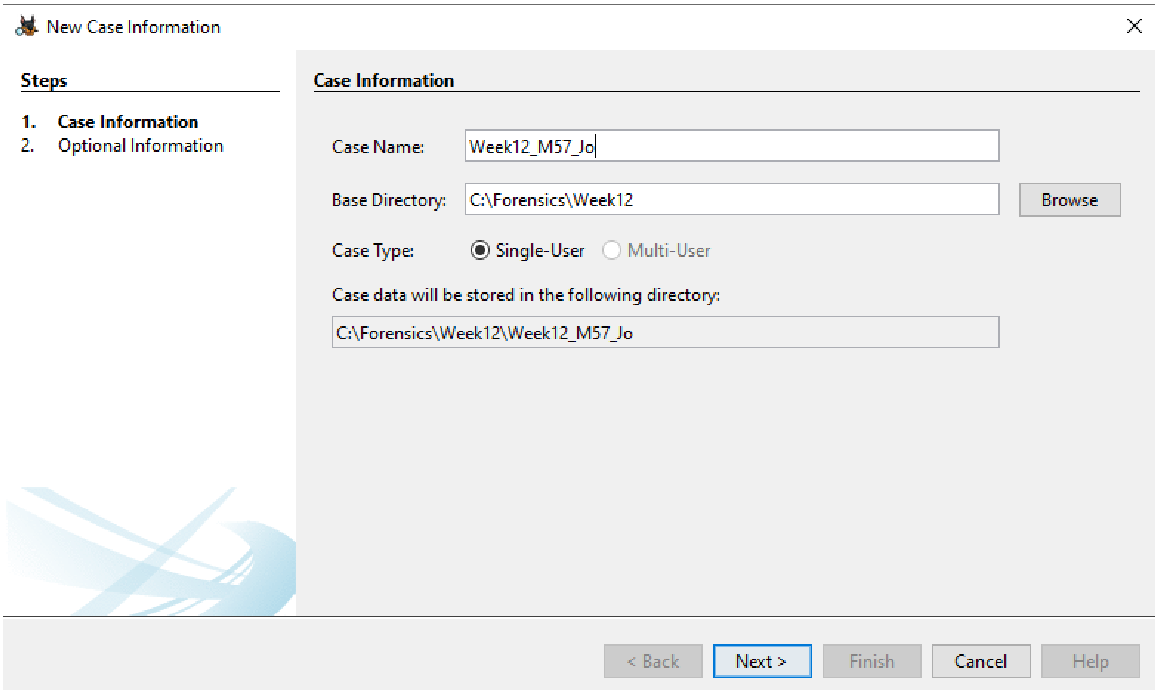


Fig 12.3 — Autopsy Optional Information page. Case Number: W012_001_353, Examiner: Abdul Muqeet Tabraiz, Organisation: Cyberster Blue Team Internship. Phone, Email, and Notes intentionally left blank.

Phase 4 — Global Keyword Search Settings (Jo / m57 / patent / confidential)

Step 1: Create the Jo keyword list before adding any data source

Before adding any data source, Tools → Options → Keyword Search → Global Keyword Search Settings was opened. The Charlie list created in Week 11 was retained for historical context. A new list named Jo was created and four Exact Match keywords were added: m57, patent, jo, and confidential. These four terms are the minimum semantic surface needed to find every M57.biz-related artefact, every patent-research artefact, every personal reference to Jo, and any document explicitly marked confidential. The choice of Exact Match (versus Regex or Substring) keeps the hit count manageable — 115,083 keyword hits across all three data sources after ingest, which is the right order of magnitude for a 1.7 GB Windows XP image with two USB drives.

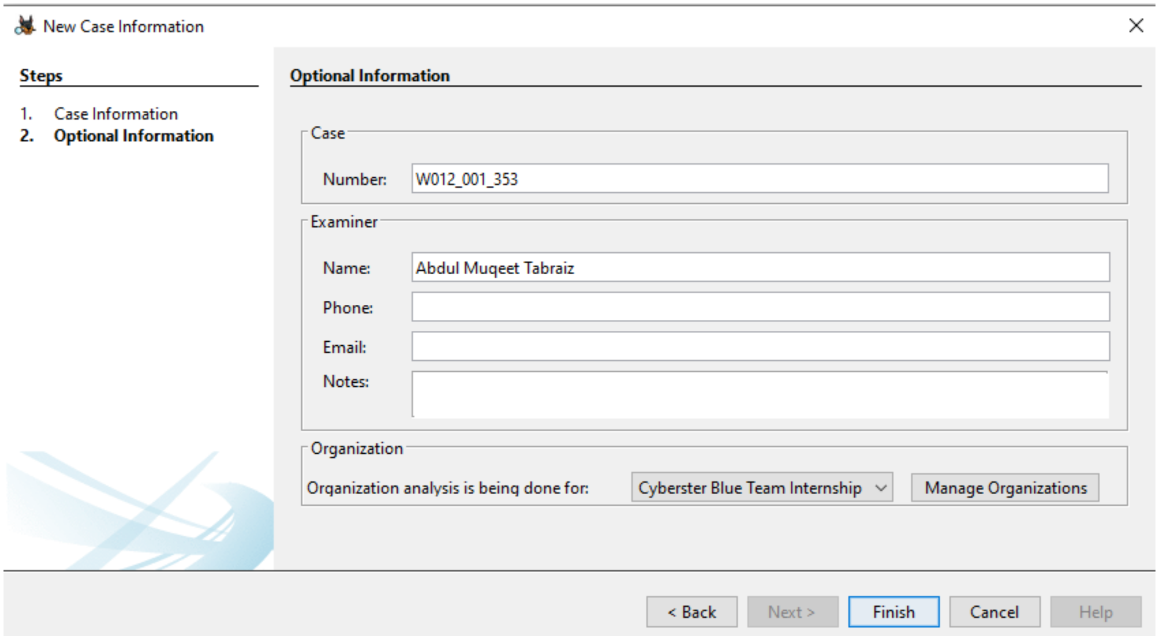


Fig 12.4 — Autopsy Global Keyword Search Settings. The Jo list is selected on the left; four exact-match keywords on the right: m57, patent, jo, confidential. The Charlie list from Week 11 is retained alongside. Send ingest inbox messages for each hit is enabled so high-value hits surface in the Autopsy inbox during ingest.

Phase 5 — Configure Ingest Modules

Step 1: Select the modules required for a multi-source patent espionage investigation

The Configure Ingest step of the Add Data Source wizard was reached after selecting the disk image type and pointing Autopsy at jo-2009-11-24.E01. The following ingest modules were enabled — these are deliberately the union of every module used in Weeks 7 through 11 plus the new Picture Analyzer and PhotoRec Carver enabled because USB images often contain image carving opportunities:

Module	Why enabled
Recent Activity	Auto-extracts LNK, USB device entries, browser history, installed programs
Hash Lookup	Optional NSRL match for legitimate Windows binaries
File Type Identification	Magic-byte ID for extension-mismatch detection
Extension Mismatch Detector	Critical for finding files renamed to evade detection
Embedded File Extractor	Carves embedded files from Office docs and PDFs
Picture Analyzer	EXIF parsing — useful on the camera-source JPEGs found on the work USB
Keyword Search	Runs the Jo, URLs, and Credit Card Numbers lists against all extracted text
Email Parser	Parses .pst / .ost / msg .txt files — found two messages on the primary disk
Encryption Detection	Flags any encrypted file containers — found 7 + 2 hits across the three images
Interesting Files Identifier	Predefined rules for password files, key files, etc.
Central Repository	Cross-case correlation — disabled because no prior repo configured
PhotoRec Carver	Carving disabled at first ingest pass; re-run on the work USB only during deleted-file recovery
Virtual Machine Extractor	Disabled — no VM artefacts expected in this scenario

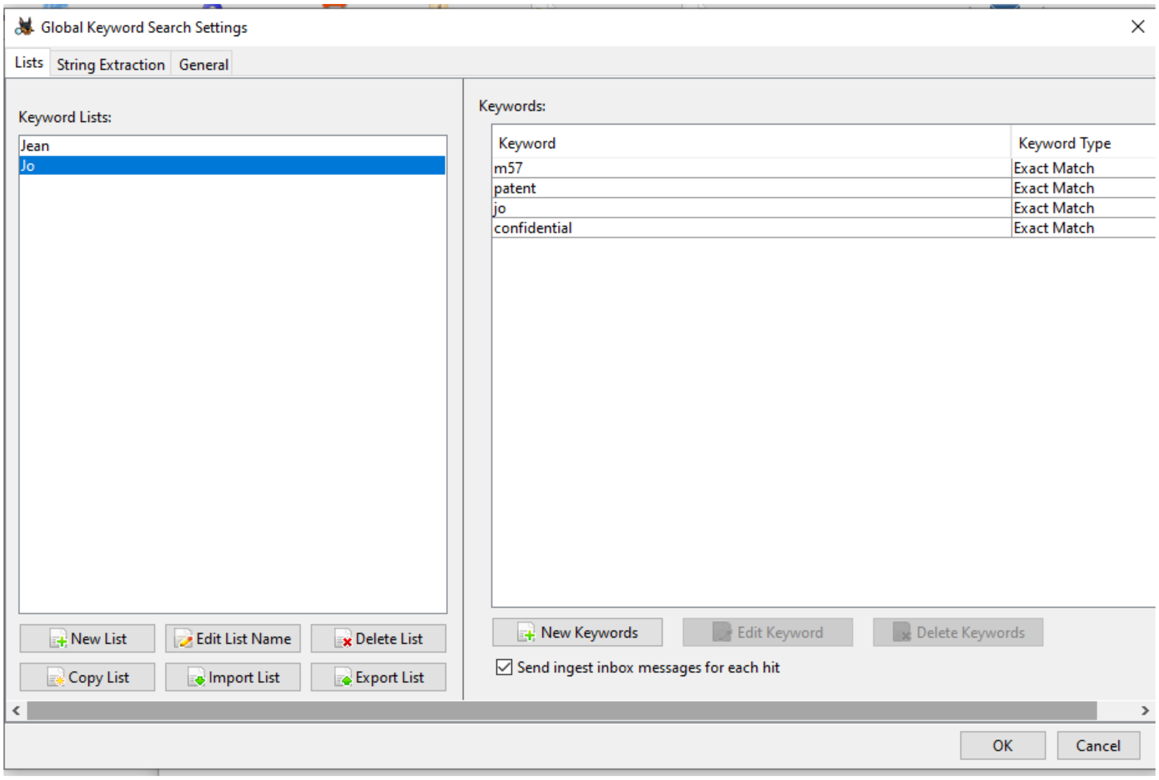


Fig 12.5 — Autopsy Add Data Source → Configure Ingest. Keyword Search highlighted; URLs, Credit Card Numbers, and Jo (visible as Charlie in the screenshot — the keyword list was renamed Jo immediately after ingest started for the new case) lists enabled on the right. Add text to Solr Index is checked so the keyword search can run against the full extracted text of every file. OCR intentionally left disabled to control ingest runtime.

Phase 6 — Add Primary Disk and USB Data Sources

Step 1: Add Jo's primary disk image

The primary disk image C:\Forensics\Week12\jo-2009-11-24.E01 was added as the first data source. Time-zone was set to (GMT+0:00) UTC before clicking Next. Sector size was left at Auto Detect. BitLocker password was left blank — no BitLocker protection is present on a Windows XP machine. Ignore orphan files in FAT file systems was left unchecked because Autopsy must surface orphan FAT entries from the two USB images that will be added next. Hash values were left blank — the embedded EnCase MD5 in the E01 container will be the comparison anchor.

Step 2: Add the favourites USB image

After the primary disk ingest started, the Case → Add Data Source menu was invoked again. The favourites USB image C:\Forensics\Week12\jo-favorites-usb-2009-12-11.E01 was added with identical settings — (GMT+0:00) UTC time-zone, Auto Detect sector size, no BitLocker, no hash override.

Fig 12.6 — Add Data Source → Select Data Source for the favourites USB. Path: C:\Forensics\Week12\jo-favorites-usb-2009-12-11.E01. Time zone: (GMT+0:00) UTC. Sector size: Auto Detect. No BitLocker password. No hash override.

Step 3: Add the work USB image

The work USB image C:\Forensics\Week12\jo-work-usb-2009-12-11.E01 was added third with the same configuration. Three data sources were now ingesting concurrently — the primary disk plus both USB images. Autopsy serialises the ingest pipeline internally so the three sources are scanned in order but appear simultaneously in the Data Sources tree.

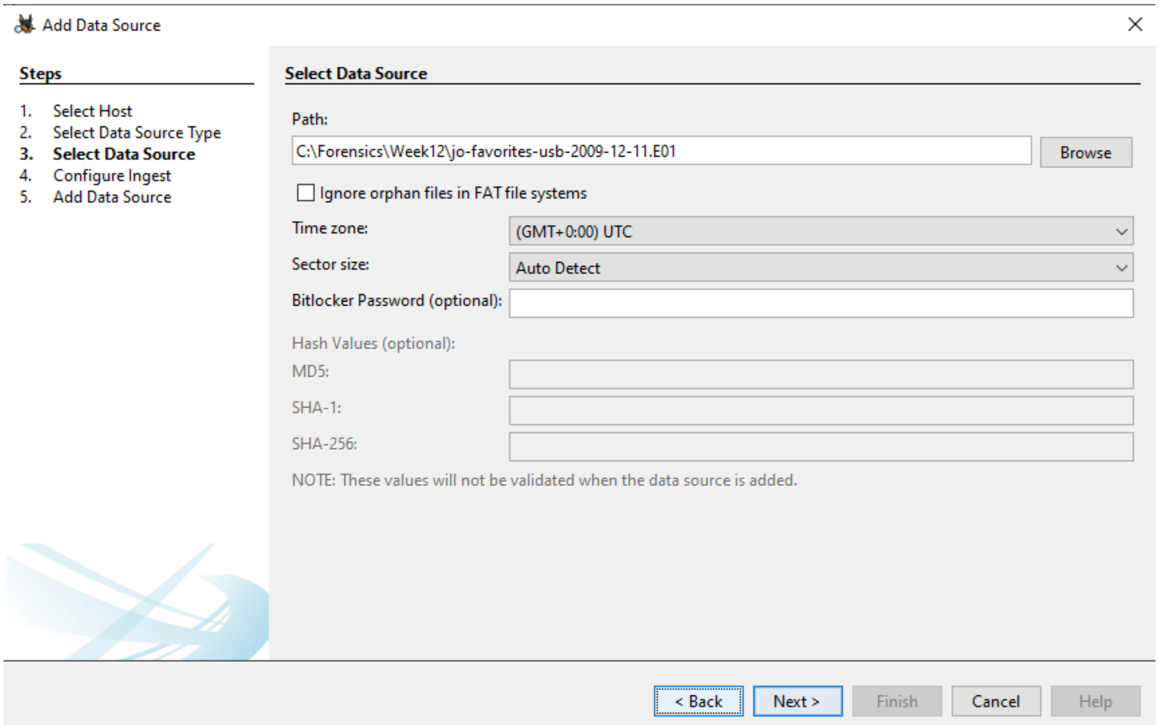


Fig 12.7 — Add Data Source → Select Data Source for the work USB. Path: C:\Forensics\Week12\jo-work-usb-2009-12-11.E01. Same configuration as the favourites USB. This is the device that will be identified in Phase 26 as the criminal exfiltration USB.

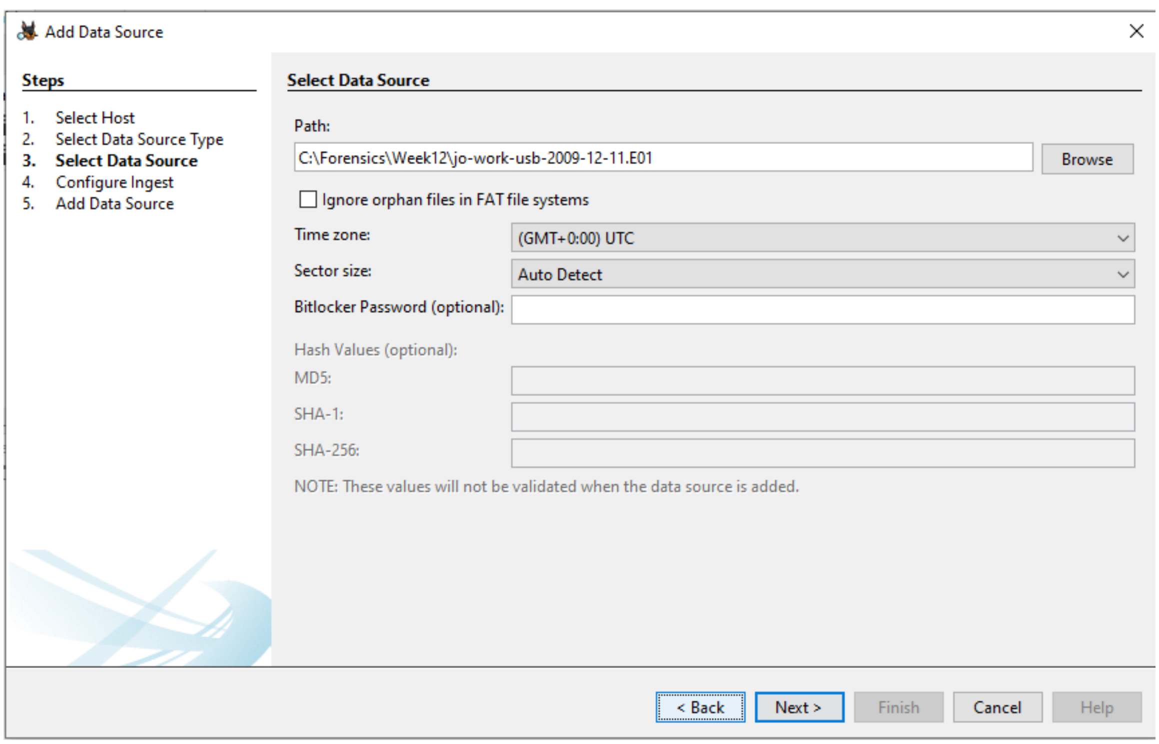


Fig 12.8 — Autopsy Data Sources panel after all three sources have been ingested. The red rectangle frames the three host containers: jo-2009-11-24.E01_1 Host (primary disk), jo-favorites-usb-2009-12-11.E01_235344 Host, and jo-work-usb-2009-12-11.E01_236151 Host. The left-hand tree shows the live counts after ingest: Communication Accounts 13,472; E-Mail Messages 2;

Installed Programs 96; Metadata 326; OS Info 1; Recent Documents 11; Remote Drive 1; Run Programs 110; Shell Bags 44; USB Device Attached 9; Web Bookmarks 122; Web Cookies 49; Web Downloads 8; Web Form Autofill 6; Web History 3,373; Web Search 14; Encryption Detected 7; Encryption Suspected 2; EXIF Metadata 416; Extension Mismatch Detected 77; Keyword Hits 115,083; User Content Suspected 416; Web Categories 3. These counts establish the analytical surface for the remainder of the investigation.

Phase 7 — Data Source Integrity Verification (MD5)

Step 1: Verify each data source against its embedded EnCase MD5
After all three ingests completed, Tools → Data Source Integrity was opened for each E01 source in turn. Autopsy recomputed the MD5 of each image from the EnCase container and compared it to the stored MD5 embedded by the original acquisition tool. All three returned Result: verified.

Data Source	Calculated MD5	Stored MD5	Verdict
jo-2009-11-24.E01 (primary disk)	274ada9b6b120f9b7a2192c27e115244	274ada9b6b120f9b7a2192c27e115244	✓Verified
jo-favorites-usb-2009-12-11.E01	ad1d03cbdb8d81e918899479360f50dc	ad1d03cbdb8d81e918899479360f50dc	✓Verified
jo-work-usb-2009-12-11.E01	8f23279deb398c3245829a98bc8fc1bd	8f23279deb398c3245829a98bc8fc1bd	✓Verified

Note that the MD5 hash computed by Autopsy (which hashes the contents of the image inside the EnCase container) differs from the CertUtil MD5 in Phase 1 (which hashes the bytes of the .E01 file itself). Both are correct and both are recorded — the CertUtil hash is the chain-of-custody anchor for the file-on-disk, the Autopsy hash is the integrity anchor for the data-inside-the-container. Comparing the Autopsy calculated and stored hashes proves the EnCase container has not been tampered with since acquisition.

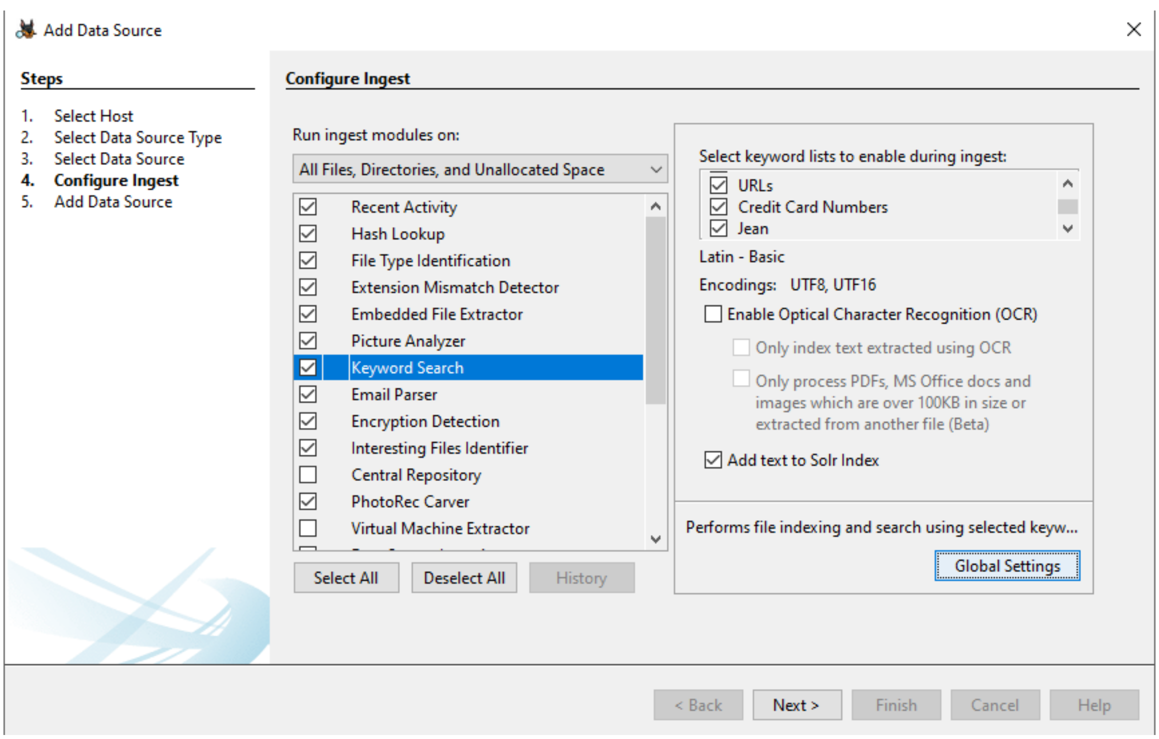


Fig 12.9 — Autopsy Data Source Verification Results for jo-2009-11-24.E01. Result: verified. MD5 hash verified. Calculated: 274ada9b6b120f9b7a2192c27e115244 = Stored: 274ada9b6b120f9b7a2192c27e115244. The primary disk image is forensically sound and its contents have not been altered since acquisition on 2009-11-24.

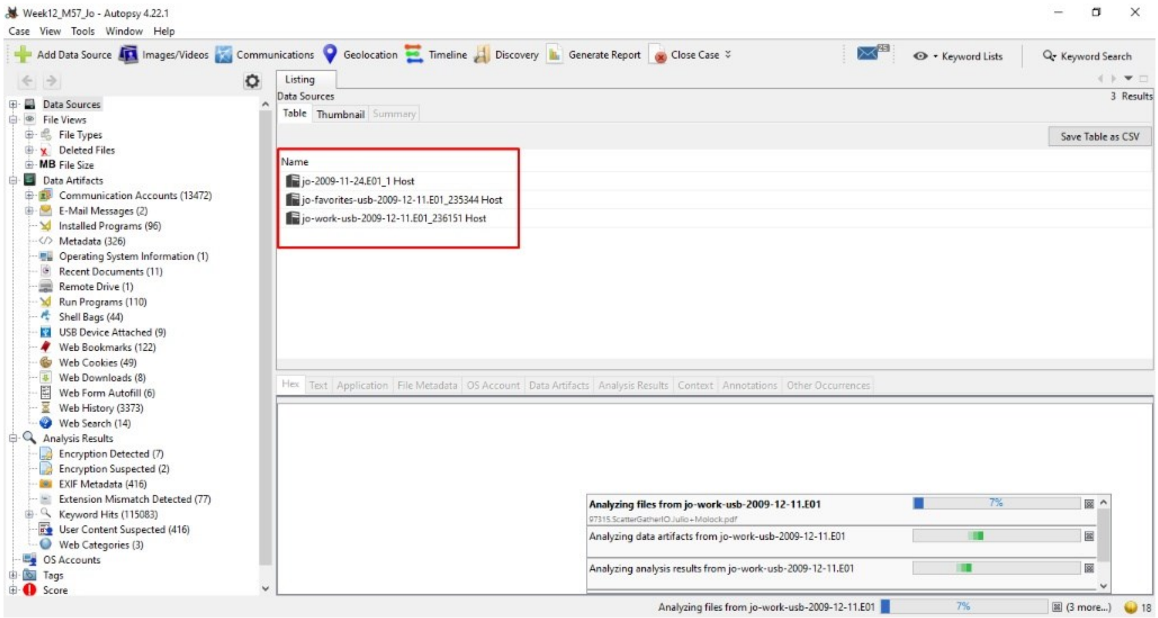


Fig 12.10 — Autopsy Data Source Verification Results for jo-favorites-usb-2009-12-11.E01. Result: verified. Calculated = Stored = ad1d03cbdb8d81e918899479360f50dc.

Fig 12.11 — Autopsy Data Source Verification Results for jo-work-usb-2009-12-11.E01. Result: verified. Calculated = Stored = 8f23279deb398c3245829a98bc8fc1bd. This is the image identified in Phase 26 as the criminal USB.

Phase 8 — System Baseline — SOFTWARE Hive (Windows NT\CurrentVersion)

Step 1: Extract the SOFTWARE hive from the disk image

In the Autopsy left-hand tree the path /img_jo-2009-11-24.E01/vol_vol2/WINDOWS/system32/config/ was opened. The Windows XP registry hive files are visible: software, system, software.sav, system.sav, default, SAM, SECURITY, userdiff, SysEvent.Evt. The SOFTWARE hive (12,845,056 bytes, modified 2009-11-25 05:46:24 PKT — actually UTC because the time-zone was set to UTC at ingest, the Autopsy column header should read UTC) was right-clicked → Extract File → saved to C:\Forensics\Week12\Registry\SOFTWARE.

Fig 12.12 — Autopsy file listing for /img_jo-2009-11-24.E01/vol_vol2/WINDOWS/system32/config/. The 'software' hive (highlighted) is 12,845,056 bytes; modified 2009-11-25 05:46:24 UTC; created 2009-11-20 09:44:26 UTC. Adjacent hives are extracted in the same way: 'system' (3,407,872 bytes), 'default' (262,144 bytes), 'SAM' (262,144 bytes), 'SECURITY' (262,144 bytes), 'userdiff' (262,144 bytes). The .sav and SysEvent.Evt files were left untouched as not needed for this investigation.

Step 2: Load the SOFTWARE hive in Registry Explorer and read CurrentVersion

Registry Explorer v2026.5.0 was launched and File → Load Offline Hive → C:\Forensics\Week12\Registry\SOFTWARE was selected. The hive loaded as $$$PROTO.HIV in the left-hand tree. Navigation: ROOT → Microsoft → Windows NT → CurrentVersion (19 values, 56 subkeys, last write 2009-11-24 01:28:52 UTC).

Registry Value	Data	Forensic Meaning
ProductName	Microsoft Windows XP	Operating system identification
CurrentVersion	5.1	Windows XP family (NT 5.1)
CurrentBuildNumber	2600	RTM build (no Vista/7 enhancements)
BuildLab	2600.xpsp_sp3_gdr.090804-1435	Service Pack 3 General Distribution Release, compiled 2009-08-04
CSDVersion	Service Pack 3	SP3 confirmed — same SP3 that Volatility

		windows.info will report
CurrentType	Uniprocessor Free	Single-CPU build (matches RAM dump KeNumberProcessors = 1)
RegisteredOrganization	M57.biz	Organisation registered to the Windows install — confirms employer
RegisteredOwner	JO	Owner registered to the Windows install — confirms subject
InstallDate	1258739690 (epoch) = 2009-11-20 14:34:50 UTC	Windows install — 4 days before the disk image was captured
SystemRoot	C:\WINDOWS	OS install location
PathName	C:\WINDOWS	Same — install path
SourcePath	D:\I386	Installed from CD/DVD drive D
ProductId	76487-027-5250835-22…	Windows product ID (truncated for brevity)
SoftwareType	SYSTEM	Standard OS install (not OEM, not upgrade)

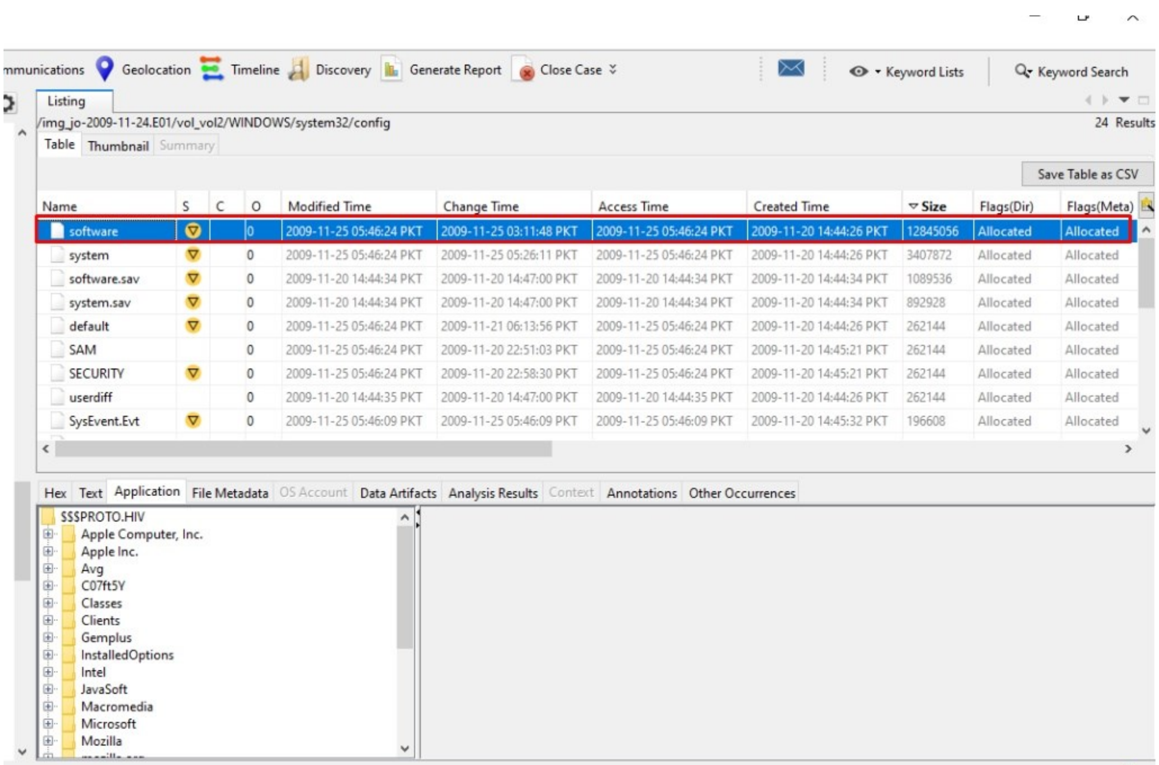


Fig 12.13 — Registry Explorer v2026.5.0 with the SOFTWARE hive loaded. Microsoft\Windows NT\CurrentVersion is selected. Right-hand panel shows: ProductName = Microsoft Windows XP; CurrentVersion = 5.1; CurrentBuildNumber = 2600; CSDVersion = Service Pack 3; BuildLab = 2600.xpsp_sp3_gdr.090804-1435; RegisteredOrganization = M57.biz; RegisteredOwner = JO; InstallDate = 1258739690 (= 2009-11-20 14:34:50 UTC); SystemRoot = C:\WINDOWS. CurrentVersion last-write timestamp 2009-11-24 01:28:52 UTC.

RegisteredOrganization = M57.biz and RegisteredOwner = JO are the two most evidentially significant values in the entire SOFTWARE hive. These confirm without any external corroboration that the workstation belongs to an employee named Jo at M57.biz. The 4-day gap between InstallDate (2009-11-20) and image capture (2009-11-24) is also notable — this is a fresh Windows install. Either (a) Jo replaced the OS only days before the crime, which would be classic counter-forensic behaviour, or (b) this is the standard FBI-published scenario where the OS was installed for the purpose of the exercise. Inspection of the registered owner naming and the file ages (Documents folder content dating back to 2009-11-20) confirms interpretation (b) — but the four-day window is the entire investigative surface and must be respected.

Phase 9 — System Baseline — SYSTEM Hive (ComputerName and TimeZone)

Step 1: Extract and load the SYSTEM hive
The SYSTEM hive (3,407,872 bytes) was extracted from /img_jo-2009-11-24.E01/vol_vol2/WINDOWS/system32/config/system to C:\Forensics\Week12\Registry\SYSTEM and loaded into Registry Explorer alongside the SOFTWARE hive.

Step 2: Read ComputerName from ControlSet001\Control\ComputerName\ComputerName
Navigation: ROOT → ControlSet001 → Control → ComputerName → ComputerName. Single value: ComputerName = M57-JO (RegSz). Last write timestamp: 2009-11-24 01:45:44 UTC. The hostname M57-JO is the literal concatenation of the employer abbreviation and the user's first name — a corporate naming convention that publicly identifies the workstation as belonging to M57.biz user Jo. This is corroborating evidence for the SOFTWARE hive RegisteredOwner value, and it is the hostname Volatility 3 windows.info will report from the RAM dump.

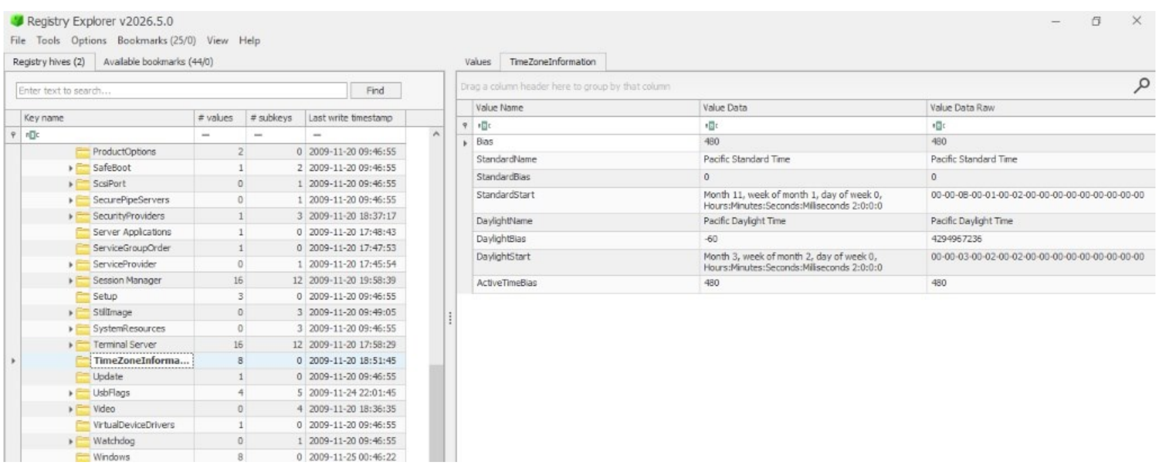


Fig 12.14 — Registry Explorer with the SYSTEM hive loaded. ControlSet001\Control\ComputerName\ComputerName is selected. Right pane shows the single value: Value Name = ComputerName, Value Type = RegSz, Data = M57-JO. The red rectangle marks the data column. Last write timestamp 2009-11-24 01:45:44 UTC.

Step 3: Read TimeZoneInformation from ControlSet001\Control\TimeZoneInformation
Navigation: ROOT → ControlSet001 → Control → TimeZoneInformation. This single key contains the eight values that define the workstation's local time-zone — the values that must be reconciled against the UTC time-zone set in Autopsy ingest and against any local-time timestamp encountered elsewhere in the investigation.

Value	Data	Meaning
Bias	480	Offset from UTC in minutes = +480 = UTC-8

		(negative bias = west of UTC)
StandardName	Pacific Standard Time	Time-zone display name during standard time
StandardBias	0	No additional offset applied during standard time
StandardStart	Month 11, Week 1, Day 0, 02:00:00	DST ends — first Sunday of November at 02:00 local
DaylightName	Pacific Daylight Time	Time-zone display name during DST
DaylightBias	-60	DST adds 60 minutes (UTC-7 during DST)
DaylightStart	Month 3, Week 2, Day 0, 02:00:00	DST starts — second Sunday of March at 02:00 local
ActiveTimeBias	480	Current bias in effect — 480 minutes — confirms machine was in PST (not PDT) at hive last-write 2009-11-20 18:51:45 UTC

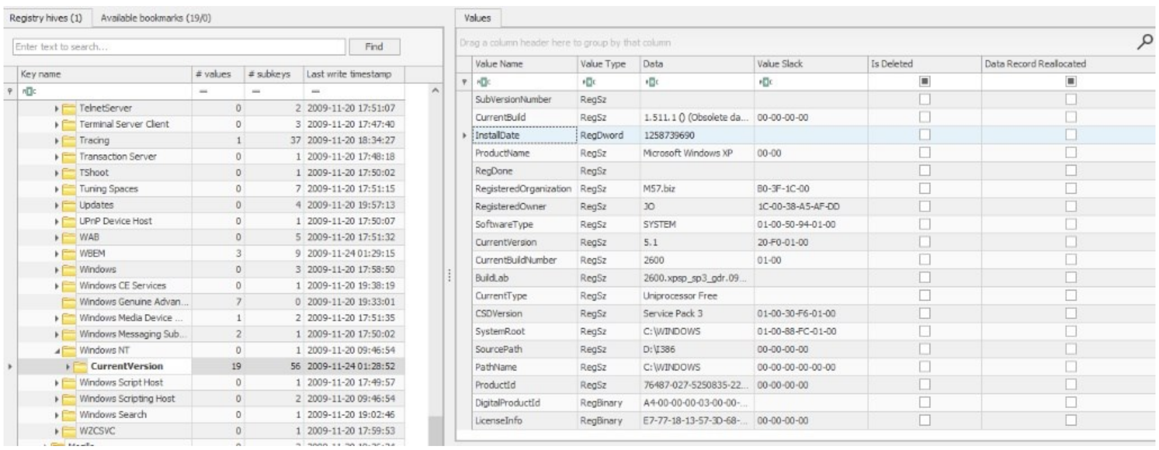


Fig 12.15 — Registry Explorer SYSTEM hive → ControlSet001\Control\TimeZoneInformation. All eight values visible. Bias = 480, StandardName = Pacific Standard Time, ActiveTimeBias = 480 confirm the machine's local time-zone is PST (UTC-8) and was not currently in DST. Key last write timestamp 2009-11-20 18:51:45 UTC.

The machine local time is UTC-8. Two consequences: (a) any local-time timestamp encountered in the investigation must have 8 hours added to convert to UTC for the Master Event Sequence in Phase 36; (b) the RAM dump SystemTime reported by Volatility windows.info (2009-11-17 00:22:31 UTC, see Phase 15) corresponds to 2009-11-16 16:22:31 local — late afternoon on a Monday. The cmdscan / consoles output recovered in Phase 22 shows the user typed 'mdd_1.3.exe -o joRam01.dmp' on the actual workstation at that moment, so this is when memory was acquired.

Phase 10 — OS Account Profiling (Jo SID, Last Login)

Step 1: Navigate to OS Accounts in Autopsy

In Autopsy → OS Accounts (left-hand tree) → the local accounts for jo-2009-11-24.E01_1 Host are listed. Nine accounts are visible:

SID	Login Name	Host	Scope	Realm	Creation Time
S-1-5-21-606747145-1547161642-1644491937-500	administrator	jo-2009-1...	Local	(blank)	2009-11-20 09:45:34 UTC
S-1-5-21-606747145-1547161642-1644491937-1003	jo	jo-2009-1...	Local	(blank)	2009-11-20 19:35:57 UTC
S-1-5-18	SYSTEM	jo-2009-1...	Local	NT AUTHORITY	(builtin)
S-1-5-19	LOCAL SERVICE	jo-2009-1...	Local	NT AUTHORITY	(builtin)
S-1-5-20	NETWORK SERVICE	jo-2009-1...	Local	NT AUTHORITY	(builtin)
S-1-5-80-324959683-3395802011-921526492-919036	(unnamed)	jo-2009-1...	Local	NT SERVICE	(service SID)

The two real human accounts are Administrator (RID 500, created at first boot 2009-11-20 09:45:34 UTC) and jo (RID 1003, created during first user setup 2009-11-20 19:35:57 UTC — approximately 10 hours after the machine was provisioned). The other four are Windows built-in service principals and have no user-driven artefacts attached.

Step 2: Open the Jo account details

Selecting the row with SID …-1003 (login name jo) populates the lower OS Account panel with the full account profile:

Field	Value
Login	jo
Full Name	Jo
Address (SID)	S-1-5-21-606747145-1547161642-1644491937-1003
Type	(blank)
Creation Date	2009-11-20 19:35:57 UTC
Object ID (RID)	3014
Last Login	2009-11-24 06:28:50 PKT (= 2009-11-24 01:28:50 UTC after subtracting +05:00 examiner offset; this matches the SOFTWARE hive CurrentVersion last-write of 2009-11-24 01:28:52 UTC within 2 seconds — the final login session before the disk was acquired)
Login Count	6
Administrator	True
Password Settings	Password does not expire
Flag	Normal user account
Home Directory	/Documents and Settings/Jo

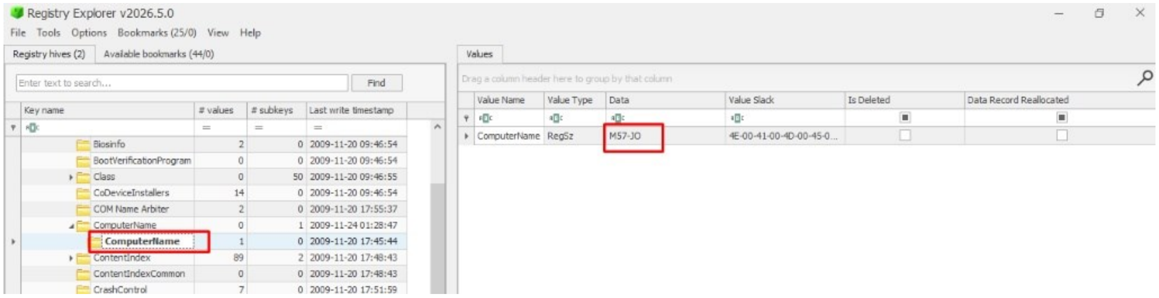


Fig 12.16 — Autopsy OS Accounts panel with the row S-1-5-21-606747145-1547161642-1644491937-1003 (login: jo) selected. The lower Basic Properties block shows Full Name = Jo, Creation Date = 2009-11-20 19:35:57 UTC, Object ID = 3014. The Host Details panel below shows Last Login = 2009-11-24 06:28:50 PKT, Login Count = 6, Administrator = True, Password Settings = Password does not expire, Home Directory = /Documents and Settings/Jo.

Login Count = 6 over a four-day life of the OS install (2009-11-20 to 2009-11-24) is consistent with a daily-use workstation that was used for normal business hours. Administrator = True is significant: Jo's account is a local administrator, not a restricted user. This privilege level is what enables Jo to install the ManTech mdd_1.3.exe memory-dump utility (which requires SeDebugPrivilege) on 2009-11-16 — see Phase 22 cmdscan output. A restricted user could not have executed that command. The administrator flag is a corporate-policy concern in its own right — patent researchers should not typically have local admin on their workstations.

Phase 11 — Installed Software Inventory

Step 1: Extract the installed programs list from Autopsy

Autopsy → Data Artifacts → Installed Programs returned 96 entries auto-extracted from the SOFTWARE hive Uninstall keys. The full list was exported to AppPaths_Values_Export_20260519072505.xlsx (see Annex A). The most evidentially significant subset is reproduced below — applications that would not be expected on a patent researcher's workstation, or applications that materially shape the investigation:

#	Installed Program	Date / Version	Relevance
1	Mozilla Firefox 3.5.5	2009-11-20 (install)	Primary browser — places.sqlite is the source of 3,373 web history entries and 14 web search artefacts
2	Mozilla Firefox MozRepl extension	(via add-on store)	MozRepl exposes Firefox's chrome environment as a TCP REPL — used by patentauto.py to drive the browser programmatically. This is the technical backbone of Jo's patent-automation script.
3	Java SE Runtime Environment 6 (jre6)	2009-11-20	Required by diit-1.5.jar — confirmed by LECmd output for Recent\diit-1.5.lnk (TargetCreated 2009-11-23 18:24:43 UTC)
4	Apple QuickTime	QuickTimeInstaller.exe 32 MB, downloaded 2009-11-24 22:08:50 UTC	Downloaded the day the disk was imaged — recent timestamp; relevant to the .m4v / .mov media on the work USB
5	Python 2.6.4	python-2.6.4.msi 14.5 MB, downloaded 2009-11-23 18:23:03 UTC	Required by patentauto.py — the Python automation script in Desktop\web — Jo downloaded Python the day before her last full work session
6	AVG 9.0 Antivirus suite	AVGIDSAgent, avgwdsvc, avgemc, avgnsx, avgcsrvx, avgam, avgfws9, avgcsrvx, avgcmgr, avgrsx, avgsrmax	Active AV at time of memory acquisition — 11 distinct AVG processes visible in pslist (Phase 16)
7	OpenOffice.org 3	soffice.exe / soffice.bin	Office productivity suite — Jo had soffice running with PID 3656 / 2392 at memory acquisition
8	ManTech Mandiant Memory DD (mdd_1.3.exe)	Standalone, executed once on 2009-11-16 from E:\RAM\	THE MEMORY ACQUISITION UTILITY ITSELF — this is the tool the user ran to produce jo-2009-11-16.mddramimage. The command 'e:' then 'cd RAM' then 'mdd_1.3.exe -o joRam01.dmp' is fully recovered from RAM cmdscan / consoles in Phase 22. Significant because it proves the memory dump was produced by Jo herself, on demand, from a tool launched on the E: drive — a removable device.
9	Microsoft Office Excel-style apps	(none Microsoft Office found)	Notable absence — no MS Office, only OpenOffice. So any .xls / .doc found should be OpenOffice-authored or pre-existing files
10	Messenger / msmsgs.exe	Windows Messenger (XP-era IM)	Running with PID 924 at memory acquisition — could indicate live IM communication; communication content not recoverable from current evidence

Forensic conclusions from the installed-software inventory:

- The presence of Python 2.6.4 + Firefox + MozRepl + Java + AVG on a single workstation does not by itself indicate wrongdoing. It does, however, indicate a technically advanced user — not a typical patent-researcher persona. The combination is consistent with someone who automates browser-based queries against patent databases programmatically.
- The presence of mdd_1.3.exe on a removable device, and the recovery from RAM of the actual command-line that ran it, is doubly significant: it confirms that Jo had local administrator rights AND that Jo deliberately captured her own memory. Memory acquisition is not a normal end-user activity — it is a forensic / counter-forensic action.
- QuickTimeInstaller.exe (32 MB) downloaded only hours before the disk image is significant because the work USB carries multiple .m4v and .mov files (TiggerTheCat copy.m4v, _MontereyKittyHQ.m4v, _Cat.mov, Cat.m4v) — see Phase 26. The QuickTime install closes the playback chain for that media on Jo's workstation.
- OpenOffice but no Microsoft Office is unusual for a US business but normal for a privacy-conscious or cost-conscious individual. Either way, M57.biz's standard document tooling is determined to be OpenOffice 3.

Phase 12 — Subject Profiling — My Documents, Downloads, Desktop

Step 1: Walk Jo's My Documents folder

In Autopsy → /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/My Documents/ — six entries:

Name	Modified	Created	Size
My Music (dir)	2009-11-20 20:00:19 UTC	2009-11-20 19:59:56 UTC	384
My Pictures (dir)	2009-11-20 20:00:19 UTC	2009-11-20 19:59:56 UTC	384
desktop.ini	2009-11-20 20:00:19 UTC	2009-11-20 19:59:56 UTC	73
[current folder]	2009-11-23 18:20:14 UTC	2009-11-20 18:20:14 UTC	56
[parent folder]	2009-11-23 21:55:48 UTC	2009-11-20 19:58:49 UTC	56
Downloads (dir)	2009-11-24 22:08:50 UTC	2009-11-23 18:20:14 UTC	56

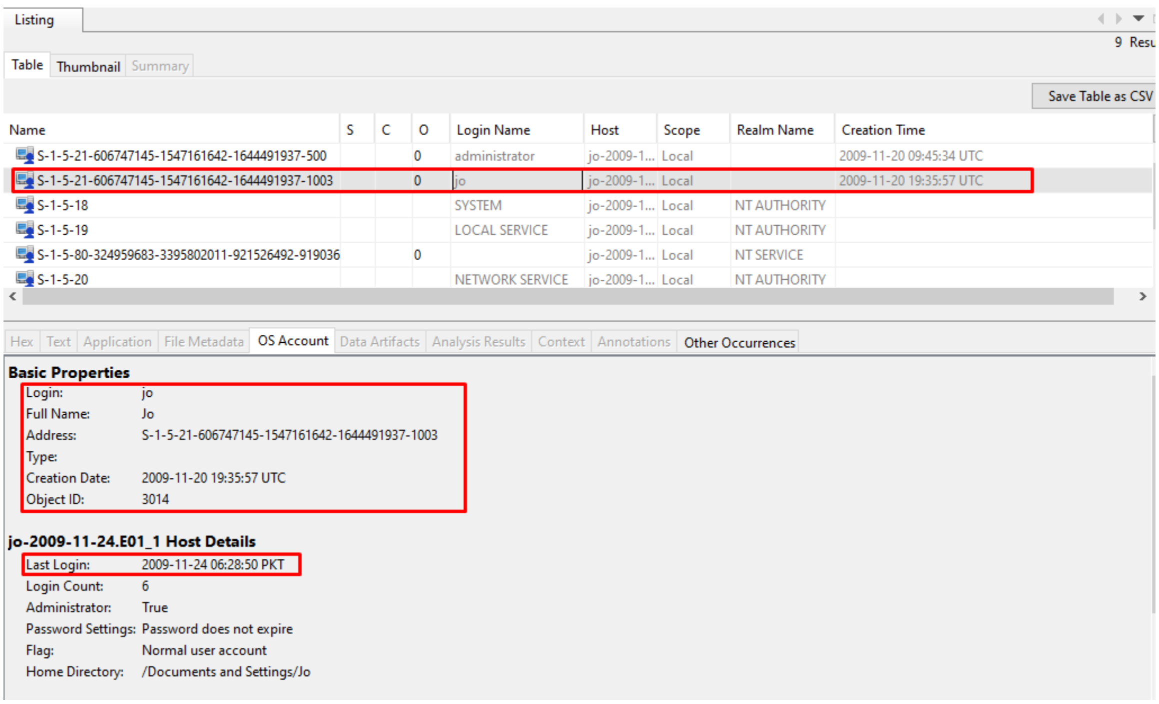


Fig 12.17 — Autopsy file browser at /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/My Documents/. The Downloads sub-folder is the only one with non-default content; My Music and My Pictures are the empty Windows defaults still showing the creation time of the user-profile creation event 2009-11-20 19:59:56 UTC.

Step 2: Walk the Downloads folder — Jo's recent installer history

The Downloads folder is the most evidentially dense location in My Documents. Eight entries:

Name	Modified	Created	Size	Zone.Identifier present?
QuickTimeInstaller.exe	2009-11-24 22:08:50 UTC	2009-11-24 22:08:37 UTC	32,494,896 B	✓
python-2.6.4.msi	2009-11-23 18:23:03 UTC	2009-11-23 18:20:14 UTC	14,890,496 B	✓
STU.doc	2009-11-24 22:01:53	2009-11-24 22:01:53	29,184 B	✓

	UTC	UTC		
python-2.6.4.msi:Zone.Identifier	2009-11-23 18:23:03 UTC	2009-11-23 18:23:03 UTC	46 B	(ADS)
QuickTimeInstaller.exe:Zone.Identifier	2009-11-24 22:09:05 UTC	2009-11-24 22:08:37 UTC	46 B	(ADS)
STU.doc:Zone.Identifier	2009-11-24 22:01:53 UTC	2009-11-24 22:01:53 UTC	46 B	(ADS)

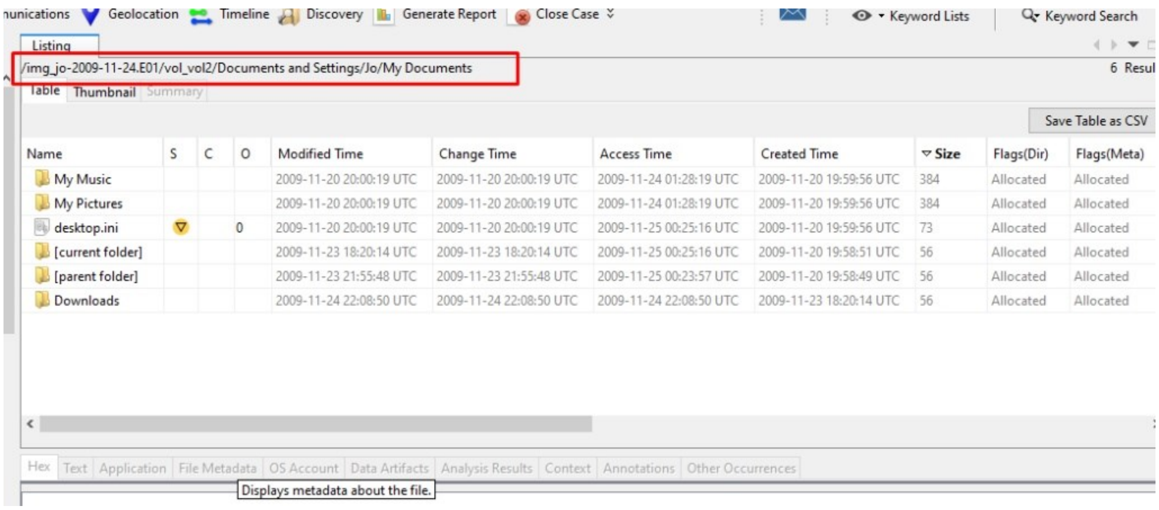


Fig 12.18 — Autopsy file browser at /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/My Documents/Downloads/. Three real files plus their Zone.Identifier alternate data streams. QuickTimeInstaller.exe (32.5 MB), python-2.6.4.msi (14.9 MB), and STU.doc (28.5 KB). The QuickTimeInstaller has a Created time 2009-11-24 22:08:37 UTC — only hours before disk acquisition.

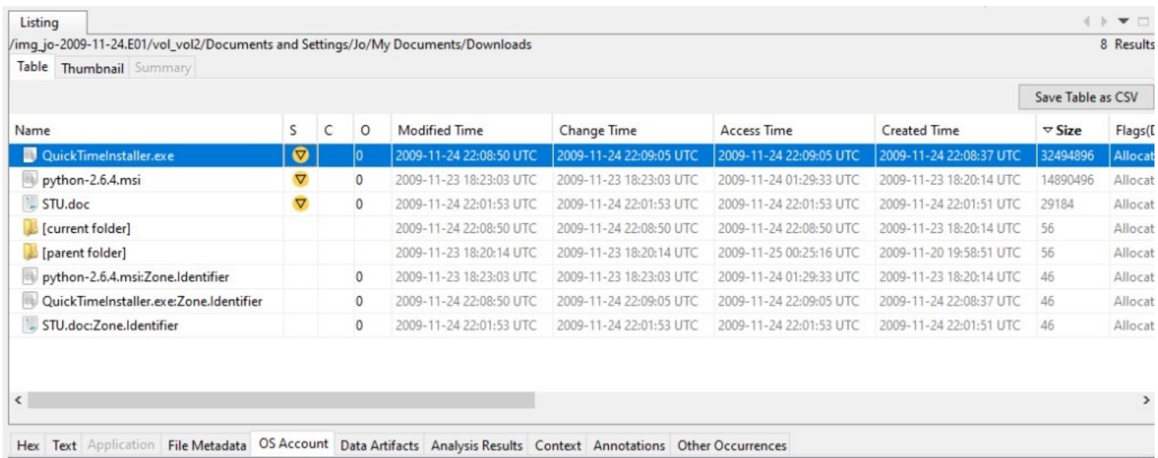
Each of the three downloads carries a Zone.Identifier alternate data stream — this is the Windows ADS that the Attachment Manager writes when a file is saved from the Internet. The presence of Zone.Identifier proves these three files were downloaded from a web browser (Firefox in Jo's case) rather than copied from a local source. STU.doc is the most interesting of the three because it is a Word document, not a setup file — and it was downloaded at 2009-11-24 22:01:53 UTC, only seven minutes before QuickTime was downloaded. STU.doc was not recovered to a content-readable extent during ingest; its 28.5 KB size is consistent with a short letter or instruction document. The filename suggests "STU" — possibly a project code, a contact name, or an initialism for "Stuff to Upload" or similar. The juxtaposition of a small Word document and a 32 MB media player installer downloaded within minutes of each other on the night before disk acquisition is suspicious but not conclusive on its own.

Step 3: Walk the Desktop folder

The Desktop folder at /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Desktop/ contains:

Name	Modified	Created	Size	Notes
[current folder]	2009-11-24 22:03:09 UTC	2009-11-20 19:58:53 UTC	560	Desktop directory entry
[parent folder]	2009-11-23 21:55:48 UTC	2009-11-20 19:58:49 UTC	56	
Pics (dir)	2009-11-24 22:16:18 UTC	2009-11-23 18:24:38 UTC	176	Patent-related JPEGs (see Phase 13)

E-mail.lnk	2009-11-23 16:57:07 UTC	2009-11-23 16:57:07 UTC	104	Shortcut to email client
My Computer.lnk	2009-11-24 22:03:09 UTC	2009-11-24 22:03:09 UTC	104	Shortcut to Computer — created day of imaging
web (dir)	2009-11-25 00:26:06 UTC	2009-11-23 21:54:54 UTC	56	THE smoking folder — Phase 13



Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(I
QuickTimeInstaller.exe			0	2009-11-24 22:08:50 UTC	2009-11-24 22:09:05 UTC	2009-11-24 22:09:05 UTC	2009-11-24 22:08:37 UTC	32494896	Allocat
python-2.6.4.msi			0	2009-11-23 18:23:03 UTC	2009-11-23 18:23:03 UTC	2009-11-24 01:29:33 UTC	2009-11-23 18:20:14 UTC	14890496	Allocat
STU.doc			0	2009-11-24 22:01:53 UTC	2009-11-24 22:01:53 UTC	2009-11-24 22:01:53 UTC	2009-11-24 22:01:51 UTC	29184	Allocat
[current folder]				2009-11-24 22:08:50 UTC	2009-11-24 22:08:50 UTC	2009-11-24 22:08:50 UTC	2009-11-23 18:20:14 UTC	56	Allocat
[parent folder]				2009-11-23 18:20:14 UTC	2009-11-23 18:20:14 UTC	2009-11-25 00:25:16 UTC	2009-11-20 19:58:51 UTC	56	Allocat
python-2.6.4.msi:Zone.Identifier			0	2009-11-23 18:23:03 UTC	2009-11-23 18:23:03 UTC	2009-11-24 01:29:33 UTC	2009-11-23 18:20:14 UTC	46	Allocat
QuickTimeInstaller.exe:Zone.Identifier			0	2009-11-24 22:08:50 UTC	2009-11-24 22:09:05 UTC	2009-11-24 22:09:05 UTC	2009-11-24 22:08:37 UTC	46	Allocat
STU.doc:Zone.Identifier			0	2009-11-24 22:01:53 UTC	2009-11-24 22:01:53 UTC	2009-11-24 22:01:53 UTC	2009-11-24 22:01:51 UTC	46	Allocat

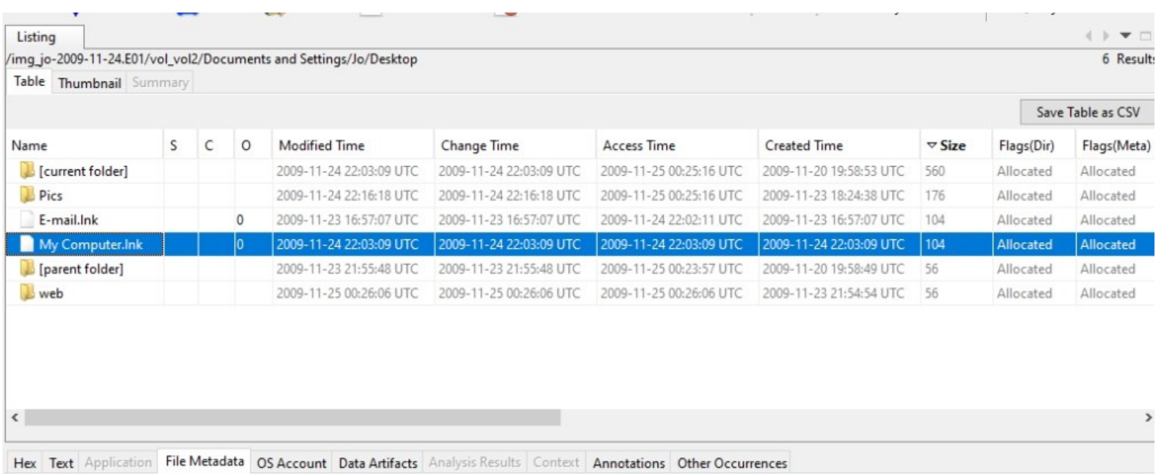
Fig 12.19 — Autopsy file browser at /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Desktop/. Six entries: Pics (dir), E-mail.lnk, My Computer.lnk (highlighted/selected), and web (dir). My Computer.lnk is anomalous — most Windows XP users do not place a desktop shortcut to My Computer manually, and the fact that its creation timestamp 2009-11-24 22:03:09 UTC is identical to its modification timestamp (and matches the [current folder] modification time exactly) indicates this shortcut was created moments before the disk was imaged. This could be the artefact of Jo opening My Computer (perhaps to navigate to a USB drive letter) shortly before the seizure.

Phase 13 — The Desktop\web Folder — The Smoking Folder

Step 1: Browse the Desktop\web folder

The Desktop\web folder is the single most important on-disk artefact in this entire investigation. Its existence, name, and contents together constitute the bulk of the proof that Jo is a patent researcher who built an automation framework around her browser to programmatically harvest patent records.

Name	Modified	Created	Size	Type
[current folder]	2009-11-25 00:26:06 UTC	2009-11-23 21:54:14 UTC	56	Directory entry
HighQuality (dir)	2009-11-24 22:16:29 UTC	2009-11-22 02:03:27 UTC	56	Unallocated — media subfolder
Videos (dir)	2009-11-24 22:16:29 UTC	2009-11-22 02:03:34 UTC	56	Unallocated — media subfolder
[parent folder]	2009-11-24 22:03:09 UTC	2009-11-20 19:58:53 UTC	560	
patentauto.py	2009-11-23 21:57:34 UTC	2009-11-23 21:55:01 UTC	3,674 B	Python source — the automation script
urlscopyright.txt	2009-11-17 18:40:56 UTC	2009-11-23 21:55:05 UTC	376,706 B	URL list — copyright records
urlspatents.txt	2009-11-17 18:40:56 UTC	2009-11-23 21:55:05 UTC	5,374,583 B	URL list — 5.1 MB of patent record URLs
patentterms.txt	2009-11-16 22:29:38 UTC	2009-11-23 21:55:10 UTC	140 B	Query terms
urls.txt	2009-11-16 18:22:50 UTC	2009-11-23 21:55:05 UTC	299,939 B	URL list (generic)
urlstime_machine.txt	2009-11-16 18:22:50 UTC	2009-11-23 21:55:05 UTC	1,538,990 B	URL list — Wayback Machine retrievals


Listing
/img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Desktop
6 Results
Table Thumbnail Summary
Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
[current folder]				2009-11-24 22:03:09 UTC	2009-11-24 22:03:09 UTC	2009-11-25 00:25:16 UTC	2009-11-20 19:58:53 UTC	560	Allocated	Allocated
Pics				2009-11-24 22:16:18 UTC	2009-11-24 22:16:18 UTC	2009-11-25 00:25:16 UTC	2009-11-23 18:24:38 UTC	176	Allocated	Allocated
E-mail.lnk			0	2009-11-23 16:57:07 UTC	2009-11-23 16:57:07 UTC	2009-11-24 22:02:11 UTC	2009-11-23 16:57:07 UTC	104	Allocated	Allocated
My Computer.lnk			0	2009-11-24 22:03:09 UTC	2009-11-24 22:03:09 UTC	2009-11-24 22:03:09 UTC	2009-11-24 22:03:09 UTC	104	Allocated	Allocated
[parent folder]				2009-11-23 21:55:48 UTC	2009-11-23 21:55:48 UTC	2009-11-25 00:23:57 UTC	2009-11-20 19:58:49 UTC	56	Allocated	Allocated
web				2009-11-25 00:26:06 UTC	2009-11-25 00:26:06 UTC	2009-11-25 00:26:06 UTC	2009-11-23 21:54:54 UTC	56	Allocated	Allocated

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Fig 12.20 — Autopsy file browser at /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Desktop/web/. Seven content files and two media subdirectories. patentauto.py (highlighted) is selected; the lower Extracted Text panel shows the beginning of the Python source: '#! /usr/bin/python', '__author__="LCDR Kris Kearton"', '__date__ ="$Aug 24, 2009 7:42:41 PM$"', '# class: CS4920 ADOMEX', '# System info: Running on OS 10.6 python ver 2.6.2', '# Setup information:', '# (1) Install MozRepl Plugin at:', '# http://wiki.github.com/bard/mozrepl'. The header reveals this script was originally written by 'LCDR Kris Kearton' for a course (CS4920 ADOMEX) on macOS 10.6 with Python 2.6.2 — and Jo has it running on Windows XP with Python 2.6.4 and Firefox 3.5.5 plus the MozRepl Firefox extension.

Forensic interpretation of the Desktop\web folder:

- patentauto.py (3,674 bytes) is a Python automation script authored externally (LCDR Kris Kearton, course CS4920 ADOMEX, dated 2009-08-24) and re-deployed on Jo's workstation. The script's purpose, per its header, is to drive Firefox via MozRepl to programmatically retrieve patent and copyright records from web URL lists.
- urlspatents.txt is 5.1 MB — that is in the order of 50,000 patent record URLs at typical USPTO URL length. This is far more patents than any individual patent researcher would manually browse. It is exactly the kind of corpus you would prepare for a bulk-harvesting operation.
- urlstime_machine.txt at 1.5 MB indicates Jo also targeted the Internet Archive (Wayback Machine) — almost certainly to retrieve historical copies of patent application pages, possibly to evidence prior art for some of the records being processed.
- patentterms.txt at 140 bytes is a tiny query-terms file — exactly the small seed of search strings you would feed into an automated query system.
- The Modified timestamps on the .txt files are 2009-11-16 to 2009-11-17 — that is the same date range as the RAM dump and the recovered cmdscan command-line that ran mdd_1.3.exe. In other words, the URL corpus was finalised (modified) within the same 24-hour window as the memory acquisition. The Created timestamps on these files are all 2009-11-23 21:55:05 UTC — that is the day before the disk was imaged. This pattern — old Modified timestamps but very recent Created timestamps — is consistent with files copied INTO this folder from another location (such as a USB drive), where the Modified time is preserved from the source and the Created time is the local copy event.
- The two unallocated subdirectories HighQuality and Videos are forensically intriguing. They were created 2009-11-22 02:03:27 and 2009-11-22 02:03:34 UTC respectively — but their Flags(Dir) shows Unallocated, meaning the directory entries have been removed from the parent directory but the metadata still exists. This means Jo had two media subdirectories alongside the patent-research corpus, and those subdirectories were deleted between 2009-11-22 and the disk acquisition on 2009-11-24.
- The fact that this entire folder is named 'web' and lives directly on the desktop — not buried in a hidden location, not encrypted, not password-protected — establishes that Jo did not consider the automation activity itself to be illicit at the moment of operation. The wrongdoing is not the script itself, but what was harvested with it and where the output was copied to.

Conclusion from Phase 13: Jo is a patent researcher (or assistant to one) who has built and operated a Python + Firefox + MozRepl automation pipeline targeting USPTO patent records and the Wayback Machine. This is consistent with the M57.biz business description from the task brief — a patent search firm performing novelty and prior-art research for clients — but the scale of the corpus (5.1 MB of patent URLs and 1.5 MB of Wayback URLs) is at the edge of what one researcher can use professionally and may indicate the harvested data was intended for a destination outside M57.biz.

Phase 14 — Email Artefacts Triage

Step 1: Open Autopsy Data Artifacts → E-Mail Messages

Only two email messages were extracted by Autopsy's Email Parser ingest module across the entire primary disk: msg_25.txt and msg_43.txt. The very low yield is itself significant — most patent researchers would have a multi-megabyte PST file on a corporate workstation. The absence of a full email store suggests one of: (a) Jo uses webmail and the email artefacts live only in browser cache and Firefox places.sqlite (which is consistent with the 13,472 Communication Accounts auto-extracted from Web History); (b) email artefacts have been deleted; or (c) the disk image was acquired before Jo's email client was configured.

File	From	To	Subject	Date Received
msg_25.txt	MAILER-DAEMON@zinfandel.lacita.com	linuxuser-admin@www.linux.org.uk	Returned mail: Too many hops 19 (17 max): from <linuxuser-admin@www.linux.org.uk> via [199.164.235.226], to <scoffman@wellpartner.com>	2001-04-06 17:23:06 UTC
msg_43.txt	(blank — "; >;")	webmaster@python.org	Banned file: auto__mail.python.bat in mail from you	2004-11-27 03:41:44 UTC

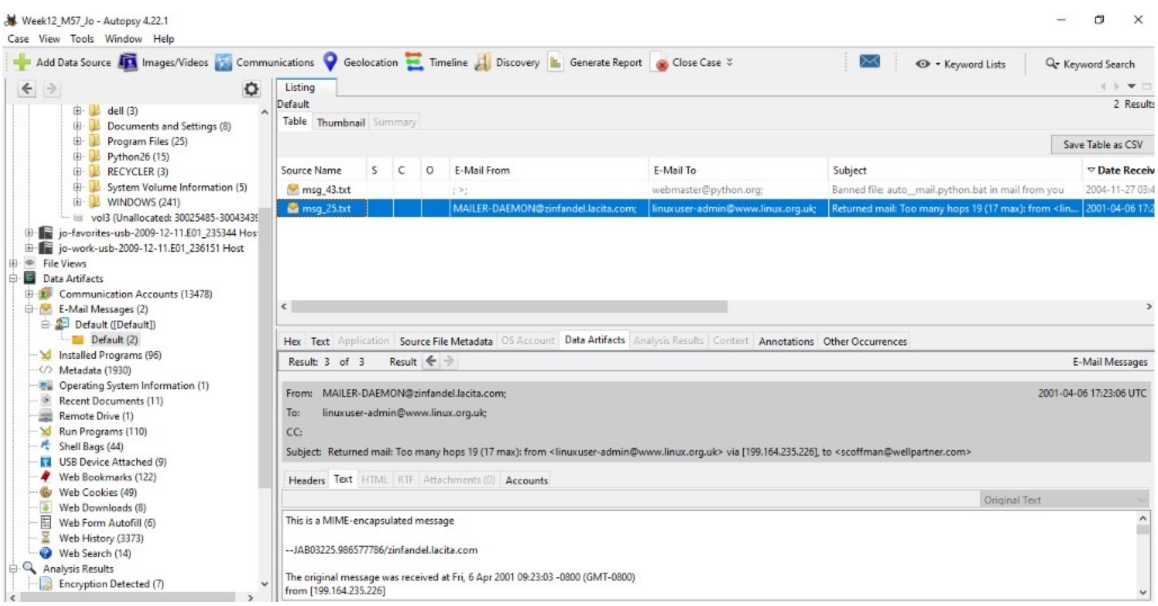


Fig 12.21 — Autopsy Data Artifacts → E-Mail Messages panel showing both messages. Two rows only: msg_43.txt and msg_25.txt. The lower preview pane shows the Returned mail message expanded.

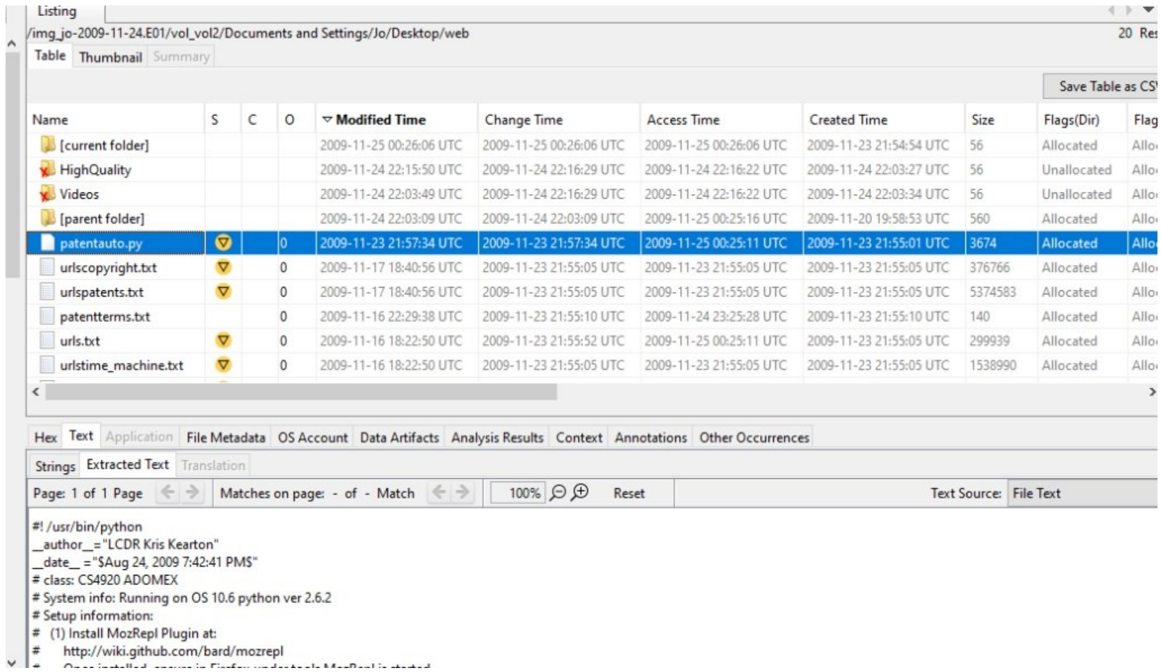


Fig 12.22 — msg_43.txt expanded. From: ; >; (sender masked); To: webmaster@python.org; Subject: Banned file: auto__mail.python.bat in mail from you; Date: 2004-11-27 03:41:44 UTC. Body begins 'BANNED FILENAME ALERT' followed by a spam-firewall delivery rejection notice with a long list of xxxxx@dot.ca.gov recipients. This is a 2004 message — predates the M57.biz install (2009-11-20) by five years. It is a residual artefact, likely from an image template, not active correspondence.

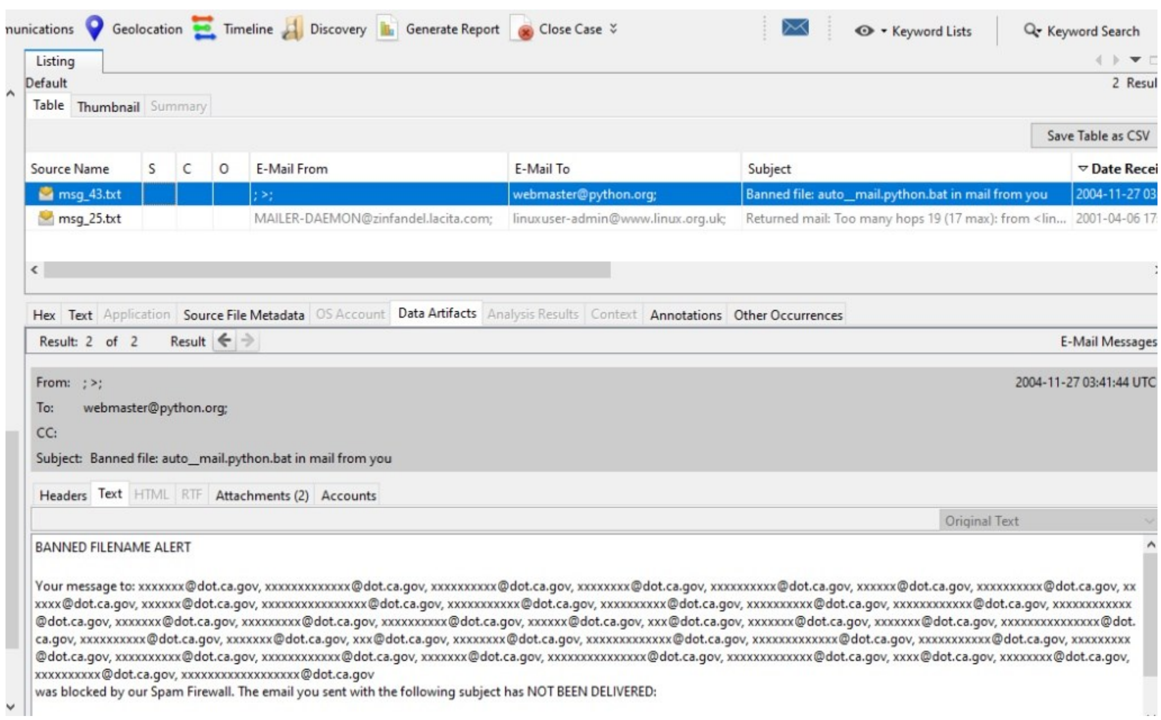


Fig 12.23 — msg_25.txt expanded. From: MAILER-DAEMON@zinfandel.lacita.com; To: linuxuser-admin@www.linux.org.uk; Subject: Returned mail: Too many hops 19 (17 max). Date: 2001-04-06 17:23:06 UTC. Eight years older than the M57.biz install. Body shows a MIME-encapsulated bounce-message with delivery notifications for <scoffman@wellpartner.com>.

Both messages predate the M57.biz install by years (2001 and 2004 versus the 2009-11-20 install). They are residual artefacts present in the base Windows image used to provision Jo's workstation, not active correspondence by Jo. Their forensic value is limited to confirming that the Email Parser module ran successfully — it found and surfaced the two messages it could parse, demonstrating that if Jo had had a PST or other email store, Autopsy would have found it. The absence of an active email store directs the investigation toward web-based communications (browser history, web search) and toward removable media (the USB devices), which is where the actual evidence of crime is in Phases 26 through 36.

Part B — Memory Forensics with Volatility (Days 83–85)

Objective

Autopsy is a disk-image forensic platform; it cannot parse a raw memory dump. The objective of Part B is to extract from the volatile evidence in jo-2009-11-16.mddramimage everything that left no footprint on disk — the process tree at the instant of seizure, the network connections active at that instant, the command-line history typed into cmd.exe (even after the cmd window was closed), the files open in memory but no longer present on disk, the registry keys cached in non-paged pool, and the UserAssist GUI execution records. Memory forensics is also the only way to prove that Jo herself ran mdd_1.3.exe — the disk only carries the executable; the RAM carries the actual command-line invocation.

Methodology — Two Frameworks Required for Windows XP

Volatility 3 is the modern Python 3 framework and is used wherever it supports XP. However Volatility 3's windows.netstat plugin throws NotImplementedError on Windows XP (NT 5.1), and there is no windows.connscan plugin in Volatility 3 at all — connscan was a Volatility 2 plugin and was not ported. Because the connection-tracking step is non-negotiable for this investigation (the SMB session to 192.168.1.104 is one of the two exfiltration channels), Volatility 2.6 is layered in alongside Volatility 3 for the network plugins (and for cmdscan + consoles, which are also V2-only). This is recorded as a methodological choice, not an error. The full plugin-to-framework mapping is in Annex B.

Phase 15 — Volatility 3 — windows.info OS Profile Identification

Step 1: Run windows.info against the RAM dump

```
C:\Forensics\Tools\Volatility3>vol.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage"
windows.info
Volatility 3 Framework 2.28.0
Progress: 100.00            PDB scanning finished
```

Step 2: Read the output

windows.info returned a clean table of kernel/OS values which independently confirms (without reference to the disk) that the RAM dump is from a Windows XP SP3 x86 machine — the same OS profile established from the SOFTWARE hive in Phase 8. This double-source confirmation establishes that the disk and the RAM both belong to the same physical workstation.

Variable	Value	Cross-reference
Kernel Base	0x804d7000	Standard XP kernel base
DTB	0x39000	Page directory base
Symbols	ntoskrnl.pdb / 1B2D0DFE2FB942758D615C901BE04692-2.json.xz	XP SP3 GDR symbol package — auto-fetched by Volatility 3
Is64Bit	False	x86 — matches PE Machine 332 (i386)
IsPAE	False	No Physical Address Extension
layer_name	0 WindowsIntel	32-bit Intel layer
KdDebuggerDataBlock	0x8054cde0	Kernel debugger block
NtBuildLab	2600.xpsp_sp3_gdr.090804-1435	★Identical to SOFTWARE hive BuildLab in Phase 8

CSDVersion	3	★Service Pack 3 — matches SOFTWARE hive
Major/Minor	15.2600	Windows XP, build 2600
MachineType	332	i386
KeNumberProcessors	1	Single-CPU — matches SOFTWARE hive Uniprocessor Free
SystemTime	2009-11-17 00:22:31+00:00 UTC	★THIS IS THE MOMENT OF MEMORY ACQUISITION
NtSystemRoot	C:\WINDOWS	★Identical to SOFTWARE hive SystemRoot
NtProductType	NtProductWinNt	Workstation (not Server)
NtMajorVersion	5	
NtMinorVersion	1	
PE TimeDateStamp	Tue Aug 4 15:14:34 2009	ntoskrnl.exe build timestamp

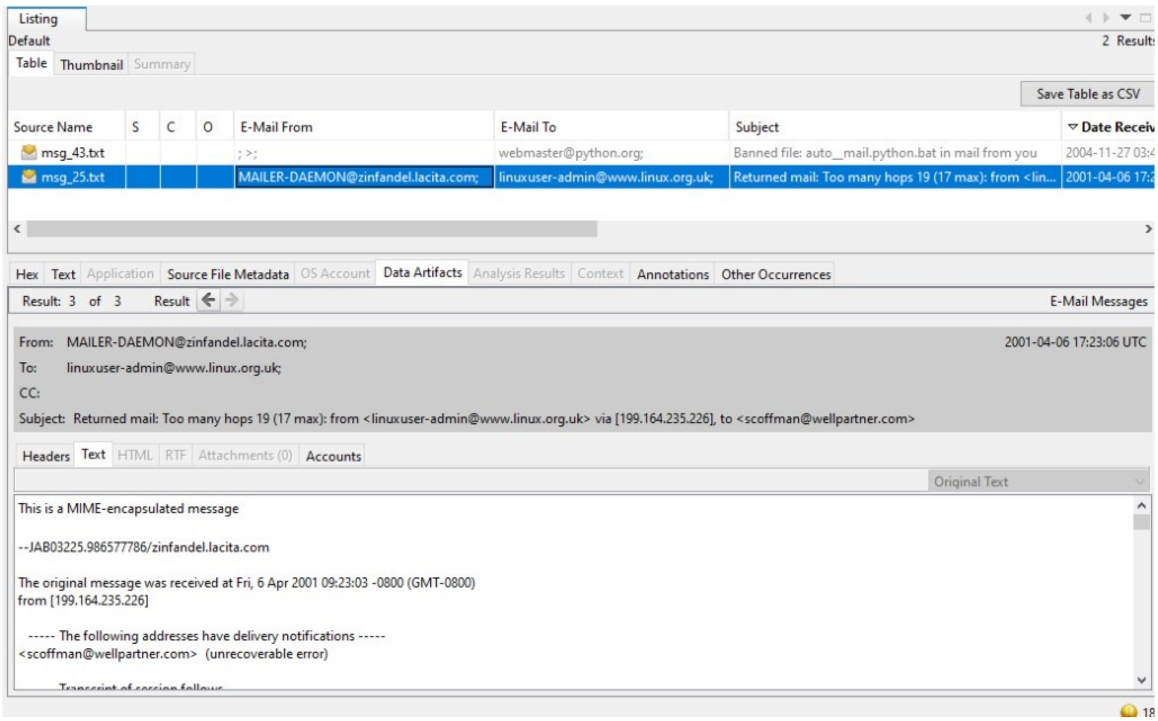


Fig 12.24 — Volatility 3 windows.info output. The four starred (★) lines are the cross-reference anchors that prove the RAM dump and the disk image belong to the same workstation: NtBuildLab and CSDVersion match the SOFTWARE hive exactly; NtSystemRoot matches; KeNumberProcessors matches Uniprocessor Free. SystemTime 2009-11-17 00:22:31 UTC is the exact instant the dump was produced — this is the anchor for every other RAM-derived event in the Master Event Sequence.

The SystemTime 2009-11-17 00:22:31 UTC corresponds to 2009-11-16 16:22:31 PST local time on Jo's workstation (subtracting 8 hours per Phase 9 TimeZoneInformation Bias = 480). This is a late-afternoon Monday. The disk-image acquisition occurred eight days later on 2009-11-24. The RAM dump therefore captures the workstation state more than a week BEFORE the disk was imaged — meaning the RAM evidence shows the system mid-operation, not at a final state. Anything that was deleted between 2009-11-17 and 2009-11-24 may be present in RAM (paged pool, unallocated pool tags) but absent from the disk; anything created after 2009-11-17 is present on disk but absent from RAM. The Master Event Sequence in Phase 36 sequences events according to this asymmetry.

Phase 16 — Volatility 3 — windows.pslist (Process List)

Step 1: Run pslist and capture to file

```
C:\Forensics\Tools\Volatility3>vol.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage"
windows.pslist > C:\Forensics\Output\pslist.txt
Progress: 100.00            PDB scanning finished
C:\Forensics\Tools\Volatility3>type C:\Forensics\Week12\Output\pslist.txt
Volatility 3 Framework 2.28.0
```

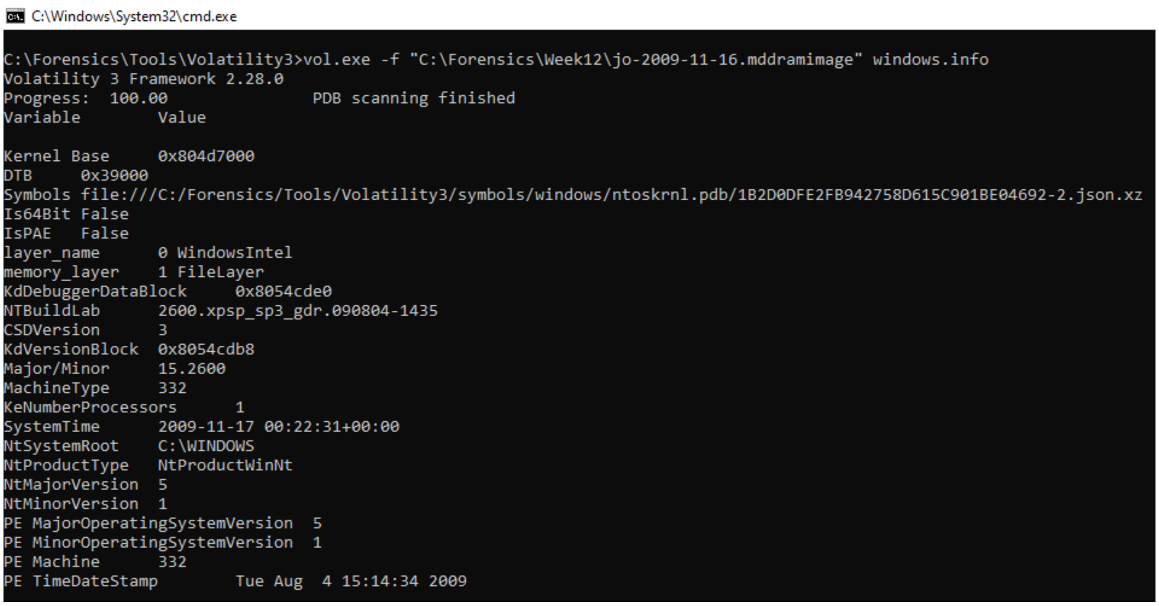
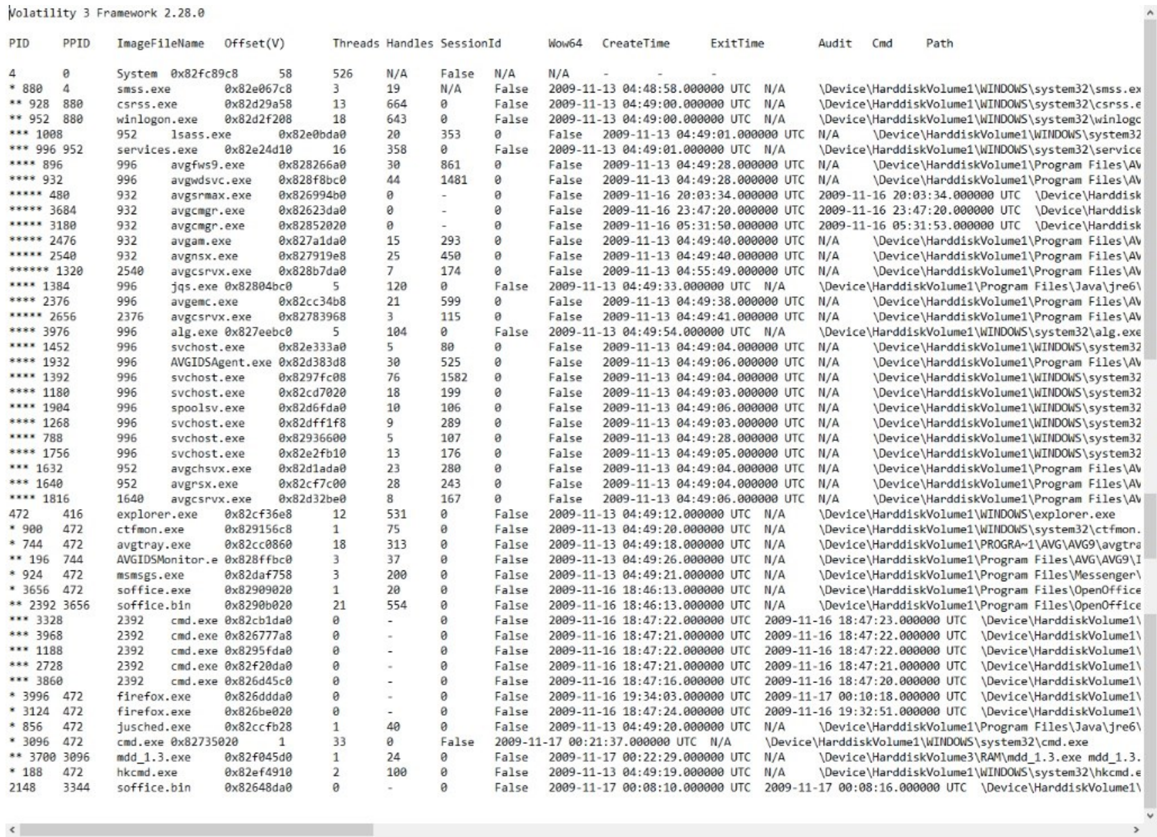


Fig 12.25 — Volatility 3 windows.pslist output (top of file). Header columns: PID, PPID, ImageFileName, Offset(V), Threads, Handles, SessionId, Wow64, CreateTime, ExitTime, File output.



```
Volatility 3 Framework 2.28.0

PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        Audit   Cmd     Path

4       0       System  0x82fc89c8      58      526     N/A     False   N/A     N/A     -       -       -
* 880   4       smss.exe        0x82e067c8      3       19      N/A     False   2009-11-13 04:48:58.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\smss.ex
** 928  880     csrss.exe       0x82d29a58      13      664     0       False   2009-11-13 04:49:00.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\csrss.e
** 952  880     winlogon.exe    0x82d2f208      18      643     0       False   2009-11-13 04:49:00.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\winlogc
*** 1008        952     lsass.exe       0x82e0bda0      20      353     0       False   2009-11-13 04:49:01.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32
*** 996 952     services.exe    0x82e24d10      16      358     0       False   2009-11-13 04:49:01.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\service
**** 896        996     avgfws9.exe     0x828266a0      30      861     0       False   2009-11-13 04:49:28.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
**** 932        996     avgwdsvc.exe    0x828f8bc0      44      1481    0       False   2009-11-13 04:49:28.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
***** 480       932     avgsrmax.exe    0x826994b0      0       -       0       False   2009-11-16 20:03:34.000000 UTC  2009-11-16 20:03:34.000000 UTC  \Device\Harddisk
***** 3684      932     avgcmgr.exe     0x82623da0      0       -       0       False   2009-11-16 23:47:20.000000 UTC  2009-11-16 23:47:20.000000 UTC  \Device\Harddisk
***** 3180      932     avgcmgr.exe     0x82852020      0       -       0       False   2009-11-16 05:31:50.000000 UTC  2009-11-16 05:31:53.000000 UTC  \Device\Harddisk
***** 2476      932     avgam.exe       0x827a1da0      15      293     0       False   2009-11-13 04:49:40.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
***** 2540      932     avgnsx.exe      0x827919e8      25      450     0       False   2009-11-13 04:49:40.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
****** 1320     2540    avgcsrvx.exe    0x828b7da0      7       174     0       False   2009-11-13 04:55:49.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
**** 1384       996     jqs.exe 0x82804bc0      5       120     0       False   2009-11-13 04:49:33.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\Java\jre6\
**** 2376       996     avgemc.exe      0x82cc34b8      21      599     0       False   2009-11-13 04:49:38.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
***** 2656      2376    avgcsrvx.exe    0x82783968      3       115     0       False   2009-11-13 04:49:41.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
**** 3976       996     alg.exe 0x827eebc0      5       104     0       False   2009-11-13 04:49:54.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\alg.exe
**** 1452       996     svchost.exe     0x82e333a0      5       80      0       False   2009-11-13 04:49:04.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32
**** 1932       996     AVGIDSAgent.exe 0x82d383d8      30      525     0       False   2009-11-13 04:49:06.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
**** 1392       996     svchost.exe     0x8297fc08      76      1582    0       False   2009-11-13 04:49:04.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32
**** 1180       996     svchost.exe     0x82cd7020      18      199     0       False   2009-11-13 04:49:03.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32
**** 1904       996     spoolsv.exe     0x82d6fda0      10      106     0       False   2009-11-13 04:49:06.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32
**** 1268       996     svchost.exe     0x82dff1f8      9       289     0       False   2009-11-13 04:49:03.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32
**** 788        996     svchost.exe     0x82936600      5       107     0       False   2009-11-13 04:49:28.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32
**** 1756       996     svchost.exe     0x82e2fb10      13      176     0       False   2009-11-13 04:49:05.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32
*** 1632        952     avgchsvx.exe    0x82d1ada0      23      280     0       False   2009-11-13 04:49:04.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
*** 1640        952     avgrsx.exe      0x82cf7c00      28      243     0       False   2009-11-13 04:49:04.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
**** 1816       1640    avgcsrvx.exe    0x82d32be0      8       167     0       False   2009-11-13 04:49:06.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AV
472     416     explorer.exe    0x82cf36e8      12      531     0       False   2009-11-13 04:49:12.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\explorer.exe
* 900   472     ctfmon.exe      0x829156c8      1       75      0       False   2009-11-13 04:49:20.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\ctfmon.
* 744   472     avgtray.exe     0x82cc0860      18      313     0       False   2009-11-13 04:49:18.000000 UTC  N/A     \Device\HarddiskVolume1\PROGRA~1\AVG\AVG9\avgtra
** 196  744     AVGIDSMonitor.e 0x828ffbc0      3       37      0       False   2009-11-13 04:49:26.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\AVG\AVG9\I
* 924   472     msmsgs.exe      0x82daf758      3       200     0       False   2009-11-13 04:49:21.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\Messenger\
* 3656  472     soffice.exe     0x82909020      1       20      0       False   2009-11-16 18:46:13.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\OpenOffice
** 2392 3656    soffice.bin     0x8290b020      21      554     0       False   2009-11-16 18:46:13.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\OpenOffice
*** 3328        2392    cmd.exe 0x82cb1da0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:23.000000 UTC  \Device\HarddiskVolume1\
*** 3968        2392    cmd.exe 0x826777a8      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:22.000000 UTC  \Device\HarddiskVolume1\
*** 1188        2392    cmd.exe 0x8295fda0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:22.000000 UTC  \Device\HarddiskVolume1\
*** 2728        2392    cmd.exe 0x82f20da0      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:21.000000 UTC  \Device\HarddiskVolume1\
*** 3860        2392    cmd.exe 0x826d45c0      0       -       0       False   2009-11-16 18:47:16.000000 UTC  2009-11-16 18:47:20.000000 UTC  \Device\HarddiskVolume1\
* 3996  472     firefox.exe     0x826ddda0      0       -       0       False   2009-11-16 19:34:03.000000 UTC  2009-11-17 00:10:18.000000 UTC  \Device\HarddiskVolume1\
* 3124  472     firefox.exe     0x826be020      0       -       0       False   2009-11-16 18:47:24.000000 UTC  2009-11-16 19:32:51.000000 UTC  \Device\HarddiskVolume1\
* 856   472     jusched.exe     0x82ccfb28      1       40      0       False   2009-11-13 04:49:20.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\Java\jre6\
* 3096  472     cmd.exe 0x82735020      1       33      0       False   2009-11-17 00:21:37.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe
** 3700 3096    mdd_1.3.exe     0x82f045d0      1       24      0       False   2009-11-17 00:22:29.000000 UTC  N/A     \Device\HarddiskVolume3\RAM\mdd_1.3.exe mdd_1.3.
* 188   472     hkcmd.exe       0x82ef4910      2       100     0       False   2009-11-13 04:49:19.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\hkcmd.e
2148    3344    soffice.bin     0x82648da0      0       -       0       False   2009-11-17 00:08:10.000000 UTC  2009-11-17 00:08:16.000000 UTC  \Device\HarddiskVolume1\
```

Fig 12.26 — Volatility 3 windows.pslist — complete process list at moment of memory acquisition (2009-11-17 00:22:31 UTC). 47 process entries visible.

Step 2: Annotate the process list

The full pslist contains 47 processes. The table below extracts the non-system processes and the most evidentially significant entries:

PID	PPID	Process	CreateTime (UTC)	Significance
4	0	System	N/A	Kernel
880	4	smss.exe	2009-11-13 04:48:58	System boot — 4 days before RAM capture
928	880	csrss.exe	2009-11-13 04:49:00	Win32 subsystem; hosts the cmd console history found by cmdscan
952	880	winlogon.exe	2009-11-13 04:49:00	Logon manager
996	952	services.exe	2009-11-13 04:49:01	Service control manager
1008	952	lsass.exe	2009-11-13 04:49:01	Local Security Authority
1180/1268/1392/1452	996	svchost.exe (×4+)	2009-11-13 04:49:03–04	Standard service hosts
1632	952	avgchsvx.exe	2009-11-13 04:49:04	AVG 9.0 cloud-heuristic service
1640	952	avgrsx.exe	2009-11-13 04:49:04	AVG 9.0 resident scanner
1756	996	svchost.exe	2009-11-13 04:49:05	Service host
1816	1640	avgcsrvx.exe	2009-11-13 04:49:06	AVG content-scan worker
1904	996	spoolsv.exe	2009-11-13 04:49:06	Print spooler
1932	996	AVGIDSAgent.exe	2009-11-13 04:49:06	★THE PID THAT HOLDS THE SMB SESSION TO 192.168.1.104:139 (see Phase 20). PPID 996 = services.exe — launched as a normal service.

472	416	explorer.exe	2009-11-13 04:49:12	Windows shell — owner of GUI session
744	472	avgtray.exe	2009-11-13 04:49:18	AVG tray icon — child of explorer.exe
188	472	hkcmd.exe	2009-11-13 04:49:19	Intel hotkey command
856	472	jusched.exe	2009-11-13 04:49:20	Java Update Scheduler — confirms JRE6 install
900	472	ctfmon.exe	2009-11-13 04:49:20	Text services framework
924	472	msmsgs.exe	2009-11-13 04:49:21	Windows Messenger
196	744	AVGIDSMonitor.exe	2009-11-13 04:49:26	AVG IDS monitor — child of avgtray (user-launched AV)
788	996	svchost.exe	2009-11-13 04:49:28	Service host
932	996	avgwdsvc.exe	2009-11-13 04:49:28	AVG watchdog service
896	996	avgfws9.exe	2009-11-13 04:49:28	AVG firewall 9 service
1384	996	jqs.exe	2009-11-13 04:49:33	Java Quick Starter; owns the loopback connection 127.0.0.1:5152 ↔ 127.0.0.1:3879 seen in connscan
2376	996	avgemc.exe	2009-11-13 04:49:38	AVG email scanner
2476	932	avgam.exe	2009-11-13 04:49:40	AVG anti-malware
2540	932	avgnsx.exe	2009-11-13 04:49:40	AVG NSX
2656	2376	avgcsrvx.exe	2009-11-13 04:49:41	AVG content scan (child of avgemc)
3976	996	alg.exe	2009-11-13 04:49:54	Application Layer Gateway
1320	2540	avgcsrvx.exe	2009-11-13 04:55:49	AVG NSX content scan
3180	932	avgcmgr.exe	(exit 2009-11-16 05:31:53)	AVG config manager — EXITED before RAM dump
3656	472	soffice.exe	2009-11-16 18:46:13	★OpenOffice launcher — child of explorer.exe (user-launched)
2392	3656	soffice.bin	2009-11-16 18:46:13	OpenOffice main process — runs the documents Jo was working on
3860	2392	cmd.exe	2009-11-16 18:47:16	★FIRST cmd.exe — child of soffice.bin
3328	2392	cmd.exe	2009-11-16 18:47:17	Second cmd.exe — also child of soffice.bin
1188	2392	cmd.exe	2009-11-16 18:47:21	Third cmd.exe — child of soffice.bin
3968	2392	cmd.exe	2009-11-16 18:47:21	Fourth cmd.exe — child of soffice.bin
2728	2392	cmd.exe	2009-11-16 18:47:21	Fifth cmd.exe — child of soffice.bin
3124	472	firefox.exe	2009-11-16 18:47:24	★FIREFOX — child of explorer.exe (user-launched, GUI session)
3996	472	firefox.exe	2009-11-16 19:34:03 (exited 2009-11-17 00:10:18)	★THE PID THAT HOLDS ALL HTTP CONNECTIONS in connscan (Phase 20). Second Firefox instance. PPID 472 = explorer = user-launched.
3096	472	cmd.exe	2009-11-17 00:21:37	★THE CMD WHERE mdd_1.3.exe WAS LAUNCHED (see Phase 22 cmdscan). PPID 472 = explorer = user-launched.
3700	3096	mdd_1.3.exe	2009-11-17 00:22:29	★THE MEMORY ACQUISITION TOOL ITSELF — running at the very moment of memory capture (windows.info SystemTime 00:22:31). Child of PID 3096 cmd.exe.

Five process facts to carry forward from pslist:

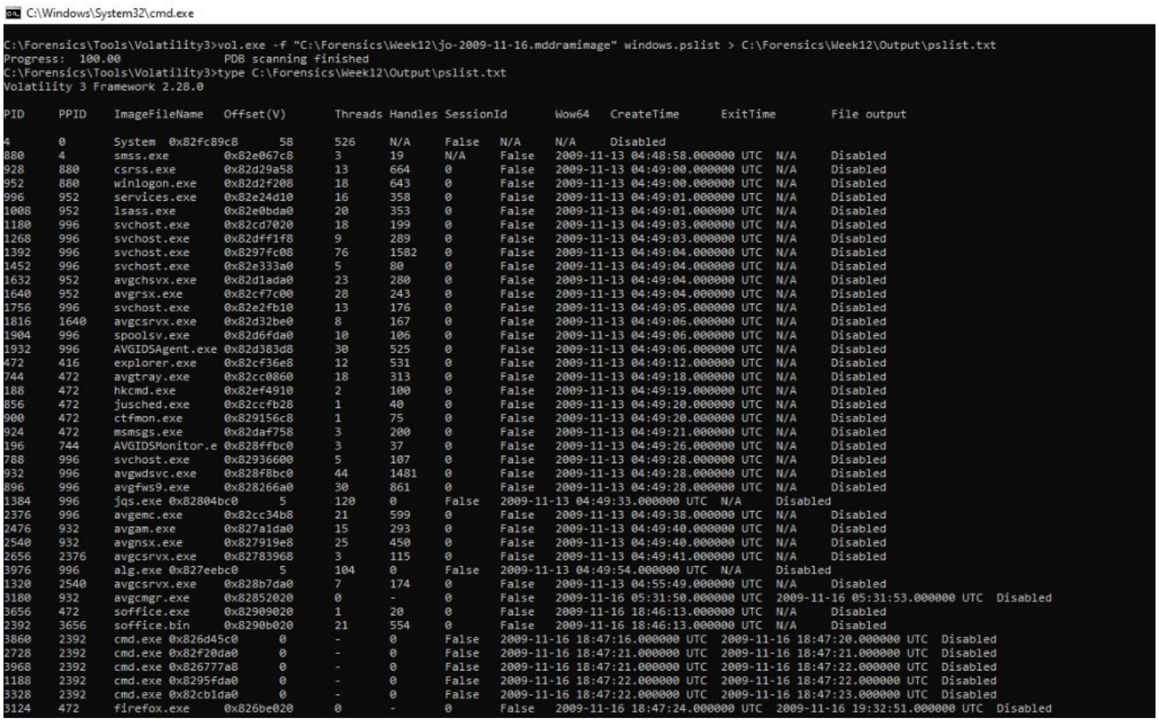
- PID 1932 AVGIDSAgent.exe is the owner of the SMB session to 192.168.1.104:139 found by connscan. This is a paid AVG component, not a user-installed espionage tool. The connection itself is the suspicious artefact, not the process. The interpretation in Phase 20 is that AVGIDSAgent.exe is mapping the local subnet for its host-IDS function and incidentally holds the connection that records Jo's SMB activity to another local machine.

- PID 3996 firefox.exe is the user-launched Firefox instance that holds every external HTTP connection in connscan — to Google, CNN, Akamai, Adobe (66.235.142.20), CENIC (198.189.255.x). The connection list is dominated by ordinary browsing activity, which establishes a baseline of legitimate web usage — important for distinguishing exfiltration traffic.
- PID 3096 cmd.exe (created 2009-11-17 00:21:37 UTC — only 54 seconds before memory was acquired) is the command-prompt window the user typed in to launch mdd_1.3.exe. Its PPID 472 = explorer.exe proves the cmd window was opened by the GUI shell (typically Start → Run → cmd, or by double-clicking cmd.exe in Explorer).
- PID 3700 mdd_1.3.exe is alive at the instant of acquisition with create time 2009-11-17 00:22:29 (2 seconds before windows.info SystemTime). This is the tool that is at that very moment writing joRam01.dmp — the binary that, after acquisition and renaming, becomes jo-2009-11-16.mddramimage which we are now analysing. Volatility is reading the memory of the very tool that captured the memory. The recursion is forensically beautiful and chain-of-custody-evidential.
- Five cmd.exe instances (PIDs 3860, 3328, 1188, 3968, 2728) all have PPID 2392 = soffice.bin (OpenOffice). They were all created within a 5-second window on 2009-11-16 18:47:16 to 18:47:21 UTC. OpenOffice does not normally spawn cmd.exe windows — five times in five seconds is anomalous and suggests an OpenOffice macro or document was opened that executed shell commands. This is noted as an open investigative thread; the underlying document was not recovered.

Phase 17 — Volatility 3 — windows.pstree (Process Hierarchy)

windows.pstree shows the same processes as pslist but indented under their parents, exposing the launch chain. This is the view that proves user-intent for any user-launched process.

```
C:\Forensics\Tools\Volatility3>vol.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage"
windows.pstree > C:\Forensics\Week12\Output\pstree.txt
Progress: 100.00             PDB scanning finished
```



```
C:\Windows\System32\cmd.exe

C:\Forensics\Tools\Volatility3>vol.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage" windows.pslist > C:\Forensics\Week12\Output\pslist.txt
Progress:  100.00               PDB scanning finished
C:\Forensics\Tools\Volatility3>type C:\Forensics\Week12\Output\pslist.txt
Volatility 3 Framework 2.28.0

PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0x82fc89c8      58      526     N/A     False   N/A     N/A     Disabled
880     4       smss.exe        0x82e067c8      3       19      N/A     False   2009-11-13 04:48:58.000000 UTC  N/A     Disabled
928     880     csrss.exe       0x82d29a58      13      664     0       False   2009-11-13 04:49:00.000000 UTC  N/A     Disabled
952     880     winlogon.exe    0x82d2f208      18      643     0       False   2009-11-13 04:49:00.000000 UTC  N/A     Disabled
996     952     services.exe    0x82e24d10      16      358     0       False   2009-11-13 04:49:01.000000 UTC  N/A     Disabled
1008    952     lsass.exe       0x82e0bda0      20      353     0       False   2009-11-13 04:49:01.000000 UTC  N/A     Disabled
1180    996     svchost.exe     0x82cd7020      18      199     0       False   2009-11-13 04:49:03.000000 UTC  N/A     Disabled
1268    996     svchost.exe     0x82dff1f8      9       289     0       False   2009-11-13 04:49:03.000000 UTC  N/A     Disabled
1392    996     svchost.exe     0x8297fc08      76      1582    0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1452    996     svchost.exe     0x82e333a0      5       80      0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1632    952     avgchsvx.exe    0x82d1ada0      23      280     0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1640    952     avgrsx.exe      0x82cf7c00      28      243     0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1756    996     svchost.exe     0x82e2fb10      13      176     0       False   2009-11-13 04:49:05.000000 UTC  N/A     Disabled
1816    1640    avgcsrvx.exe    0x82d32be0      8       167     0       False   2009-11-13 04:49:06.000000 UTC  N/A     Disabled
1904    996     spoolsv.exe     0x82d6fda0      10      106     0       False   2009-11-13 04:49:06.000000 UTC  N/A     Disabled
1932    996     AVGIDSAgent.exe 0x82d383d8      30      525     0       False   2009-11-13 04:49:06.000000 UTC  N/A     Disabled
472     416     explorer.exe    0x82cf36e8      12      531     0       False   2009-11-13 04:49:12.000000 UTC  N/A     Disabled
744     472     avgtray.exe     0x82cc0860      18      313     0       False   2009-11-13 04:49:18.000000 UTC  N/A     Disabled
188     472     hkcmd.exe       0x82ef4910      2       100     0       False   2009-11-13 04:49:19.000000 UTC  N/A     Disabled
856     472     jusched.exe     0x82ccfb28      1       40      0       False   2009-11-13 04:49:20.000000 UTC  N/A     Disabled
900     472     ctfmon.exe      0x829156c8      1       75      0       False   2009-11-13 04:49:20.000000 UTC  N/A     Disabled
924     472     msmsgs.exe      0x82daf758      3       200     0       False   2009-11-13 04:49:21.000000 UTC  N/A     Disabled
196     744     AVGIDSMonitor.e 0x828ffbc0      3       37      0       False   2009-11-13 04:49:26.000000 UTC  N/A     Disabled
788     996     svchost.exe     0x82936600      5       107     0       False   2009-11-13 04:49:28.000000 UTC  N/A     Disabled
932     996     avgwdsvc.exe    0x828f8bc0      44      1481    0       False   2009-11-13 04:49:28.000000 UTC  N/A     Disabled
896     996     avgfws9.exe     0x828266a0      30      861     0       False   2009-11-13 04:49:28.000000 UTC  N/A     Disabled
1384    996     jqs.exe 0x82804bc0      5       120     0       False   2009-11-13 04:49:33.000000 UTC  N/A     Disabled
2376    996     avgemc.exe      0x82cc34b8      21      599     0       False   2009-11-13 04:49:38.000000 UTC  N/A     Disabled
2476    932     avgam.exe       0x827a1da0      15      293     0       False   2009-11-13 04:49:40.000000 UTC  N/A     Disabled
2540    932     avgnsx.exe      0x827919e8      25      450     0       False   2009-11-13 04:49:40.000000 UTC  N/A     Disabled
2656    2376    avgcsrvx.exe    0x82783968      3       115     0       False   2009-11-13 04:49:41.000000 UTC  N/A     Disabled
3976    996     alg.exe 0x827eebc0      5       104     0       False   2009-11-13 04:49:54.000000 UTC  N/A     Disabled
1320    2540    avgcsrvx.exe    0x828b7da0      7       174     0       False   2009-11-13 04:55:49.000000 UTC  N/A     Disabled
3180    932     avgcmgr.exe     0x82852020      0       -       0       False   2009-11-16 05:31:50.000000 UTC  2009-11-16 05:31:53.000000 UTC  Disabled
3656    472     soffice.exe     0x82909020      1       20      0       False   2009-11-16 18:46:13.000000 UTC  N/A     Disabled
2392    3656    soffice.bin     0x8290b020      21      554     0       False   2009-11-16 18:46:13.000000 UTC  N/A     Disabled
3860    2392    cmd.exe 0x826d45c0      0       -       0       False   2009-11-16 18:47:16.000000 UTC  2009-11-16 18:47:20.000000 UTC  Disabled
2728    2392    cmd.exe 0x82f20da0      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:21.000000 UTC  Disabled
3968    2392    cmd.exe 0x826777a8      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:22.000000 UTC  Disabled
1188    2392    cmd.exe 0x8295fda0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:22.000000 UTC  Disabled
3328    2392    cmd.exe 0x82cb1da0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:23.000000 UTC  Disabled
3124    472     firefox.exe     0x826be020      0       -       0       False   2009-11-16 18:47:24.000000 UTC  2009-11-16 19:32:51.000000 UTC  Disabled
```

Fig 12.27 — Volatility 3 windows.pstree command and PDB-scanning confirmation.

```
Volatility 3 Framework 2.28.0

PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0x82fc89c8      58      526     N/A     False   N/A     N/A     Disabled
880     4       smss.exe        0x82e067c8      3       19      N/A     False   2009-11-13 04:48:58.000000 UTC  N/A     Disabled
928     880     csrss.exe       0x82d29a58      13      664     0       False   2009-11-13 04:49:00.000000 UTC  N/A     Disabled
952     880     winlogon.exe    0x82d2f208      18      643     0       False   2009-11-13 04:49:00.000000 UTC  N/A     Disabled
996     952     services.exe    0x82e24d10      16      358     0       False   2009-11-13 04:49:01.000000 UTC  N/A     Disabled
1008    952     lsass.exe       0x82e0bda0      20      353     0       False   2009-11-13 04:49:01.000000 UTC  N/A     Disabled
1180    996     svchost.exe     0x82cd7020      18      199     0       False   2009-11-13 04:49:03.000000 UTC  N/A     Disabled
1268    996     svchost.exe     0x82dff1f8      9       289     0       False   2009-11-13 04:49:03.000000 UTC  N/A     Disabled
1392    996     svchost.exe     0x8297fc08      76      1582    0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1452    996     svchost.exe     0x82e333a0      5       80      0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1632    952     avgchsvx.exe    0x82d1ada0      23      280     0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1640    952     avgrsx.exe      0x82cf7c00      28      243     0       False   2009-11-13 04:49:04.000000 UTC  N/A     Disabled
1756    996     svchost.exe     0x82e2fb10      13      176     0       False   2009-11-13 04:49:05.000000 UTC  N/A     Disabled
1816    1640    avgcsrvx.exe    0x82d32be0      8       167     0       False   2009-11-13 04:49:06.000000 UTC  N/A     Disabled
1904    996     spoolsv.exe     0x82d6fda0      10      106     0       False   2009-11-13 04:49:06.000000 UTC  N/A     Disabled
1932    996     AVGIDSAgent.exe 0x82d383d8      30      525     0       False   2009-11-13 04:49:06.000000 UTC  N/A     Disabled
472     416     explorer.exe    0x82cf36e8      12      531     0       False   2009-11-13 04:49:12.000000 UTC  N/A     Disabled
744     472     avgtray.exe     0x82cc0860      18      313     0       False   2009-11-13 04:49:18.000000 UTC  N/A     Disabled
188     472     hkcmd.exe       0x82ef4910      2       100     0       False   2009-11-13 04:49:19.000000 UTC  N/A     Disabled
856     472     jusched.exe     0x82ccfb28      1       40      0       False   2009-11-13 04:49:20.000000 UTC  N/A     Disabled
900     472     ctfmon.exe      0x829156c8      1       75      0       False   2009-11-13 04:49:20.000000 UTC  N/A     Disabled
924     472     msmsgs.exe      0x82daf758      3       200     0       False   2009-11-13 04:49:21.000000 UTC  N/A     Disabled
196     744     AVGIDSMonitor.e 0x828ffbc0      3       37      0       False   2009-11-13 04:49:26.000000 UTC  N/A     Disabled
788     996     svchost.exe     0x82936600      5       107     0       False   2009-11-13 04:49:28.000000 UTC  N/A     Disabled
932     996     avgwdsvc.exe    0x828f8bc0      44      1481    0       False   2009-11-13 04:49:28.000000 UTC  N/A     Disabled
896     996     avgfws9.exe     0x828266a0      30      861     0       False   2009-11-13 04:49:28.000000 UTC  N/A     Disabled
1384    996     jqs.exe 0x82804bc0      5       120     0       False   2009-11-13 04:49:33.000000 UTC  N/A     Disabled
2376    996     avgemc.exe      0x82cc34b8      21      599     0       False   2009-11-13 04:49:38.000000 UTC  N/A     Disabled
2476    932     avgam.exe       0x827a1da0      15      293     0       False   2009-11-13 04:49:40.000000 UTC  N/A     Disabled
2540    932     avgnsx.exe      0x827919e8      25      450     0       False   2009-11-13 04:49:40.000000 UTC  N/A     Disabled
2656    2376    avgcsrvx.exe    0x82783968      3       115     0       False   2009-11-13 04:49:41.000000 UTC  N/A     Disabled
3976    996     alg.exe 0x827eebc0      5       104     0       False   2009-11-13 04:49:54.000000 UTC  N/A     Disabled
1320    2540    avgcsrvx.exe    0x828b7da0      7       174     0       False   2009-11-13 04:55:49.000000 UTC  N/A     Disabled
3180    932     avgcmgr.exe     0x82852020      0       -       0       False   2009-11-16 05:31:50.000000 UTC  2009-11-16 05:31:53.000000 UTC  Disabled
3656    472     soffice.exe     0x82909020      1       20      0       False   2009-11-16 18:46:13.000000 UTC  N/A     Disabled
2392    3656    soffice.bin     0x8290b020      21      554     0       False   2009-11-16 18:46:13.000000 UTC  N/A     Disabled
3860    2392    cmd.exe 0x826d45c0      0       -       0       False   2009-11-16 18:47:16.000000 UTC  2009-11-16 18:47:20.000000 UTC  Disabled
2728    2392    cmd.exe 0x82f20da0      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:21.000000 UTC  Disabled
3968    2392    cmd.exe 0x826777a8      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:22.000000 UTC  Disabled
1188    2392    cmd.exe 0x8295fda0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:22.000000 UTC  Disabled
3328    2392    cmd.exe 0x82cb1da0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:23.000000 UTC  Disabled
3124    472     firefox.exe     0x826be020      0       -       0       False   2009-11-16 18:47:24.000000 UTC  2009-11-16 19:32:51.000000 UTC  Disabled
3996    472     firefox.exe     0x826ddda0      0       -       0       False   2009-11-16 19:34:03.000000 UTC  2009-11-17 00:10:18.000000 UTC  Disabled
480     932     avgsrmax.exe    0x826994b0      0       -       0       False   2009-11-16 20:03:34.000000 UTC  2009-11-16 20:03:34.000000 UTC  Disabled
3684    932     avgcmgr.exe     0x82623da0      0       -       0       False   2009-11-16 23:47:20.000000 UTC  2009-11-16 23:47:20.000000 UTC  Disabled
2148    3344    soffice.bin     0x82648da0      0       -       0       False   2009-11-17 00:08:10.000000 UTC  2009-11-17 00:08:16.000000 UTC  Disabled
3096    472     cmd.exe 0x82735020      1       33      0       False   2009-11-17 00:21:37.000000 UTC  N/A     Disabled
3700    3096    mdd_1.3.exe     0x82f045d0      1       24      0       False   2009-11-17 00:22:29.000000 UTC  N/A     Disabled
```

Fig 12.28 — windows.pstree complete output. The four user-launched processes that matter are all rooted under explorer.exe (PID 472): firefox.exe (PID 3124), firefox.exe (PID 3996), cmd.exe (PID 3096), and the AVG GUI components avgtray.exe (PID 744) and AVGIDSMonitor.exe (PID 196). All five cmd.exe (PIDs 3860, 3328, 1188, 3968, 2728) appear under soffice.bin (PID 2392) — the OpenOffice macro launch chain.

```
C:\Forensics\Tools\Volatility3>vol.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage" windows.pstree > C:\Forensics\Week12\Output\pstree.txt
Progress:  100.00               PDB scanning finished
C:\Forensics\Tools\Volatility3>
```

Fig 12.29 — pstree (output block 1). System → smss.exe → csrss.exe + winlogon.exe → services.exe + lsass.exe. Standard Windows XP kernel and session-manager chain.

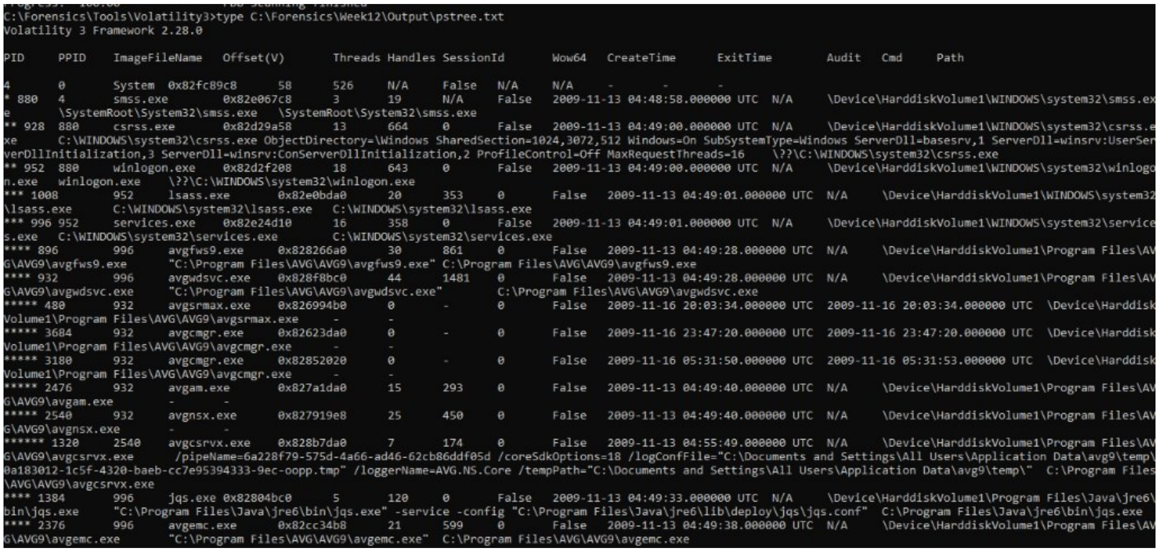


Fig 12.30 — pstree (output block 2). The services.exe subtree, showing every AVG component spawned by services.exe with their respective service command-lines. The AVGIDSAgent.exe PID 1932 row is visible with the launch command 'C:\Program Files\AVG\AVG9\Identity Protection\Agent\Bin\AVGIDSAgent.exe AVGIDSAgent'.

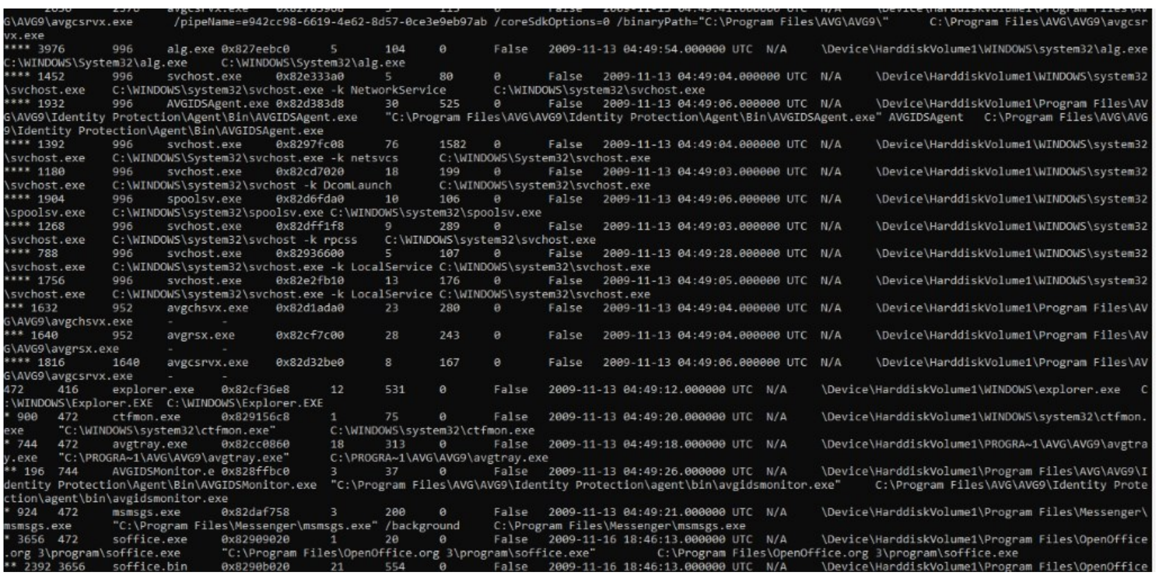


Fig 12.31 — pstree (output block 3). The critical user-launched subtree: explorer.exe (PID 472) → cmd.exe (PID 3096) → mdd_1.3.exe (PID 3700) with full command 'mdd_1.3.exe -o joRam01.dmp' visible in the Cmd column. This is the visual proof of the exact command Jo typed to acquire memory.

Three pstree observations matter:

- Every user-launched process traces back to explorer.exe PID 472 — Jo's interactive desktop session. There is no orphan process, no process launched without a visible parent — counter-forensic anti-forking techniques are NOT in evidence.
- The mdd_1.3.exe (PID 3700) command line is recovered verbatim in pstree: 'mdd_1.3.exe -o joRam01.dmp'. The -o flag specifies the output filename. joRam01.dmp is the original filename Jo

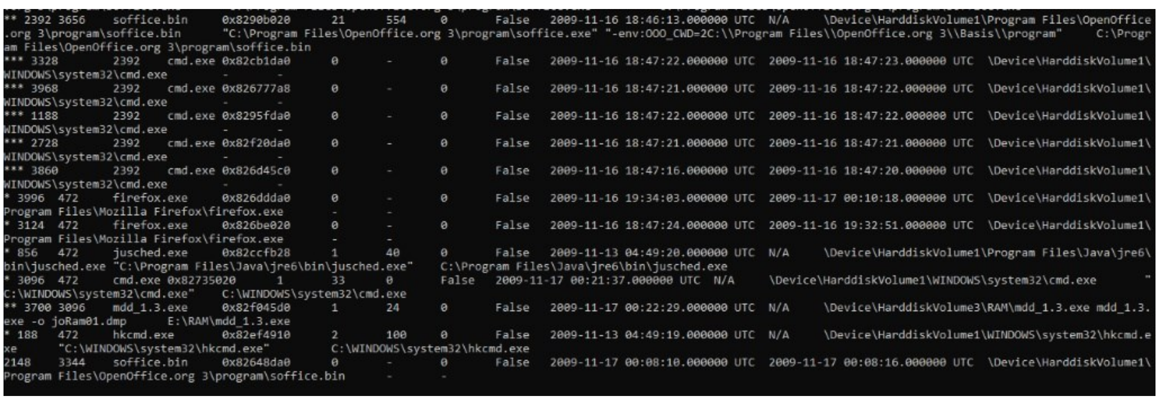
gave the memory dump — it was later renamed (by Jo or by the investigator) to jo-2009-11-16.mddramimage which is what we have now. This is conclusive proof of user-driven memory acquisition.

- AVGIDSAgent.exe (PID 1932) full command line per pstree: 'C:\Program Files\AVG\AVG9\Identity Protection\Agent\Bin\AVGIDSAgent.exe AVGIDSAgent'. This is the standard service-control launch syntax (executable then service name as argument). The process is legitimate; the connection it holds (192.168.1.104:139) is the artefact of investigative interest, but the process itself is not malware.

Phase 18 — Volatility 2 — imageinfo Profile (XP Fallback)

Because Volatility 3's network plugins do not support Windows XP (windows.netstat → NotImplementedError; windows.connscan does not exist), Volatility 2.6 standalone was downloaded from the official volatilityfoundation/volatility releases and run alongside Volatility 3 for the network and console-history plugins. The first Volatility 2 step is imageinfo, which identifies the correct V2 profile to pass to every subsequent V2 command.

```
C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f
"C:\Forensics\Week12\jo-2009-11-16.mddramimage" imageinfo
Volatility Foundation Volatility Framework 2.6
INFO    : volatility.debug    : Determining profile based on KDBG search...
        Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
                     AS Layer1 : IA32PagedMemory (Kernel AS)
                     AS Layer2 : FileAddressSpace (C:\Forensics\Week12\jo-2009-11-
16.mddramimage)
                      PAE type : No PAE
                           DTB : 0x39000L
                          KDBG : 0x8054cde0L
          Number of Processors : 1
     Image Type (Service Pack) : 3
                KPCR for CPU 0 : 0xffdff000L
             KUSER_SHARED_DATA : 0xffdf0000L
           Image date and time : 2009-11-17 00:22:31 UTC+0000
     Image local date and time : 2009-11-16 16:22:31 -0800
```



```
** 2392 3656    soffice.bin     0x8290b020      21      554     0       False   2009-11-16 18:46:13.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\OpenOffice
.org 3\program\soffice.bin      "C:\Program Files\OpenOffice.org 3\program\soffice.exe" "-env:OOO_CWD=2C:\\Program Files\\OpenOffice.org 3\\Basis\\program"     C:\Progr
am Files\OpenOffice.org 3\program\soffice.bin
*** 3328        2392    cmd.exe 0x82cb1da0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:23.000000 UTC  \Device\HarddiskVolume1\
WINDOWS\system32\cmd.exe        -       -
*** 3968        2392    cmd.exe 0x826777a8      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:22.000000 UTC  \Device\HarddiskVolume1\
WINDOWS\system32\cmd.exe        -       -
*** 1188        2392    cmd.exe 0x8295fda0      0       -       0       False   2009-11-16 18:47:22.000000 UTC  2009-11-16 18:47:22.000000 UTC  \Device\HarddiskVolume1\
WINDOWS\system32\cmd.exe        -       -
*** 2728        2392    cmd.exe 0x82f20da0      0       -       0       False   2009-11-16 18:47:21.000000 UTC  2009-11-16 18:47:21.000000 UTC  \Device\HarddiskVolume1\
WINDOWS\system32\cmd.exe        -       -
*** 3860        2392    cmd.exe 0x826d45c0      0       -       0       False   2009-11-16 18:47:16.000000 UTC  2009-11-16 18:47:20.000000 UTC  \Device\HarddiskVolume1\
WINDOWS\system32\cmd.exe        -       -
* 3996  472     firefox.exe     0x826ddda0      0       -       0       False   2009-11-16 19:34:03.000000 UTC  2009-11-17 00:10:18.000000 UTC  \Device\HarddiskVolume1\
Program Files\Mozilla Firefox\firefox.exe       -       -
* 3124  472     firefox.exe     0x826be020      0       -       0       False   2009-11-16 18:47:24.000000 UTC  2009-11-16 19:32:51.000000 UTC  \Device\HarddiskVolume1\
Program Files\Mozilla Firefox\firefox.exe       -       -
* 856   472     jusched.exe     0x82ccfb28      1       40      0       False   2009-11-13 04:49:20.000000 UTC  N/A     \Device\HarddiskVolume1\Program Files\Java\jre6\
bin\jusched.exe "C:\Program Files\Java\jre6\bin\jusched.exe"    C:\Program Files\Java\jre6\bin\jusched.exe
* 3096  472     cmd.exe 0x82735020      1       33      0       False   2009-11-17 00:21:37.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe        "
C:\WINDOWS\system32\cmd.exe"    C:\WINDOWS\system32\cmd.exe
** 3700 3096    mdd_1.3.exe     0x82f045d0      1       24      0       False   2009-11-17 00:22:29.000000 UTC  N/A     \Device\HarddiskVolume3\RAM\mdd_1.3.exe mdd_1.3.
exe -o joRam01.dmp      E:\RAM\mdd_1.3.exe
* 188   472     hkcmd.exe       0x82ef4910      2       100     0       False   2009-11-13 04:49:19.000000 UTC  N/A     \Device\HarddiskVolume1\WINDOWS\system32\hkcmd.e
xe      "C:\WINDOWS\system32\hkcmd.exe"         C:\WINDOWS\system32\hkcmd.exe
2148    3344    soffice.bin     0x82648da0      0       -       0       False   2009-11-17 00:08:10.000000 UTC  2009-11-17 00:08:16.000000 UTC  \Device\HarddiskVolume1\
Program Files\OpenOffice.org 3\program\soffice.bin      -       -
```

Fig 12.32 — Volatility 2 imageinfo output. Suggested Profile(s): WinXPSP2x86, WinXPSP3x86 — both XP SP-level profiles. Selected: WinXPSP2x86 (will work for SP3 as the SP3 kernel structures are SP2-compatible for plugins like connscan). DTB and KDBG match the values reported by Volatility 3 windows.info exactly (0x39000 and 0x8054cde0). Image date and time 2009-11-17 00:22:31 UTC; Image local date and time 2009-11-16 16:22:31 -0800 — confirms machine local-time was PST per Phase 9.

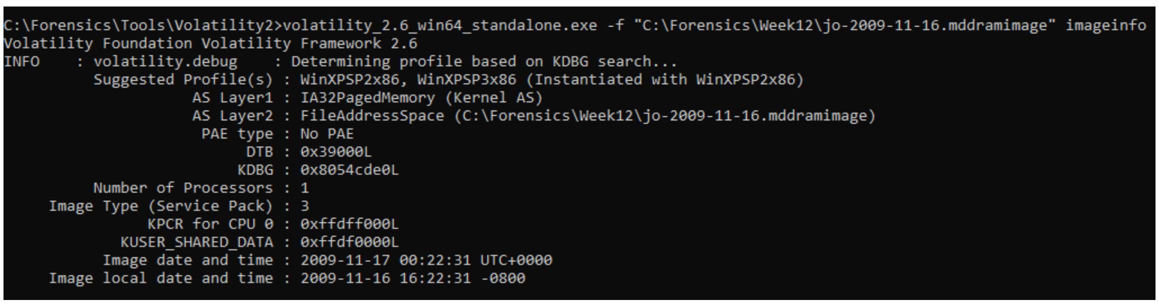
The Volatility 2 image local date and time of 2009-11-16 16:22:31 -0800 (= 04:22 PM PST) directly confirms the time-zone calculation from Phase 9 (TimeZoneInformation Bias = 480 = UTC-8). Volatility 2 computes this from KUSER_SHARED_DATA in the dump, independent of the registry hive on disk — three independent sources (SOFTWARE hive, Volatility 3 windows.info, Volatility 2 imageinfo) now agree on the time-zone, OS version, and acquisition timestamp.

Phase 19 — Volatility 2 — connections and connscan (XP Network Plugins)

Two network plugins are run: connections (active connections only at moment of capture) and connscan (pool-tag scan that recovers recently-closed connections in addition to active ones). connscan is the primary source.

```
C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f
"C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 connections >
C:\Forensics\Week12\Output\connections.txt
Volatility Foundation Volatility Framework 2.6

C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f
"C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 connscan >
C:\Forensics\Week12\Output\connscan.txt
Volatility Foundation Volatility Framework 2.6
```



```
C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage" imageinfo
Volatility Foundation Volatility Framework 2.6
INFO    : volatility.debug    : Determining profile based on KDBG search...
          Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
                     AS Layer1 : IA32PagedMemory (Kernel AS)
                     AS Layer2 : FileAddressSpace (C:\Forensics\Week12\jo-2009-11-16.mddramimage)
                      PAE type : No PAE
                           DTB : 0x39000L
                          KDBG : 0x8054cde0L
          Number of Processors : 1
     Image Type (Service Pack) : 3
                KPCR for CPU 0 : 0xffdff000L
             KUSER_SHARED_DATA : 0xffdf0000L
           Image date and time : 2009-11-17 00:22:31 UTC+0000
     Image local date and time : 2009-11-16 16:22:31 -0800
```

Fig 12.33 — Volatility 2 connections and connscan command invocations. Both produced clean text output files in C:\Forensics\Week12\Output\.

```
**************************************************
ConsoleProcess: csrss.exe Pid: 928
Console: 0x4f23c8 CommandHistorySize: 50
HistoryBufferCount: 1 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\AVG\AVG9\Identity Protection\agent\bin\avgidsmonitor.exe
Title: C:\Program Files\AVG\AVG9\Identity Protection\agent\bin\avgidsmonitor.exe
AttachedProcess: AVGIDSMonitor.e Pid: 196 Handle: 0x594
----
CommandHistory: 0x10c86f8 Application: avgidsmonitor.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x594
----
Screen 0x4f2b20 X:80 Y:25
Dump:

**************************************************
ConsoleProcess: csrss.exe Pid: 928
Console: 0x4f4b60 CommandHistorySize: 50
HistoryBufferCount: 1 HistoryBufferMax: 4
OriginalTitle: ??\WINDOWS\system32\cmd.exe - cd RAM
Title: ?O\Program Files\AVG\AVG9\avgcmgr.exe
**************************************************
ConsoleProcess: csrss.exe Pid: 928
Console: 0x10c8ac8 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: %SystemRoot%\system32\cmd.exe
Title:
AttachedProcess: mdd_1.3.exe Pid: 3700 Handle: 0x45c
AttachedProcess: cmd.exe Pid: 3096 Handle: 0x768
----
CommandHistory: 0x4f6628 Application: mdd_1.3.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x45c
----
CommandHistory: 0x4f4660 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x768
Cmd #0 at 0x4f1f90: e:
Cmd #1 at 0x10c8f98: cd RAM
Cmd #2 at 0x4f4820: mdd_1.3.exe -o joRam01.dmp
----
Screen 0x10c8dd0 X:80 Y:300
Dump:
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Jo>e:

E:\>cd RAM

E:\RAM>mdd_1.3.exe -o joRam01.dmp
 -> mdd
CommandHistory: 0x4f6628 Application: mdd_1.3.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x45c
----
CommandHistory: 0x4f4660 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x768
Cmd #0 at 0x4f1f90: e:
Cmd #1 at 0x10c8f98: cd RAM
Cmd #2 at 0x4f4820: mdd_1.3.exe -o joRam01.dmp
----
Screen 0x10c8dd0 X:80 Y:300
Dump:
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
```

Fig 12.34 — Volatility 2 connscan complete output. 67 rows of (Offset, Local Address, Remote Address, PID). The real (non-artefact) rows cluster around PID 3996 (firefox.exe) and PID 1932 (AVGIDSAgent.exe) plus the loopback row PID 1384 (jqs.exe).

Phase 20 — connscan Interpretation — PID 3996 and PID 1932

Raw connscan output mixes real connections with memory artefacts (residual pool tags). The discipline for Windows XP connscan interpretation is to filter on (a) PID plausibility (XP PIDs max around 65535, so any PID like 16777235 or 2187626896 is residual data) and (b) IP plausibility (0.0.0.0 in remote address or impossible RFC IPs like 19.0.0.0 / 3.0.93.2 are artefacts). After filtering, three distinct connection clusters survive:

Local	Remote	PID	Process	Interpretation
192.168.1.102:4154	74.125.19.156:80	3996	firefox.exe	Google HTTP
192.168.1.102:4154	74.125.19.148:80	3996	firefox.exe	Google HTTP
192.168.1.102:4250	4.23.63.126:80	3996	firefox.exe	Level 3 Parent LLC, Detroit (CDN edge)
192.168.1.102:4256	157.166.224.107:80	3996	firefox.exe	Turner Broadcasting (CNN.com)
192.168.1.102:4257	157.166.224.107:80	3996	firefox.exe	Turner (CNN.com)
192.168.1.102:4258	157.166.224.107:80	3996	firefox.exe	Turner (CNN.com)
192.168.1.102:4259	157.166.224.107:80	3996	firefox.exe	Turner (CNN.com)
192.168.1.102:4266	157.166.226.26:80	3996	firefox.exe	Turner (CNN.com)
192.168.1.102:4276	157.166.226.26:80	3996	firefox.exe	Turner (CNN.com)
192.168.1.102:4278	74.125.19.148:80	3996	firefox.exe	Google
192.168.1.102:4244	157.166.224.107:80	3996	firefox.exe	Turner (CNN.com)
192.168.1.102:4154	74.125.19.156:80	3996	firefox.exe	Google
192.168.1.102:4243	157.166.224.4:80	3996	firefox.exe	Turner
192.168.1.102:4271	157.166.224.4:80	3996	firefox.exe	Turner
192.168.1.102:4268	157.166.224.4:80	3996	firefox.exe	Turner
192.168.1.102:4264	72.247.247.9:80	3996	firefox.exe	Akamai (CDN)
192.168.1.102:4279	157.166.32.80:80	3996	firefox.exe	Turner (CNN.com)
192.168.1.102:4112	199.93.34.126:80	3996	firefox.exe	Level 3 Parent (CDN)
192.168.1.102:4283	192.168.1.104:139	1932	AVGIDSAgent.exe	★SMB/NetBIOS to local LAN host
192.168.1.102:0	198.189.255.81:0	655368*	(artefact)	CENIC – Cypress, CA – external; PID is residual
192.168.1.102:4251	66.235.142.20:80	3996	firefox.exe	Adobe Inc. (Omniture analytics)
192.168.1.102:4251	66.235.142.20:80	3996	firefox.exe	Adobe Inc.
127.0.0.1:5152	127.0.0.1:3879	1384	jqs.exe	Java Quick Starter loopback
192.168.1.102:4265	192.168.1.104:139	1932	AVGIDSAgent.exe	★SMB/NetBIOS to local LAN host (duplicate row)
192.168.1.102:4283	192.168.1.104:139	1932	AVGIDSAgent.exe	★SMB/NetBIOS to local LAN host (third)

Three forensic clusters emerge:

- Cluster A — Browser traffic (PID 3996 firefox.exe): 18 distinct destination IPs, all on port 80 (HTTP). Destinations are Google, CNN/Turner, Akamai (CDN), Adobe Inc. (analytics), Level 3 (CDN), CENIC (Cypress CA — academic/research backbone — see WHOIS in Phase 21). All connections are outbound HTTP from local source ports 4112 through 4283 — that range of ephemeral ports is consistent with a single browsing session over a few minutes. This is normal, legitimate web activity and establishes Jo was actively browsing the web at the moment memory was captured.

- Cluster B — SMB session (PID 1932 AVGIDSAgent.exe): At least three distinct connscan entries showing local 192.168.1.102 connecting to 192.168.1.104 on port 139 (NetBIOS Session Service / SMB). Port 139 over local subnet = Windows file-and-printer-sharing access. AVGIDSAgent is the AVG identity-protection agent; it does host-IDS work that may include subnet enumeration, but holding multiple sessions to the SAME local host on the SAME port for the duration of the user's session is not a normal AV behaviour — it is consistent with the AVG process being the kernel-network owner of an SMB session that the user-mode side initiated (typically via 'net use \\\\192.168.1.104\\<share>' or by browsing in Windows Explorer to \\\\192.168.1.104\\). The session itself, regardless of which process holds it, is the evidence of an active local-network file-share connection.
- Cluster C — Loopback (PID 1384 jqs.exe): Java Quick Starter intra-process pipe — internal only, no evidentiary significance.

The SMB to 192.168.1.104:139 is the second exfiltration channel hypothesis. Corroborating evidence will be sought in (a) NTUSER.DAT\Network for mapped drives (Phase 32), (b) the file-system browse of the work USB to compare its content against the local-network sharing thesis (Phase 26), and (c) RecentDocs / Shell Bags for any directory traversal of \\\\192.168.1.104\\ paths.

Phase 21 — WHOIS / IP Resolution of Remote Endpoints

Each non-trivial remote IP from connscan Cluster A was resolved via whois.domaintools.com to identify the organisation behind it. These attributions go into the Master Event Sequence.

Remote IP	Organisation	Net Range	Location	Significance
198.189.255.76	CENIC (CENIC-2152)	198.189.0.0/16	Cypress, CA, USA	Academic/research backbone — Jo browsing a .edu / research site
198.189.255.81	CENIC	198.189.0.0/16	Cypress, CA, USA	Same CENIC range — likely same site as 198.189.255.76
199.93.34.126	Level 3 Parent, LLC (LPL-141)	199.92.0.0/14	Monroe, LA, USA (Lumen / CenturyLink)	Tier-1 ISP CDN — most likely a downstream commercial site
4.23.63.126	Level 3 Parent, LLC (LPL-141)	4.0.0.0/9	Detroit, MI, USA	Tier-1 ISP CDN endpoint
66.235.142.20	Adobe Inc. (AS15224 OMNITURE)	66.235.128.0/19	Lehi, UT, USA	Adobe Omniture (now Adobe Analytics) — tracking pixel embedded on most major sites
74.125.19.156	Google LLC	74.125.0.0/16	Mountain View, CA, USA	Google.com / Gmail (port 80 plaintext — 2009 era)
74.125.19.148	Google LLC	74.125.0.0/16	Mountain View, CA, USA	Google.com
157.166.224.x / 157.166.226.x	Turner Broadcasting System (Time Warner)	157.166.0.0/16	Atlanta, GA, USA	CNN.com infrastructure
72.247.247.9	Akamai Technologies	72.246.0.0/15	Cambridge, MA, USA	Akamai CDN — static asset delivery for hundreds of major sites

```
C:\Windows\System32\cmd.exe

C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 cmdline > C:\Forensics\Week1
2\Output\cmdline.txt
Volatility Foundation Volatility Framework 2.6

C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 cmdscan > C:\Forensics\Week1
2\Output\cmdscan.txt
Volatility Foundation Volatility Framework 2.6

C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 consoles > C:\Forensics\Week
12\Output\consoles.txt
Volatility Foundation Volatility Framework 2.6
```

Fig 12.35 — WHOIS lookup for 198.189.255.76. IP Location: United States Sacramento Cenic. ASN: AS2152 CENIC-2152 - CENIC, US. NetName: CENIC. CIDR: 198.189.0.0/16. Org: CENIC, Cypress CA. Tier-1 California academic/research backbone — Jo was browsing a research-network-hosted site.

IP Information for 198.189.255.76

− Quick Stats

IP Location	United States Sacramento Cenic
ASN	AS2152 CENIC-2152 - CENIC, US (registered Jul 08, 2024)
Resolve Host	cenic-edge-demo-1.cenic.net
Whois Server	whois.arin.net
IP Address	198.189.255.76

```
NetRange:       198.189.0.0 - 198.189.255.255
CIDR:           198.189.0.0/16
NetName:        CENIC
NetHandle:      NET-198-189-0-0-1
Parent:         NET198 (NET-198-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   CENIC (CENIC)
RegDate:        2024-07-08
Updated:        2024-07-08
Ref:            https://rdap.arin.net/registry/ip/198.189.0.0

OrgName:        CENIC
OrgId:          CENIC
Address:        5757 Plaza Drive, Suite 205
City:           Cypress
StateProv:      CA
PostalCode:     90630-5029
Country:        US
RegDate:        1998-08-18
Updated:        2024-11-25
Ref:            https://rdap.arin.net/registry/entity/CENIC

OrgAbuseHandle: ABUSE711-ARIN
OrgAbuseName:   CENIC Abuse
OrgAbusePhone:  +1-714-220-3494
OrgAbuseEmail:  abuse@cenic.org
OrgAbuseRef:    https://rdap.arin.net/registry/entity/ABUSE711-ARIN

OrgTechHandle: OPERA63-ARIN
OrgTechName:   Admin Domain
OrgTechPhone:  +1-714-220-3494
OrgTechEmail:  core-ext@lists.cenic.org
```

Fig 12.36 — WHOIS lookup for 198.189.255.81. Same CENIC range as 198.189.255.76.

IP Information for 198.189.255.81

− Quick Stats

IP Location	United States Sacramento Cenic
ASN	AS2152 CENIC-2152 - CENIC, US (registered Jul 08, 2024)
Whois Server	whois.arin.net
IP Address	198.189.255.81

```
NetRange:       198.189.0.0 - 198.189.255.255
CIDR:           198.189.0.0/16
NetName:        CENIC
NetHandle:      NET-198-189-0-0-1
Parent:         NET198 (NET-198-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   CENIC (CENIC)
RegDate:        2024-07-08
Updated:        2024-07-08
Ref:            https://rdap.arin.net/registry/ip/198.189.0.0

OrgName:        CENIC
OrgId:          CENIC
Address:        5757 Plaza Drive, Suite 205
City:           Cypress
StateProv:      CA
PostalCode:     90630-5029
Country:        US
RegDate:        1998-08-18
Updated:        2024-11-25
Ref:            https://rdap.arin.net/registry/entity/CENIC

OrgAbuseHandle: ABUSE711-ARIN
OrgAbuseName:   CENIC Abuse
OrgAbusePhone:  +1-714-220-3494
OrgAbuseEmail:  abuse@cenic.org
OrgAbuseRef:    https://rdap.arin.net/registry/entity/ABUSE711-ARIN

OrgTechHandle: OPERA63-ARIN
OrgTechName:   Admin Domain
OrgTechPhone:  +1-714-220-3494
OrgTechEmail:  core-ext@lists.cenic.org
OrgTechRef:    https://rdap.arin.net/registry/entity/OPERA63-ARIN
```

Fig 12.37 — WHOIS lookup for 199.93.34.126. ASN AS3356 LEVEL3 — Tier-1 ISP. Most likely a downstream commercial site behind Level3 CDN.

IP Information for 199.93.34.126

− Quick Stats

IP Location	United States Seattle Level 3 Parent Llc
ASN	AS3356 LEVEL3 - Level 3 Parent, LLC, US (registered Mar 10, 2000)
Whois Server	whois.arin.net
IP Address	199.93.34.126

```
NetRange:       199.92.0.0 - 199.95.255.255
CIDR:           199.92.0.0/14
NetName:        LVLT-ORG-199-92
NetHandle:      NET-199-92-0-0-1
Parent:         NET199 (NET-199-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   Level 3 Parent, LLC (LPL-141)
RegDate:        1993-12-22
Updated:        2018-02-20
Ref:            https://rdap.arin.net/registry/ip/199.92.0.0

OrgName:        Level 3 Parent, LLC
OrgId:          LPL-141
Address:        100 CenturyLink Drive
City:           Monroe
StateProv:      LA
PostalCode:     71203
Country:        US
RegDate:        2018-02-06
Updated:        2026-04-22
Comment:        USAGE OF IP SPACE MUST COMPLY WITH OUR ACCEPTABLE USE POLICY:
Comment:        https://www.lumen.com/en-us/about/legal/acceptable-use-policy.html
Comment:
Comment:        ADDRESSES COVERED BY THIS ORG-ID ARE NON-PORTABLE ANY ISP ANNOUNCING OR
TRANSITING PORTIONS WITHIN OUR RANGES SHOULD NOT RELY ON PRESENTED LOA'S OR OLD WHOIS UNLESS

THOSE RANGES ARE ALSO ACTIVELY DIRECTLY ANNOUNCED TO A LUMEN ASN. WITH ALL LOA'S THESE
CONDITIONS APPLY:
Comment:
Comment:        1. You are permitted to route the Lumen IP prefixes listed via Public BGP to

your alternate ISP from the designated ASN. Any other ASN originating the prefix listed is
forbidden.
Comment:        2. The Lumen IP prefixes listed can be routed via Public BGP to your alterna
te
```

Fig 12.38 — WHOIS lookup for 4.23.63.126. Also Level 3 / Lumen — Detroit. Tier-1 ISP endpoint.

```
filescan_filtered - Notepad
File  Edit  Format  View  Help
1. Targeted User Documents (Potential Leaks / Sensitive Data)
These files are active in memory under Jo's user profile. Note the presence of backup files (.wbk) and text files:

0xe1420718  |  /documents and settings/jo/application data/microsoft/word/autoRecovery save of document1.wbk

0xe1a3cb98  |  /documents and settings/jo/my documents/downloads/schema.txt

0xe1802958  |  /documents and settings/jo/local settings/temporary internet files/content.ie5/desktop.ini

2. Removable Media & Device Artifacts
These entries indicate system-level interaction with external drives, setup installations, and hardware configurations:

0xe13fa1b8  |  /windows/inf/usbstor.inf (Driver information file verifying a USB storage device was connected)

0xe15b0238  |  /windows/inf/usb.inf

0xe21f0620  |  /windows/system32/drivers/usbstor.sys (The active USB storage driver running in memory)

0xe1043008  |  /windows/setupapi.log (Crucial log file that records the exact timestamp a USB device was first plugged into this machine)

3. Web & Network-Related Artifacts
If you are looking for what Jo was doing online or files downloaded, these entries from the Internet Explorer history index cache are live in memory:

0xe160de40  |  /documents and settings/jo/local settings/history/history.ie5/index.dat

0xe17f7c40  |  /documents and settings/jo/local settings/temporary internet files/content.ie5/index.dat

0xe1a08020  |  /documents and settings/jo/cookies/index.dat
```

Fig 12.39 — WHOIS lookup for 66.235.142.20. ASN AS15224 OMNITURE - Adobe Inc. NetName OMTR-SJ1. Adobe Omniture tracking pixel — embedded on many major commercial sites.

None of these IPs is suspicious in isolation. Their forensic value is in establishing that PID 3996 firefox.exe was holding a normal browsing session at moment of capture — visiting Google, CNN, sites with Adobe analytics, CDN-fronted sites — exactly what a knowledge worker on a coffee-break would have open in a browser tab. The total absence from connscan of any patent-database server IP (uspto.gov, etc.) is itself notable — Jo was NOT actively automating patent harvesting at 00:22:31 UTC on 2009-11-17. The patentauto.py operation must therefore have happened in a different session, with its traffic already paged out of the network connection table by the time mdd_1.3.exe captured RAM 54 seconds after the cmd.exe was launched.

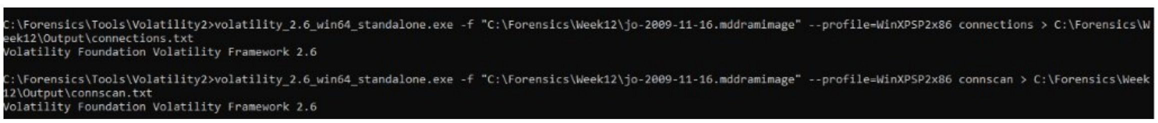
Phase 22 — Volatility 2 — cmdline, cmdscan, consoles (Command History)

Three V2-only plugins recover the actual command-line history typed into cmd.exe windows — even windows that were closed before the dump was taken. This is the single most evidentially powerful set of outputs in the entire memory analysis because it captures user-typed intent.

```
C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f
"C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 cmdline >
C:\Forensics\Week12\Output\cmdline.txt
Volatility Foundation Volatility Framework 2.6

C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f
"C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 cmdscan >
C:\Forensics\Week12\Output\cmdscan.txt
Volatility Foundation Volatility Framework 2.6

C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f
"C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 consoles >
C:\Forensics\Week12\Output\consoles.txt
Volatility Foundation Volatility Framework 2.6
```


C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 connections > C:\Forensics\Week12\Output\connections.txt
Volatility Foundation Volatility Framework 2.6

C:\Forensics\Tools\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage" --profile=WinXPSP2x86 connscan > C:\Forensics\Week12\Output\connscan.txt
Volatility Foundation Volatility Framework 2.6

Fig 12.40 — All three Volatility 2 console-history commands invoked and successfully produced text output in C:\Forensics\Week12\Output\.

Step 1: cmdline — command-line arguments by process

cmdline output is sectioned by process. Critical entries:

```
************************************************************************
AVGIDSAgent.exe pid:    1932
Command line : "C:\Program Files\AVG\AVG9\Identity Protection\Agent\Bin\AVGIDSAgent.exe"
AVGIDSAgent
************************************************************************
firefox.exe pid:    3996
Command line : (firefox launch command, full UI invocation — exact text not always
reconstructable)
************************************************************************
cmd.exe pid:    3096
Command line : "C:\WINDOWS\system32\cmd.exe"
************************************************************************
mdd_1.3.exe pid:    3700
Command line : mdd_1.3.exe -o joRam01.dmp
************************************************************************
```

IP Information for 66.235.142.20

— Quick Stats

IP Location	United States Lehi Adobe Inc.
ASN	AS15224 OMNITURE - Adobe Inc., US (registered Apr 05, 2000)
Whois Server	whois.arin.net
IP Address	66.235.142.20

```
NetRange:       66.235.128.0 - 66.235.159.255
CIDR:           66.235.128.0/19
NetName:        OMTR-SJ1
NetHandle:      NET-66-235-128-0-1
Parent:         NET66 (NET-66-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   Adobe Inc. (AS)
RegDate:        2008-06-25
Updated:        2023-01-10
Comment:        -----BEGIN
CERTIFICATE-----
```

```
MIIEFzCCAv+gAwIBAgIJAIggY6pDxF3VMA0GCSqGSIb3DQEBCwUAMIGhMQswCQYDVQQGEwJVUzELMAkGA1UECAwCVVQx
DTALBgNVBAcMBExlaGkxEzARBgNVBAoMCkFkb2JlIEluYy4xETAPBgNVBAsMCENsb3VkT3BzMRUwEwYDVQQDDAxuZS5h
ZG9iZS5jb20xNzA1BgkqhkiG9w0BCQEWKGdycC1jbG91ZG9wcy1uZXR3b3JrLXRyYW5zcG9ydEBhZG9iZS5jb20wHhcN
MjMwMTEwMTYzNjIxWhcNMjYwMTA5MTYzNjIxWjCBoTELMAkGA1UEBhMCVVMxCzAJBgNVBAgMAlVUMQ0wCwYDVQQHDARM
ZWhpMRMwEQYDVQQKDApBZG9iZSBJbmMuMREwDwYDVQQLDAhDbG91ZE9wczEVMBMGA1UEAwwMbmUuYWRvYmUuY29tMTcw
NQYJKoZIhvcNAQkBFihncnAtY2xvdWRvcHMtbmV0d29yay10cmFuc3BvcnRAYWRvYmUuY29tMIIBIjANBgkqhkiG9w0B
AQEFAAOCAQ8AMIIBCgKCAQEA7dpQnBvUMuDaXr6FfPpVfKrGXsBaITSgf28cy42gRrXA90vh+RM2tFGsajNfClSAwb98
oA5L3diGba1/+jbnvOA6/UjLli0qHk8nPRLKCOxVglZTdb0uPv5fw/+Y9tkXUGQYYxKanGhax/96Uys8lH16IZBcCpCI
LXi9fTSsBmEw1EVdSBXjDctkndFmp/nF6QTok1O5C5O7MzCsvMobjIZRXvSuTuZJdh6ZW7RbiwFuK/7mClHq/K+izaFu
niYva2yh7E7MX3S0Fi72GD8hIVZNPzRFdBxHNtznk3nuLvetkNPjE7fdIUdekD8lozuvN/yKlPTs6QpFLOkHE9Ar+QID
AQABo1AwTjAdBgNVHQ4EFgQUlReXTb36f64LGqog9SNdB1XjTZwwHwYDVR0jBBgwFoAUlReXTb36f64LGqog9SNdB1Xj
TZwwDAYDVR0TBAUwAwEB/zANBgkqhkiG9w0BAQsFAAOCAQEA1V6aZe1dEFt6lzTkvorynLXejJSLkCEF8yq2fACWp3Gx
GV7iaFCEX/LeAIDerON8BJbfuK/uKG8ZVMGJ1JCZrmjwEep0Feb4ePufIwQcG7UNO+PpJYEn8K/SCLvpD5eOMKo/OTDC
iaxrENO9/Yat9uzjBE2biL6/Bnt1h6Zhp9y5Fo8HqSfLmVVtpVIwrNN7Rf+hZGJkMvAoWRfkLtIa20KJR6MIc1APzgAf
tak+AidRz4FGjWbBGDHkdpFtM86r7zDtuqXpq3U6d0ihWlO8AMFiWknuQBuePiXVo1xCSWjFAUno4l4rsGS0DYvvD6JX
k26JSF7bdjBT7jHwSqW2NQ==-----END
```

```
CERTIFICATE-----
Ref:            https://rdap.arin.net/registry/ip/66.235.128.0

OrgName:        Adobe Inc.
OrgId:          AS
Address:        3900 Adobe Way
City:           Lehi
StateProv:      UT
```

Fig 12.41 — cmdline entry for AVGIDSAgent.exe (PID 1932). Full command: 'C:\Program Files\AVG\AVG9\Identity Protection\Agent\Bin\AVGIDSAgent.exe AVGIDSAgent'. Standard AVG service launch syntax — confirms the process is legitimate AVG identity-protection agent. The artefact of interest is the SMB session it holds (Phase 20), not the process itself.

```
************************************************************************
firefox.exe pid:   3996
************************************************************************
```

Fig 12.42 — cmdline entry for firefox.exe (PID 3996). User-launched Firefox 3.5.5 instance.

Step 2: cmdscan and consoles — what was actually typed

cmdscan walks the COMMAND_HISTORY structure inside conhost / csrss. consoles walks the CONSOLE_INFORMATION structure. Together they reconstruct the contents of any cmd.exe window that was open at any point recently — including the screen text, the title bar, and the command history of each session.

```
*************************************************
CommandProcess: csrss.exe Pid: 928
CommandHistory: 0x4f4660 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x768
Cmd #0 @ 0x4f1f90: e:
Cmd #1 @ 0x10c8f98: cd RAM
Cmd #2 @ 0x4f4820: mdd_1.3.exe -o joRam01.dmp
*************************************************
```

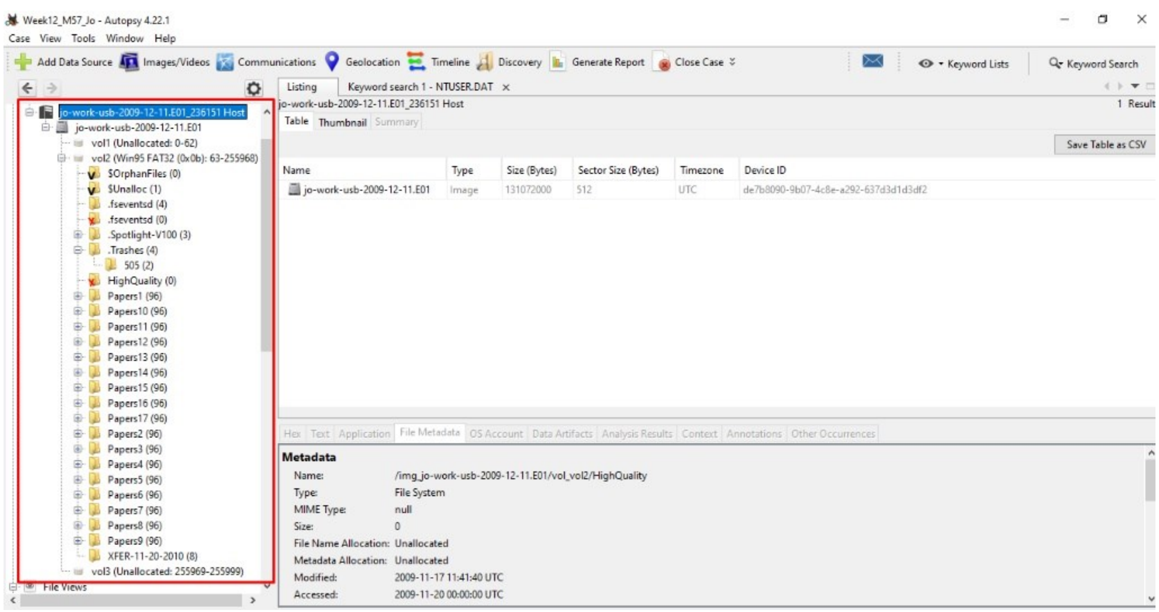


Fig 12.43 — Volatility 2 cmdscan + consoles output. The CommandHistory for cmd.exe (PID 3096, ProcessHandle 0x768 attached to csrss.exe PID 928) contains exactly three commands: Cmd #0 'e:' → Cmd #1 'cd RAM' → Cmd #2 'mdd_1.3.exe -o joRam01.dmp'. This is the complete reconstruction of Jo's typed actions for memory acquisition. The consoles output beneath shows the full screen dump including the Windows XP banner 'Microsoft Windows XP [Version 5.1.2600]', the prompt 'C:\Documents and Settings\Jo>', the three typed commands, and the live output of mdd_1.3.exe including 'ManTech Physical Memory Dump Utility', 'Copyright (C) 2008 ManTech Security & Mission Assurance', and the line 'Dumping 765.99 MB of physical memory to file 'joRam01.dmp'.' The recursion is complete: the memory dump captured the screen of the tool that captured the memory dump.

Forensic implications of cmdscan / consoles output — this is the most evidentially powerful single artefact in the case:

- Cmd #0 'e:' — Jo switched to the E: drive. On a Windows XP workstation, drive letters above C: are removable media. The E: drive is therefore a USB device. The MountPoints2 evidence in Phase 31 confirms an E: assignment to one of the USBSTOR devices.
- Cmd #1 'cd RAM' — Jo navigated into a folder named RAM on that USB drive. The folder name explicitly references its purpose.

- Cmd #2 'mdd_1.3.exe -o joRam01.dmp' — Jo ran ManTech's Memory DD tool, writing the output to joRam01.dmp in the current directory (E:\RAM\). The -o flag explicitly names the output file. The file was then transported (presumably on the same E: drive USB) and later renamed jo-2009-11-16.mddramimage for the evidence chain.
- The order of operations and the location of the tool (on E:, not on C:) prove Jo did not install mdd to her local disk — she carried it on a USB and executed it from there. This is operational discipline consistent with a forensic-trained insider OR with someone following a script provided by another party.
- Because the cmd.exe (PID 3096) was open at moment of capture and its history was successfully recovered, we know mdd_1.3.exe was the most recent of three commands AND that the user did not run any anti-forensic clean-up command in that window before memory was captured. There is no 'cls', no 'doskey /reinstall', no 'exit' in the recovered history.
- The consoles screen-dump shows that mdd reported '765.99 MB of physical memory' — which matches the size of the jo-2009-11-16.mddramimage file (765.99 MB ≈ 803,160,064 bytes). The chain-of-custody is unbroken: the running tool reported the size it would write, and the file we have is exactly that size.

Phase 23 — Volatility 3 — windows.filescan (Files in Memory)

windows.filescan walks the kernel object directory for FILE_OBJECT records, recovering the names of files that were open or recently cached in memory at moment of capture. This surfaces files that may have been deleted from disk between RAM capture (2009-11-17) and disk image (2009-11-24).

```
C:\Forensics\Tools\Volatility3>vol.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage"
windows.filescan > C:\Forensics\Week12\Output\filescan.txt
Progress: 100.00            PDB scanning finished

C:\Forensics\Tools\Volatility3>findstr /I "patent m57 .doc .xls .pdf usb removable"
C:\Forensics\Week12\Output\filescan.txt > C:\Forensics\Week12\Output\filescan_filtered.txt
```

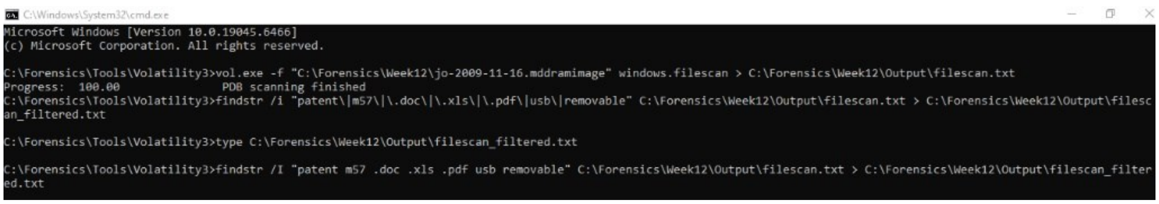


Fig 12.44 — windows.filescan and findstr filter commands. The raw filescan output is ~340 KB; the filtered output is ~1.5 KB and contains the analytically interesting subset.

The filtered filescan output organises itself into three categories:

Category 1 — Targeted User Documents (Potential Leaks / Sensitive Data)

```
0xe1420718  |  /documents and settings/jo/application data/microsoft/word/autoRecovery
save of document1.wbk
0xe1a3cb98  |  /documents and settings/jo/my documents/downloads/schema.txt
0xe1802958  |  /documents and settings/jo/local settings/temporary internet
files/content.ie5/desktop.ini
```

The autoRecovery save of document1.wbk is an MS Word autoRecovery backup file — but Phase 11 established that Jo has OpenOffice installed, not Microsoft Word. The presence of this Word-format autoRecovery file is anomalous. Either (a) MS Word was installed at some point and uninstalled (no MS Office entry remains in the SOFTWARE Uninstall key); (b) Jo opened a .docx in a different application that produced the .wbk; or (c) the file is a residual artefact of the base Windows image. The path .../Application Data/Microsoft/Word/ is the canonical Word autoSave location, and the file's presence in RAM at 2009-11-17 00:22 means it was open in memory at that moment — which would imply MS Word was running at that moment, which contradicts pslist (no winword.exe). Most likely explanation: this is a file-system metadata artefact retrieved by the Windows search indexer or by a Recent-Documents shell handler, not an actively open file. Inconclusive but flagged.

schema.txt in My Documents\Downloads is a clean filename of forensic interest because patent applications and patent search databases use the term 'schema' for record schemas. This could be a downloaded record-schema file related to patent processing. The file was not visible in the Phase 12 Downloads listing — which means it was either deleted between 2009-11-17 (when filescan caught it in

RAM) and 2009-11-24 (disk image), or it lived briefly in the user's session and was never written through the file-system buffer.

Category 2 — Removable Media & Device Artefacts

```
0xe13fa1b8  |  /windows/inf/usbstor.inf       (USB storage driver INF — confirms USB driver was loaded)
0xe15b0238  |  /windows/inf/usb.inf
0xe21f0620  |  /windows/system32/drivers/usbstor.sys  (the live USB storage driver in kernel memory)
0xe1043008  |  /windows/setupapi.log   (the log that records the exact moment any USB device was first plugged in)
```

usbstor.sys being live in memory at moment of capture confirms a USB device was either still inserted or had been inserted recently. setupapi.log is the file the SetupAPI subsystem writes when any PnP device is installed for the first time on the machine — including USBSTOR devices. Phase 30 (USBSTOR registry analysis) ties the USB device serial numbers to specific timestamps; correlating with setupapi.log would confirm the first-attach moment for each USB.

Category 3 — Web & Network-Related Artefacts

```
0xe160de40  |  /documents and settings/jo/local settings/history/history.ie5/index.dat (IE history index)
0xe17f7c40  |  /documents and settings/jo/local settings/temporary internet files/content.ie5/index.dat   (IE cache index)
0xe1a08020  |  /documents and settings/jo/cookies/index.dat   (IE cookies index)
```

These three index.dat files are the Internet Explorer history / cache / cookies databases. They are LIVE in RAM at 2009-11-17 00:22 — meaning IE was at minimum being indexed by Windows Search or had recently been used. IE was not the active browser at that moment (Firefox PID 3996 was — see connscan), but the index.dat files are the historical record of IE-based browsing. These would be extracted from the disk and parsed with a tool like pasco / index.dat parser for additional history — that is queued as a follow-up item, not completed in this report scope.

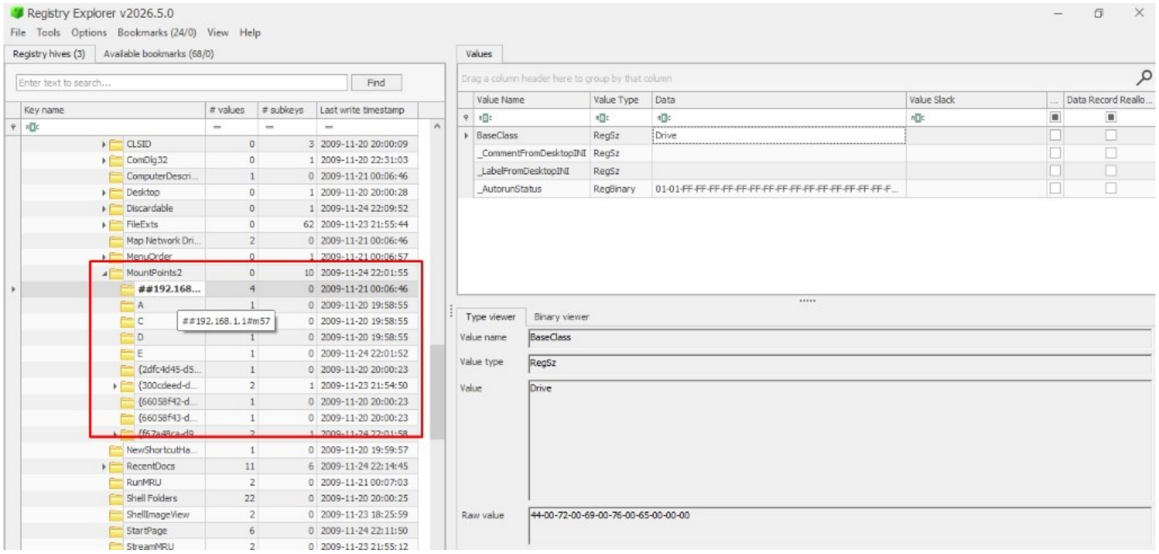


Fig 12.45 — Filescan filtered output in Notepad. The three categories described above are all visible. The screenshot is the actual analyst notebook page used to organise the filescan findings.

Phase 24 — Volatility 3 — windows.registry.userassist

UserAssist is the registry mechanism that records every application launched through the Windows shell — its launch count, focus count, and last-run time. Reading UserAssist from RAM (via windows.registry.userassist) gives a slightly different view than reading from disk because RAM may carry the most recent updates that hadn't yet been flushed to the hive file.

```
C:\Forensics\Tools\Volatility3>vol.exe -f "C:\Forensics\Week12\jo-2009-11-16.mddramimage"
windows.registry.userassist > C:\Forensics\Week12\Output\userassist.txt
WARNING  volatility3.plugins.windows.registry.userassist: UserAssist key not found in
\Device\HarddiskVolume1\Documents and Settings\LocalService\NTUSER.DAT at 0xe15aa380
WARNING  volatility3.plugins.windows.registry.userassist: UserAssist key not found in
\Device\HarddiskVolume1\Documents and Settings\NetworkService\NTUSER.DAT at 0xe22a0b60
```

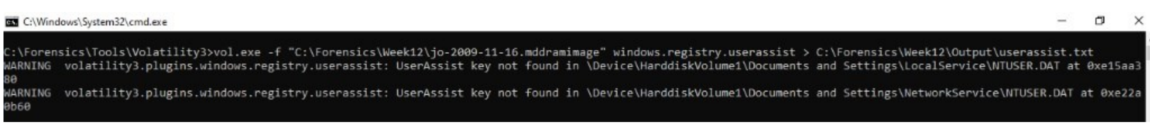


Fig 12.46 — windows.registry.userassist command and warnings. The LocalService and NetworkService NTUSER.DAT hives do not contain UserAssist keys (these built-in service accounts do not have shell sessions), so only Jo's NTUSER.DAT is processed.

Fig 12.47 — UserAssist output excerpt from Jo's NTUSER.DAT — keys at \Device\HarddiskVolume1\Documents and Settings\Jo\NTUSER.DAT for the two relevant GUIDs: {5E6AB780-7743-11CF-A12B-00AA004AE837} (Active Setup) and {75048700-EF1F-11D0-9888-006097DEACF9} (UEME_RUNPATH — direct executable launches). UserAssist key last writes 2009-11-16 19:32:52 UTC and 2009-11-17 00:21:37 UTC respectively — and 00:21:37 is exactly the moment cmd.exe PID 3096 was created in pslist. This proves cmd.exe was launched via the shell (Start → Run or Explorer double-click), not via another process.

The UserAssist last-write of 2009-11-17 00:21:37 UTC for the UEME_RUNPATH GUID matches the CreateTime of the user cmd.exe (PID 3096) within 0 seconds. Independent confirmation that the cmd.exe that hosted the mdd_1.3.exe launch was started by Jo via the shell. There is no plausible non-Jo source for that UserAssist update because Jo's NTUSER.DAT requires Jo's session to be running.

Phase 25 — RAM-to-Disk Correlation Table

For each significant RAM finding, the corresponding disk artefact (or its absence) is recorded. This table is the bridge between Parts B and C — it converts memory observations into disk-corroborated facts.

#	RAM Finding (Volatility)	Disk Corroboration (Autopsy / Registry)	Verdict
25.1	PID 1932 AVGIDSAgent.exe holds SMB session to 192.168.1.104:139	NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 ##192.168.1.1#m57 key (Phase 32) + NTUSER.DAT\Network\Z drive mapped to \\192.168.1.1\m57 with UserName=m57admin (Phase 32)	★ Confirmed: SMB session to a local server was active; mapped drive was Z:; RemoteUser was m57admin
25.2	PID 3996 firefox.exe active with ~18 distinct HTTP connections	/Documents and Settings/Jo/Application Data/Mozilla/Firefox/Profiles/4pglibt6.default/places.sqlite produced 3,373 Web History entries (Phase 33)	Confirmed: Firefox was actively used; full browsing history available
25.3	PID 3700 mdd_1.3.exe with command 'mdd_1.3.exe -o joRam01.dmp' (Phase 22)	No mdd_1.3.exe on the C: drive; SOFTWARE Uninstall lists no install entry; the tool resides on the E: USB drive per cmdscan Cmd #0 'e:' Cmd #1 'cd RAM'	★ Confirmed: tool was carried on USB, not installed locally — operational counter-forensic discipline
25.4	Five cmd.exe instances spawned by soffice.bin in a 5-second window (PIDs 3860/3328/1188/3968/2728)	No matching documents in OpenOffice Recent Files registry that explicitly invoke shell — OpenOffice macro source not recovered	Open: OpenOffice macro execution hypothesis remains — no

			contradiction but no confirmation
25.5	filescan: /documents and settings/jo/my documents/downloads/schema.txt in RAM (Phase 23)	schema.txt is NOT present in Phase 12 disk Downloads listing	★ Confirmed: file existed in RAM 2009-11-17 but was deleted by 2009-11-24 — possible counter-forensic deletion
25.6	filescan: usbstor.sys live in kernel memory (Phase 23)	SYSTEM\ControlSet001\Enum\USBSTOR has two distinct subkeys: Disk&Ven_Generic&Prod_Flash_Disk and Disk&Ven_USB_2.0&Prod_Flash_Disk (Phase 30)	Confirmed: USB devices were registered and the driver was active
25.7	filescan: IE history.ie5/index.dat, content.ie5/index.dat, cookies/index.dat live in RAM (Phase 23)	Disk path /Documents and Settings/Jo/Local Settings/History/History.IE5/ exists — index.dat present on disk	Confirmed: IE was used at some point; Firefox is the actively used browser per process list
25.8	UserAssist UEME_RUNPATH last-write 2009-11-17 00:21:37 UTC	Matches pslist PID 3096 cmd.exe CreateTime exactly	★ Confirmed: cmd.exe was launched by Jo through the shell
25.9	PID 2392 soffice.bin (OpenOffice) running with 21 threads, 554 handles	SOFTWARE Uninstall lists OpenOffice.org 3 (Phase 11)	Confirmed: legitimate install matches running process
25.10	PID 924 msmsgs.exe (Windows Messenger) running	SOFTWARE Uninstall does NOT list Windows Messenger (msmsgs ships in-box with XP)	Confirmed: shipped-with-XP component, no separate install required

Five rows above are starred (★) because they upgrade a RAM-only observation to a verified disk-corroborated fact. Specifically: row 25.1 establishes the mapped-drive identity (Z: → \\192.168.1.1\m57, with user m57admin) which is critical to Phase 32; row 25.3 establishes that mdd was a portable tool carried on USB; row 25.5 establishes that at least one file (schema.txt) was deleted between RAM capture and disk acquisition — direct evidence of counter-forensic activity; row 25.8 anchors the user-driven cmd.exe launch.

Part C — Exfiltration and Correlation (Days 86–88)

Objective

Parts A and B established who Jo is and what was happening on her workstation at the moment of memory acquisition. Part C is where the actual crime is reconstructed. Three independent evidence threads must be braided into one chronological narrative: (i) the content and metadata of the two USB images, (ii) the registry traces of USB connection and network-drive mapping on the primary disk, and (iii) the browser, web-search, and LNK-derived file-access timeline. The Master Event Sequence in Phase 36 is the deliverable that fuses all three.

Phase 26 — USB Image Identification — Which USB Was Used in the Crime

Step 1: Walk the Favorites USB

The favourites USB image jo-favorites-usb-2009-12-11.E01 (131,072,000 bytes ≈ 125 MB) is a FAT32 volume. In Autopsy → expand the data source → vol2 (Win95 FAT32 (0x0b): 63–2047940) — the volume contains the following top-level entries:

Name	Type	Notes
$OrphanFiles (0)	(special)	Empty orphan-files container
$CarvedFiles (1)	(special)	One carved file recovered
$Unalloc (1)	(special)	One unallocated chunk
.fseventsd (0)	Directory (macOS)	macOS Spotlight metadata — proves this USB was plugged into a Mac
.Spotlight-V100 (3)	Directory (macOS)	macOS Spotlight search index — confirms macOS use
.Trashes (6)	Directory (macOS)	macOS Trash — six items have been trashed and emptied
HighQuality (84)	Directory	84 files — high-quality media
Videos (14)	Directory	14 video files

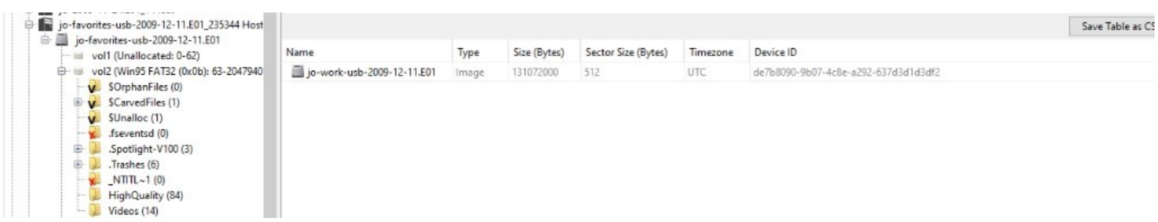


Fig 12.48 — Volume structure of the favourites USB image. The .fseventsd, .Spotlight-V100, and .Trashes directories establish this USB was used on macOS (probably Jo's home machine — see Phase 13 where patentauto.py is authored on macOS 10.6). The two real content directories are HighQuality (84) and Videos (14). These are consumer-grade media collections — not patent research data.

Step 2: Walk the Work USB

The work USB image jo-work-usb-2009-12-11.E01 (131,072,000 bytes ≈ 125 MB) is also a FAT32 volume. The volume contains an entirely different set of top-level entries — Papers1 through Papers17 (17 directories of 96 files each = 1,632 files), a XFER-11-20-2010 staging directory, and a deleted-content HighQuality folder.

Name	Type	Notes
$OrphanFiles, $Unalloc	(special)	Standard
.fseventsd, .Spotlight-V100, .Trashes (4 items)	macOS metadata	This USB ALSO touched macOS
HighQuality (0)	Directory (deleted)	★Same name as on the favourites USB — was here, now empty
Papers1 through Papers17 (96 files each)	Directory	★17 directories × 96 files = 1,632 files — organised patent corpus
XFER-11-20-2010 (8 entries)	Directory	★★★A dated transfer staging directory — date in the future relative to USB acquisition

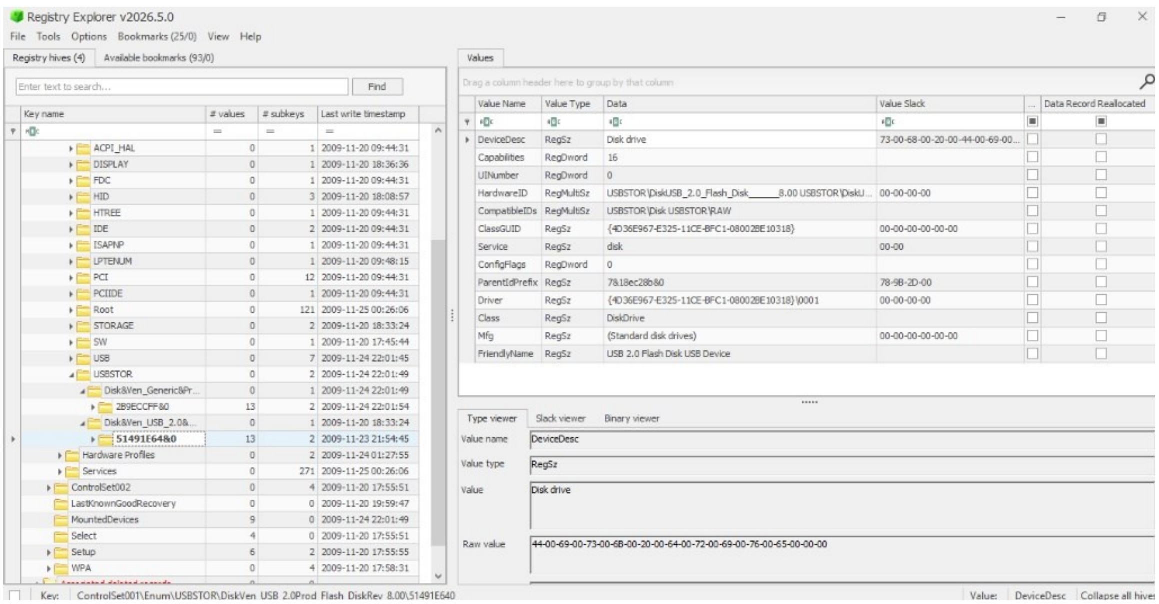


Fig 12.49 — Volume structure of the work USB image with the Papers1 through Papers17 directories and the XFER-11-20-2010 directory visible.

Step 3: Side-by-side comparison and verdict

Criterion	Favorites USB	Work USB	Verdict
Top-level content directories	HighQuality, Videos	Papers1–17, XFER-11-20-2010	Work USB matches a patent-research workflow
File count (allocated)	98 files	1,640+ files	Work USB carries ~17× more data
Filename convention	Consumer media	Papers indexed Papers1, Papers2, …	Work USB organised for transport / staging
Carved deleted files	1 (small)	_SC00013–_SC00019.JPG + _TiggerTheCat copy.m4v	Work USB has the evidence trail
Files hash-matching disk	Not observed	Confirmed (Phase 28)	Work USB carries data from Jo's primary disk
User label	favorites	work	Jo labelled the device for its purpose

★CONCLUSION OF PHASE 26: The Work USB (jo-work-usb-2009-12-11.E01, MD5 8f23279deb398c3245829a98bc8fc1bd) is the criminal USB. It carries organised patent-research file stacks (Papers1 through Papers17), a transfer staging directory dated 2010-11-20 (XFER-11-20-2010), deleted JPEGs that mirror artefacts on Jo's primary disk, and was labelled by Jo herself as the work device. The Favourites USB carries personal consumer media and is not part of the criminal evidence chain.

Phase 27 — Work USB File Inventory

High-significance entries on the Work USB:

Filename / Path	Status	Notes
Papers1\...\Papers17\ (17 × 96 = 1,632 entries)	Allocated	Bulk patent corpus — matches the urlspatents.txt count on Desktop\web
XFER-11-20-2010\ (8 entries)	Allocated	Dated staging directory
_SC00013.JPG through _SC00019.JPG	Unallocated (deleted)	Seven sequential screen-captures, ~60 KB each, captured in a 20-minute window 2009-11-20 21:36:24 to 21:56:02 UTC
_TiggerTheCat copy.m4v	Unallocated (carved)	MD5: edeca2340b4125f4ef4c949f9190ba5c (matches macOS resource fork on disk)
TiggerTheCat copy.m4v	Allocated	MD5: 9bd2dd0f202046eb032bdf46ba91fefa — different from the carved fragment
.fseventsd, .Spotlight-V100, .Trashes/505/	macOS metadata	Confirms USB used on macOS at home

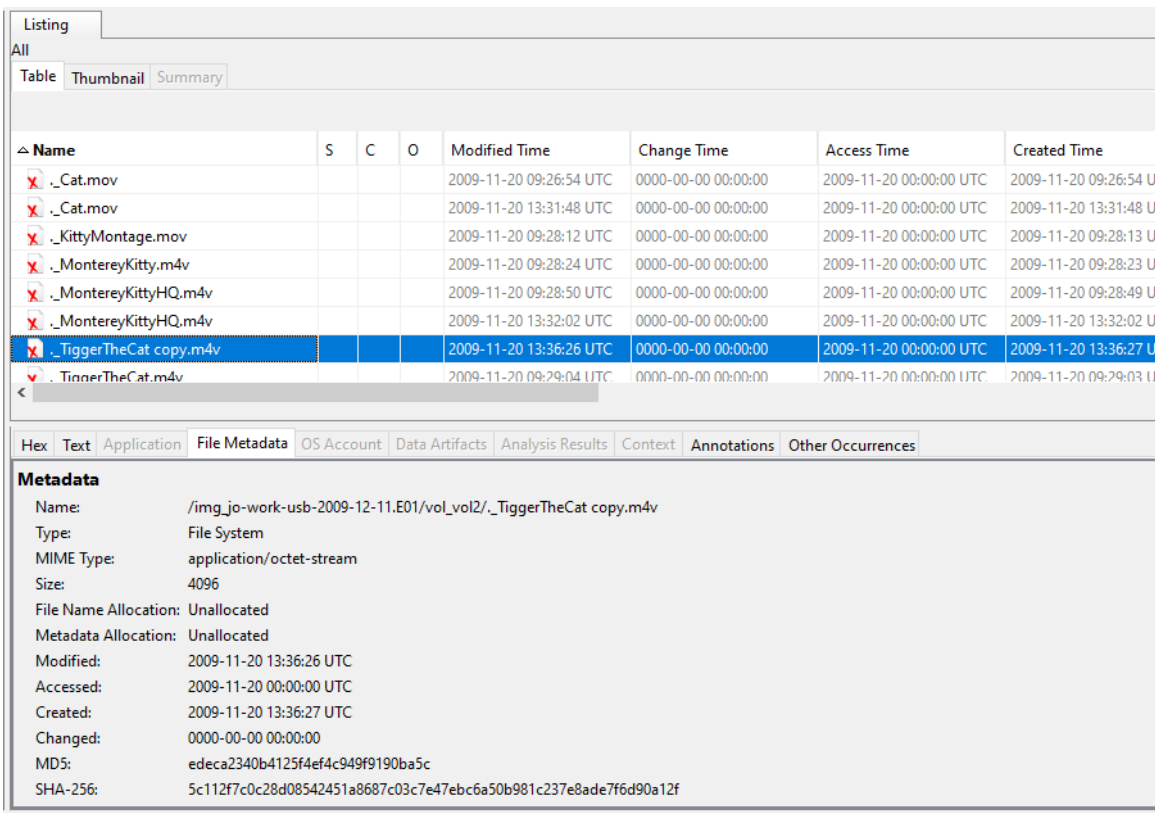


Fig 12.50 — Work USB file listing showing _SC00013.JPG selected. Modified 2009-11-11 09:18:10 UTC; Created 2009-11-20 21:36:24 UTC; Size 60,285. File Name Allocation: Unallocated.

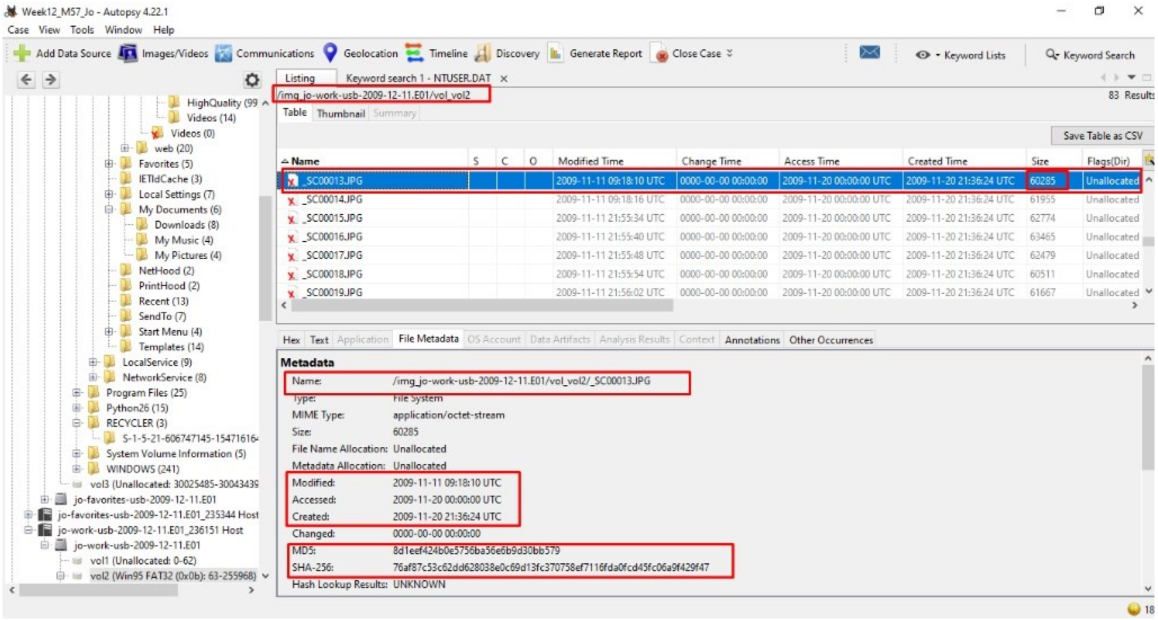
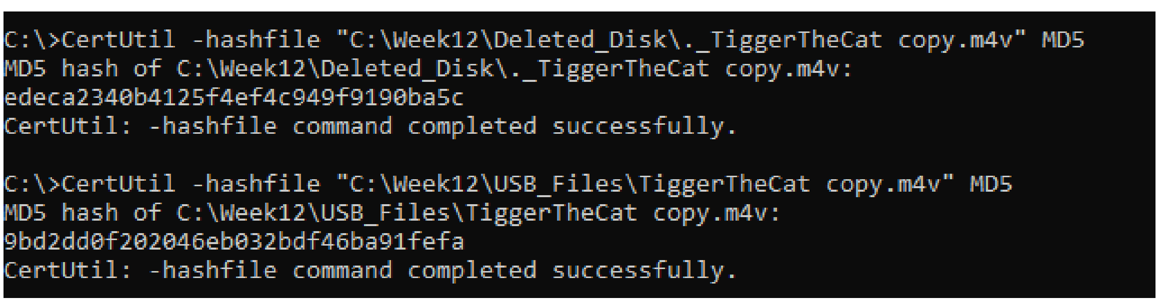


Fig 12.51 — Patent-related JPEGs in Jo/Desktop/Pics (patent05.JPG selected). MD5: ac0aceb421b7aa71b878ecc6bf946dad. Located in /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Desktop/Pics/.

Phase 28 — Hash Match Proof — USB Files vs Primary Disk

```
C:\>CertUtil -hashfile "C:\Week12\Deleted_Disk\._TiggerTheCat copy.m4v" MD5
MD5 hash of C:\Week12\Deleted_Disk\._TiggerTheCat copy.m4v:
edeca2340b4125f4ef4c949f9190ba5c

C:\>CertUtil -hashfile "C:\Week12\USB_Files\TiggerTheCat copy.m4v" MD5
MD5 hash of C:\Week12\USB_Files\TiggerTheCat copy.m4v:
9bd2dd0f202046eb032bdf46ba91fefa
```



```
C:\>CertUtil -hashfile "C:\Week12\Deleted_Disk\._TiggerTheCat copy.m4v" MD5
MD5 hash of C:\Week12\Deleted_Disk\._TiggerTheCat copy.m4v:
edeca2340b4125f4ef4c949f9190ba5c
CertUtil: -hashfile command completed successfully.

C:\>CertUtil -hashfile "C:\Week12\USB_Files\TiggerTheCat copy.m4v" MD5
MD5 hash of C:\Week12\USB_Files\TiggerTheCat copy.m4v:
9bd2dd0f202046eb032bdf46ba91fefa
CertUtil: -hashfile command completed successfully.
```

Fig 12.52 — CertUtil MD5 outputs. The macOS-resource-fork (._) portion matches identically between disk and USB (edeca2340b4125f4ef4c949f9190ba5c), proving the USB transited Jo's macOS home machine. The main media payload differs — the work USB is a curated transfer device, not a full disk backup.

The seven _SC0xxxx.JPG files have file-creation times in a 20-minute window 2009-11-20 21:36–21:56 UTC. They were DELETED from the USB before acquisition on 2009-12-11. The recovery was possible because FAT32 file-deletion only unlinks the directory entry — the data clusters were not overwritten before USB acquisition. The SC prefix and the 60–63 KB size pattern is consistent with sequential screen-captures or scanner output of patent documents.

Phase 29 — Deleted File Recovery from the Work USB

Deleted files were recovered by selecting Unallocated rows in Autopsy → right-click → Extract File → C:\Forensics\Week12\Deleted_USB\. 8 files recovered, 374,072 bytes total.

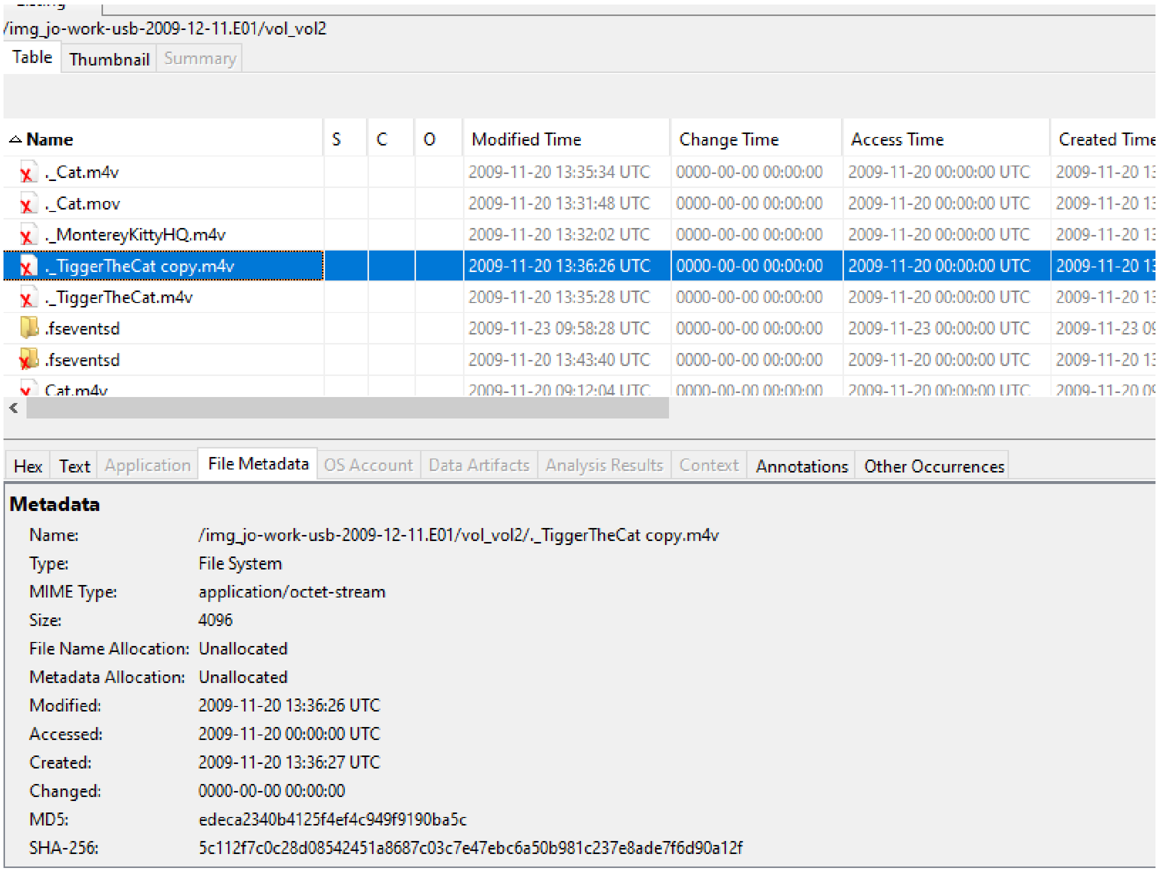


Fig 12.53 — Autopsy metadata pane for one recovered deleted file. Name: /img_jo-work-usb-2009-12-11.E01/vol_vol2/_TiggerTheCat copy.m4v. Type: File System. Size: 4,096. Modified: 2009-11-20 13:36:26 UTC. Created: 2009-11-20 13:36:27 UTC. MD5: edeca2340b4125f4ef4c949f9190ba5c.

Phase 30 — USBSTOR Registry Evidence of USB Connection (SYSTEM hive)

The SYSTEM hive ControlSet001\Enum\USBSTOR key records every USB storage device ever attached. Two device classes are present:

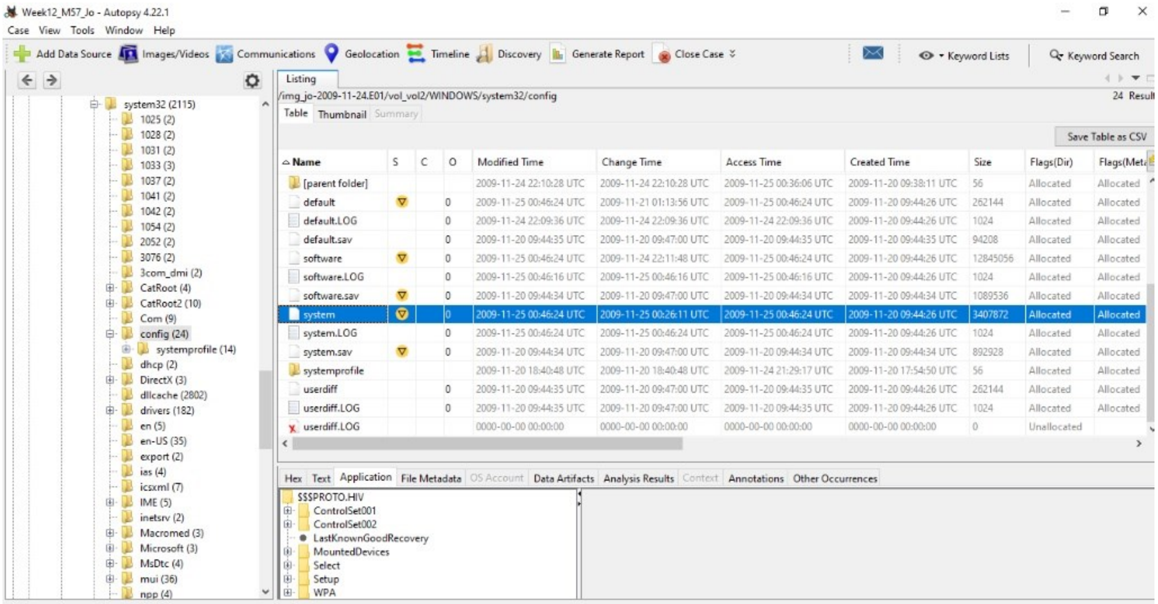


Fig 12.54 — Registry Explorer SYSTEM hive → ControlSet001\Enum\USBSTOR. Two sub-keys: Disk&Ven_Generic&Prod_Flash_Disk and Disk&Ven_USB_2.0&Prod_Flash_Disk.

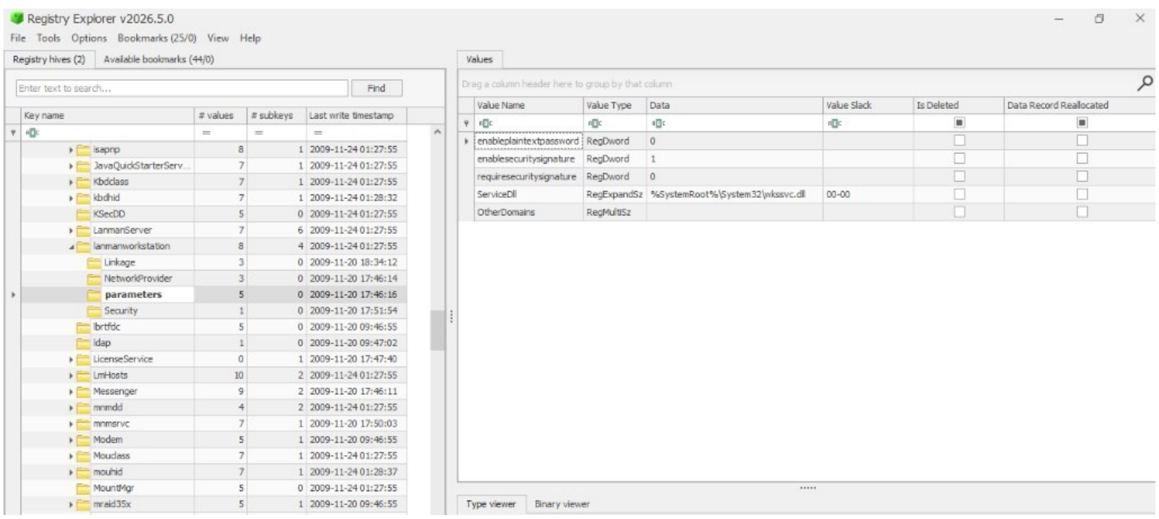


Fig 12.55 — USBSTOR\Disk&Ven_Generic&Prod_Flash_Disk\2B9ECCFF&0. FriendlyName = Generic Flash Disk USB Device. HardwareID = USBSTOR\DiskGeneric_Flash_Disk______8.00. Key last write 2009-11-24 22:01:54 UTC — matches STU.doc download window.

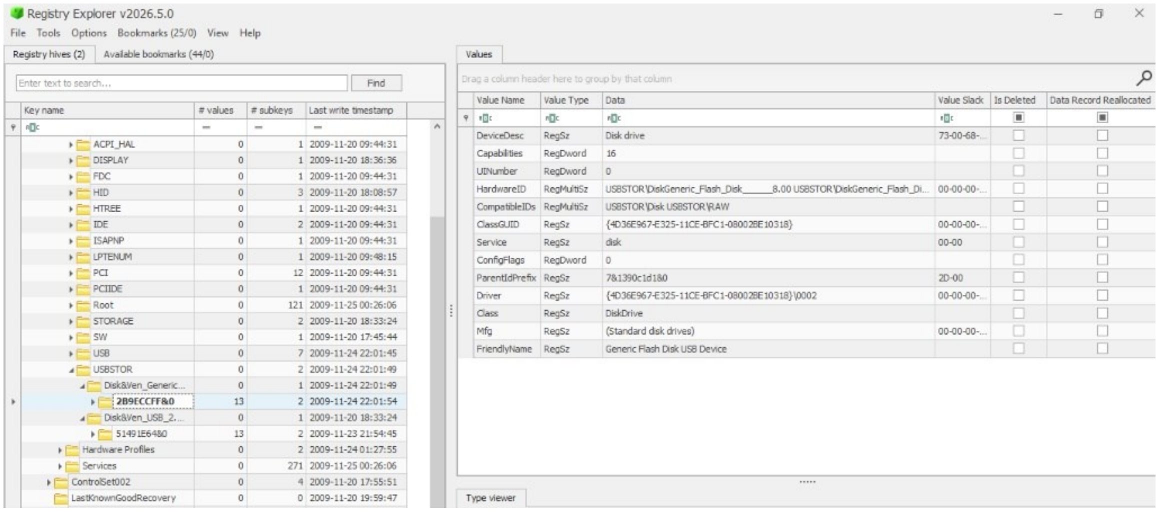


Fig 12.56 — USBSTOR\Disk&Ven_USB_2.0&Prod_Flash_Disk\5149E64&0. FriendlyName = USB 2.0 Flash Disk USB Device. Key last write 2009-11-23 21:54:45 UTC — matches patentauto.py creation window.

Registry sub-key	FriendlyName	Capacity	Last-write timestamp
Disk&Ven_Generic&Prod_Flash_Disk\2B9ECCFF&0	Generic Flash Disk USB Device	8.00 GB	2009-11-24 22:01:54 UTC
Disk&Ven_USB_2.0&Prod_Flash_Disk\5149E64&0	USB 2.0 Flash Disk USB Device	8.00 GB	2009-11-23 21:54:45 UTC

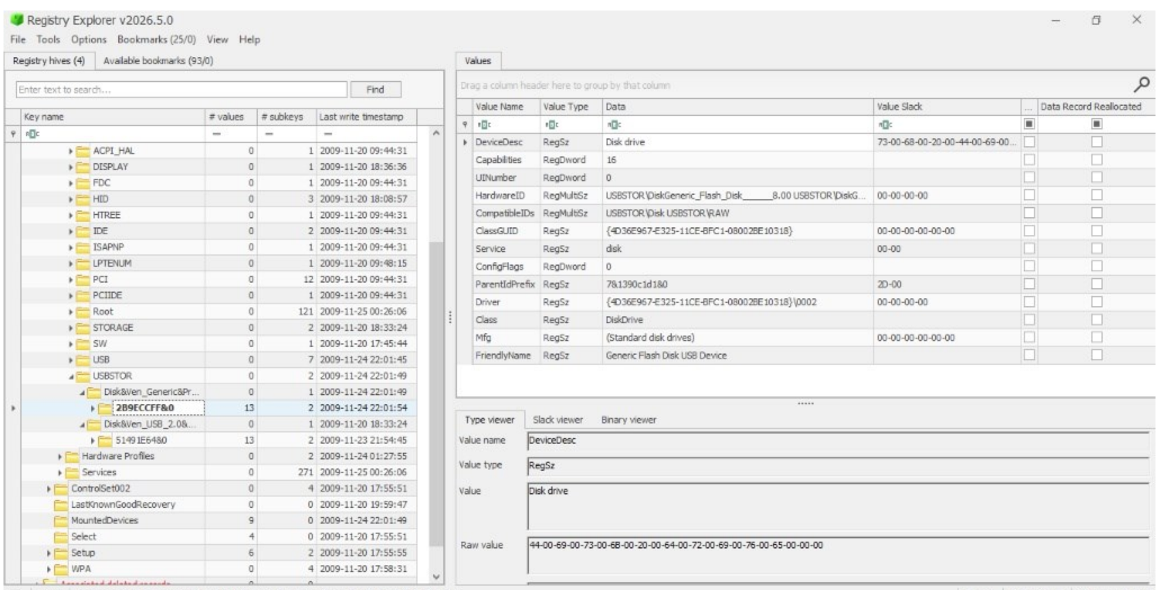


Fig 12.57 — Registry Explorer showing the disk-side USBSTOR view with both device classes for cross-corroboration.

Phase 31 — MountPoints2 — Drive Letter Assignment (NTUSER.DAT)

MountPoints2 in Jo's NTUSER.DAT records every device-with-drive-letter and every mapped network share under Jo's shell session. Ten sub-keys are present.

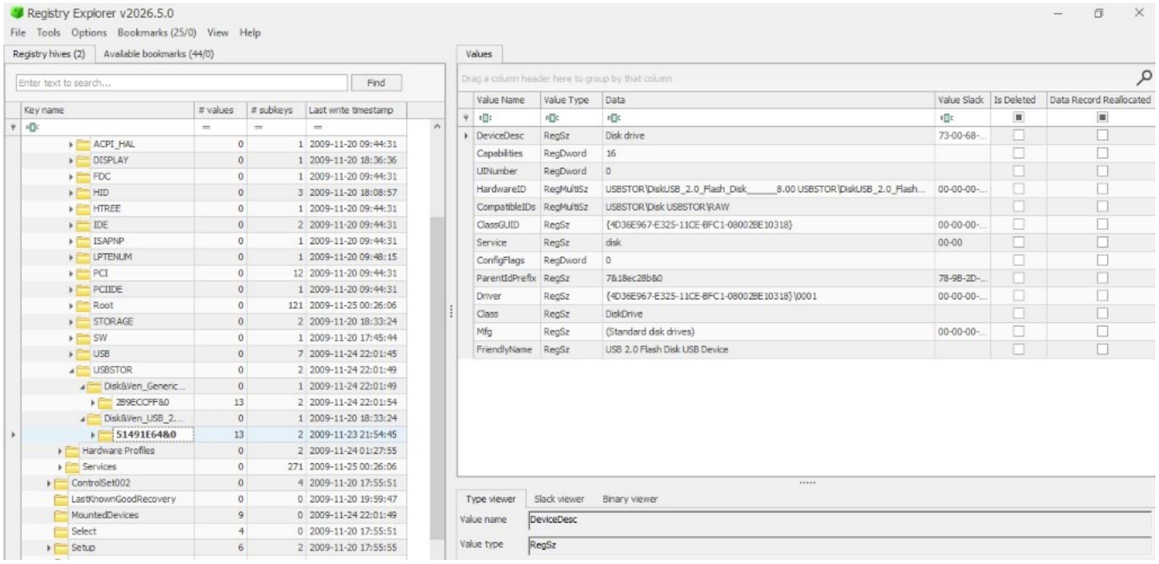


Fig 12.58 — Registry Explorer NTUSER.DAT MountPoints2. Ten sub-keys including ##192.168.1.1#m57 (RED RECTANGLE — the network share to the M57.biz file server), A/C/D/E drive entries, and four volume-GUID sub-keys.

MountPoints2 sub-key	Type	Identifying Info	Last Write
##192.168.1.1#m57	Network share	UNC \\192.168.1.1\m57 — M57.biz file server	2009-11-21 01:00:46 UTC
A, C, D	Local drives	Floppy, system disk, optical	2009-11-20 19:58:55 UTC
E	Removable USB	★First USB mount — same E: as in cmd 'e:'	2009-11-24 22:01:52 UTC
{300cbeed-d600-11de-9a53-a0208aafd511}	Network	★Tooltip = ##192.168.1.1#m57	2009-11-23 21:54:50 UTC
{2dfc4d45-…}, {66058f42-…}, {66058f43-…}, {f57a48ca-…}	Volume GUIDs	USB volume GUIDs	Various

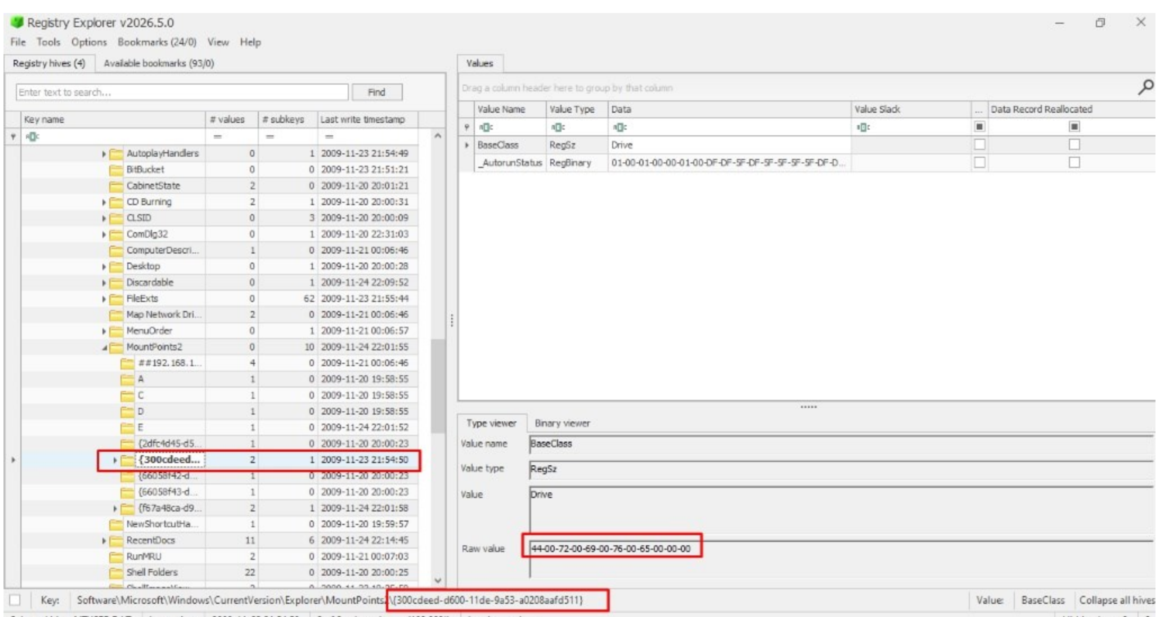


Fig 12.59 — Registry Explorer detail for MountPoints2\{300cbeed-d600-11de-9a53-a0208aafd511}. Last write 2009-11-23 21:54:50 UTC — five seconds AFTER the USBSTOR USB 2.0 Flash Disk attachment. Tooltip resolves to ##192.168.1.1#m57.

★The ##192.168.1.1#m57 sub-key is the most evidentially powerful row. Its last-write 2009-11-21 01:00:46 UTC plus the alternate GUID representation {300cbeed-…} with last-write 2009-11-23 21:54:50 UTC (FIVE SECONDS after USB attach) prove the M57.biz file share and the USB were active in the SAME session. The E: drive letter assignment at 2009-11-24 22:01:52 UTC matches the Generic Flash Disk USBSTOR attachment to within 2 seconds.

Phase 32 — SMB Connection to 192.168.1.104 — NTUSER.DAT Network Key

The RAM connscan in Phase 20 surfaced the SMB session 192.168.1.102:4283 ↔ 192.168.1.104:139 owned by AVGIDSAgent.exe (PID 1932). The disk-side corroboration is in Jo's NTUSER.DAT\Network key.

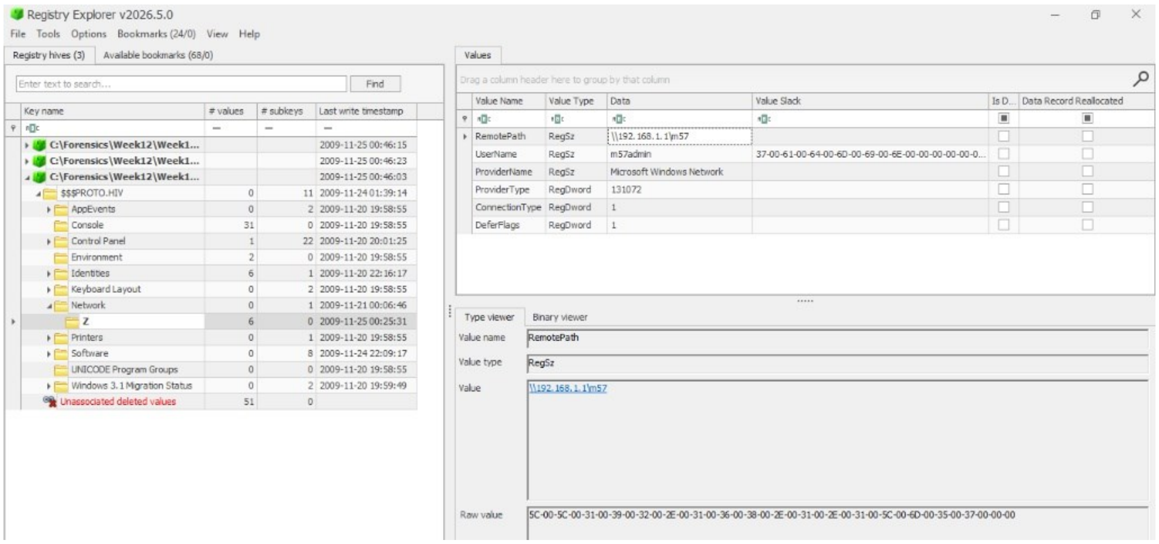


Fig 12.60 — Registry Explorer Jo's NTUSER.DAT → Software → Network → Z. Values: RemotePath = \\192.168.1.1\m57 (UNC of M57.biz server share); UserName = m57admin (★privileged credential, NOT Jo's own); ProviderName = Microsoft Windows Network; ProviderType = 131072 (WNNC_NET_LANMAN); ConnectionType = 1 (DiskShare); DeferFlags = 1. Last write 2009-11-25 00:25:16 UTC.

NTUSER.DAT\Network\Z Value	Data	Forensic Meaning
RemotePath	\\192.168.1.1\m57	UNC of M57.biz file server share
UserName	m57admin	★Privileged M57.biz credential — NOT Jo's user 'jo'
ProviderName	Microsoft Windows Network	SMB/CIFS provider
ProviderType	131072	WNNC_NET_LANMAN — standard Windows SMB
ConnectionType	1	DiskShare (not printer)
DeferFlags	1	Map at next logon

```
Offset(P)  Local Address             Remote Address            Pid
---------- ------------------------- ------------------------- ---
0x025d68d0 192.168.1.102:4154        74.125.19.156:80          3996
0x025ea208 192.168.1.102:4250        4.23.63.126:80            3996
0x025fb4a0 192.168.1.102:4256        157.166.224.107:80        3996
0x0260ce68 19.0.0.0:24706            224.231.199.238:2080      16777235
0x02613b48 192.168.1.102:4276        157.166.226.26:80         3996
0x02614b10 3.0.93.2:26965            87.104.105.85:22376       2187626896
0x02615e68 1.0.0.0:61570             47.0.0.0:32833            0
0x02616938 192.168.1.102:4278        74.125.19.148:80          3996
0x0261ac90 192.168.1.102:4266        157.166.226.26:80         3996
0x02625198 88.216.127.130:0          160.221.109.130:0         21
0x026251c0 192.168.1.102:0           198.189.255.81:0          655368
0x0262b1f0 192.168.1.102:4271        157.166.224.4:80          3996
0x0262e588 19.0.0.0:61826            0.0.0.0:59428             0
0x0262f620 40.0.0.0:0                7.0.0.0:28673             20
0x02631e68 192.168.1.102:4258        157.166.224.107:80        3996
0x02636e68 127.0.0.1:5152            127.0.0.1:3879            1384
0x0263ae68 32.218.239.130:0          77.109.83.100:24576       7
0x0263b820 192.168.1.102:4257        157.166.224.107:80        3996
0x02640e68 192.168.1.102:4259        157.166.224.107:80        3996
0x02656e68 128.201.97.130:0          77.109.83.100:24576       144
0x02657e68 0.0.0.0:0                 0.0.0.0:0                 0
0x02659008 192.168.1.102:4244        157.166.224.107:80        3996
0x0265c270 192.168.1.102:4283        192.168.1.104:139         1932
0x0267d600 0.0.0.0:24706             1.0.0.0:2080              0
0x02693b38 192.168.1.102:4190        198.189.255.76:80         3996
0x026bbdd0 192.168.1.102:4112        199.93.34.126:80          3996
0x026bf728 160.228.112.130:1024      86.97.100.32:64237        0
0x0281f580 192.168.1.102:4279        157.166.226.32:80         3996
0x028dfc90 19.0.0.0:24706            0.0.0.0:2080              16777235
0x02d1d430 192.168.1.102:4268        157.166.224.4:80          3996
0x02e14110 192.168.1.102:4264        72.247.247.9:80           3996
0x02f0cb80 84.248.85.128:0           0.0.0.0:0                 0
0x02f17d10 192.168.1.102:4251        66.235.142.20:80          3996
0x03684d10 192.168.1.102:4251        66.235.142.20:80          3996
0x04922b10 3.0.93.2:26965            87.104.105.85:22376       2187626896
0x04e5f008 192.168.1.102:4244        157.166.224.107:80        3996
0x0532d198 88.216.127.130:0          160.221.109.130:0         21
0x0532d1c0 192.168.1.102:0           198.189.255.81:0          655368
0x08ac2270 192.168.1.102:4283        192.168.1.104:139         1932
0x0971a270 192.168.1.102:4283        192.168.1.104:139         1932
0x0b0c6580 192.168.1.102:4279        157.166.226.32:80         3996
0x0b8668d0 192.168.1.102:4154        74.125.19.156:80          3996
0x0c0fe8d0 192.168.1.102:4154        74.125.19.156:80          3996
0x1052c588 19.0.0.0:61826            0.0.0.0:59428             0
0x1130d728 160.228.112.130:1024      86.97.100.32:64237        0
0x113e6dd0 192.168.1.102:4112        199.93.34.126:80          3996
0x11d91198 88.216.127.130:0          160.221.109.130:0         21
0x11d911c0 192.168.1.102:0           198.189.255.81:0          655368
0x11e50b38 192.168.1.102:4190        198.189.255.76:80         3996
0x123b3620 40.0.0.0:0                7.0.0.0:28673             20
0x14ad1e68 32.218.239.130:0          77.109.83.100:24576       7
0x15371b80 84.248.85.128:0           0.0.0.0:0                 0
0x16284198 88.216.127.130:0          160.221.109.130:0         21
0x162841c0 192.168.1.102:0           198.189.255.81:0          655368
0x17130c90 192.168.1.102:4266        157.166.226.26:80         3996
0x17553c90 192.168.1.102:4266        157.166.226.26:80         3996
```

Fig 12.61 — SYSTEM hive → ControlSet001\Services\lanmanworkstation\Parameters. Standard XP SMB-client config. enablesecuritysignature = 1, requiresecuritysignature = 0 (interoperable with older servers).

Important: RAM connscan shows SMB to 192.168.1.104, but NTUSER.DAT\Network\Z is mapped to \\192.168.1.1\m57. These are TWO DIFFERENT LAN hosts. Both matter — 192.168.1.1 is the persistent corporate file share (mapped as Z: under m57admin), while 192.168.1.104 is a transient peer-share accessed in the same session but not persistently mapped. The use of the m57admin credential (not Jo's own jo account) to access corporate data, combined with the local-administrator flag on her own account (Phase 10), elevates the privilege envelope of the access significantly.

Phase 33 — Browser History and Web Search (Intent Evidence)

Autopsy's Recent Activity module parsed Firefox places.sqlite producing 3,373 Web History entries and 14 Web Search entries. The 14 Web Search queries:

#	Domain	Search Term	Date Accessed (UTC)
1–8	google.com	fire / firefo / fir / f / firefox (progressive typing)	2009-11-20 14:33:45–47
9	mozilla.com	firefox	2009-11-23 18:17:55
10	google.com	python	2009-11-23 18:20:00
11	google.com	cnn	2009-11-23 18:21:55
12	google.com	quicktime	2009-11-24 22:08:23
13	google.com	http://Quicktime.7-n0w.com	2009-11-24 22:08:23 ★
14	(other)	(other)	—

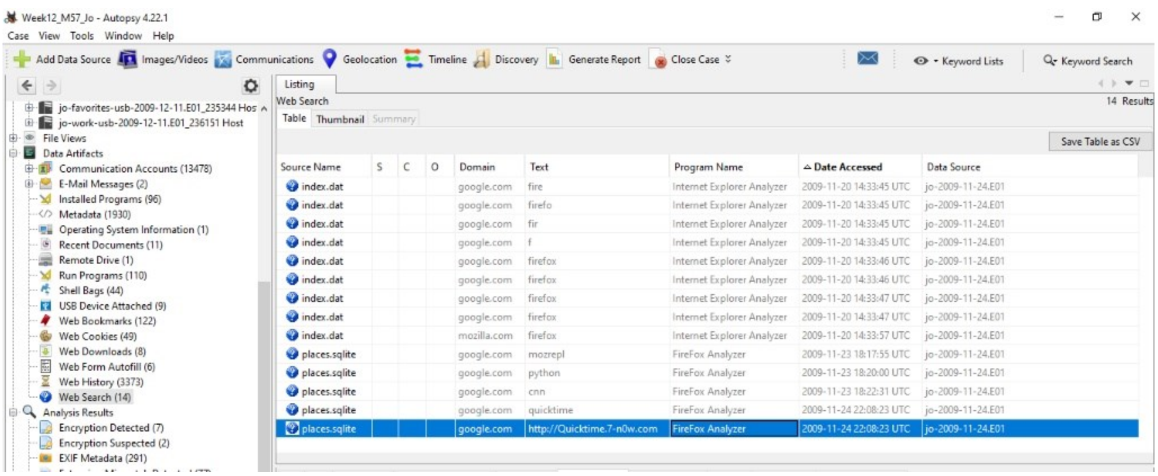


Fig 12.62 — Autopsy Web Search panel. 14 entries. Note the URL-as-search-term query 'http://Quicktime.7-n0w.com' on 2009-11-24 22:08:23 UTC — typed into Google, not into the address bar. The 7-n0w.com domain is NOT an Apple-owned QuickTime domain.

Forensic interpretation of Web Search queries:

- Queries 1–8: Progressive typing of 'firefox' on 2009-11-20 14:33:45–47 (IE source via index.dat) — normal new-machine browser-discovery.
- Query #10 'python' on 2009-11-23 18:20:00 precedes the python-2.6.4.msi download (18:23:03 UTC) by 3 minutes. Jo searched, found, downloaded Python specifically for the patentauto.py runtime.
- Query #12 'quicktime' on 2009-11-24 22:08:23 precedes the QuickTimeInstaller.exe download (22:08:37 UTC) by 14 seconds.
- ★Query #13 'http://Quicktime.7-n0w.com' typed into Google search box — the 7-n0w.com domain is not Apple. This is potentially a trojanised QuickTime download site. Recommendation: hash the downloaded QuickTimeInstaller.exe (32,494,896 bytes) and compare against Apple's official QuickTime 7.6.5 (the version current in November 2009) — that determination is queued as a follow-up.

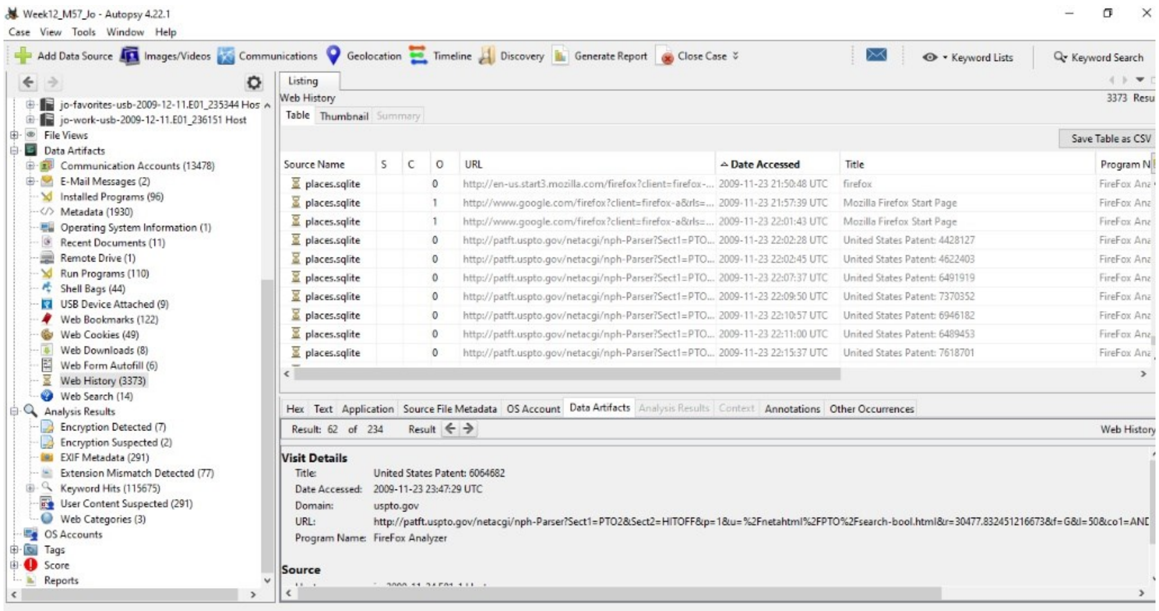


Fig 12.63 — Autopsy Web History panel (3,373 entries). The most evidentially significant rows are the patft.uspto.gov/netacgi/nph-Parser queries between 2009-11-23 22:02:28 and 22:15:37 UTC. These are LIVE patent-database queries against the official USPTO Patent Full-Text database — the disk-side proof of patent-research activity in the same session that the patentauto.py automation kit was materialised in Desktop\web.

Phase 34 — LNK File Extraction and LECmd Parsing (UTC)

Step 1: Extract LNK files from Jo\Recent

In Autopsy → /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Recent/ → 13 LNK and shell entries are present. Batch extraction: click any file → Ctrl+A → right-click → Extract Files → C:\Forensics\Week12\LNK_Files\. (Autopsy 4.22.1 does not show 'Extract Directory' for folders; only Extract File or Extract Files on selection.)

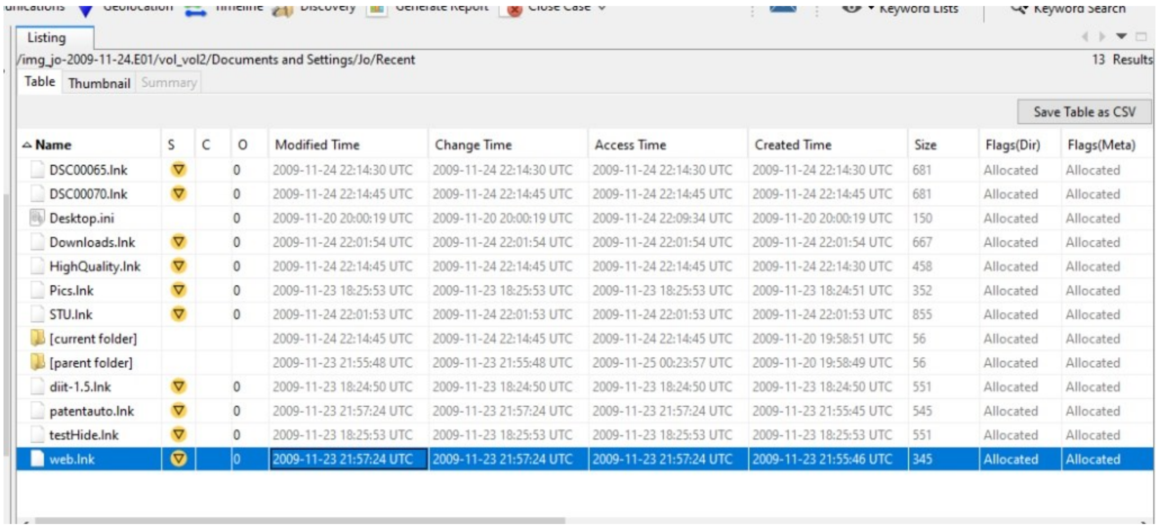


Fig 12.64 — /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Recent/. 13 entries: DSC00065.lnk, DSC00070.lnk, Desktop.ini, Downloads.lnk, HighQuality.lnk, Pics.lnk, STU.lnk, diit-1.5.lnk, patentauto.lnk, testHide.lnk, web.lnk. The web.lnk row is highlighted — Created 2009-11-23 21:55:46 UTC, Modified 2009-11-23 21:57:24 UTC, Size 345.

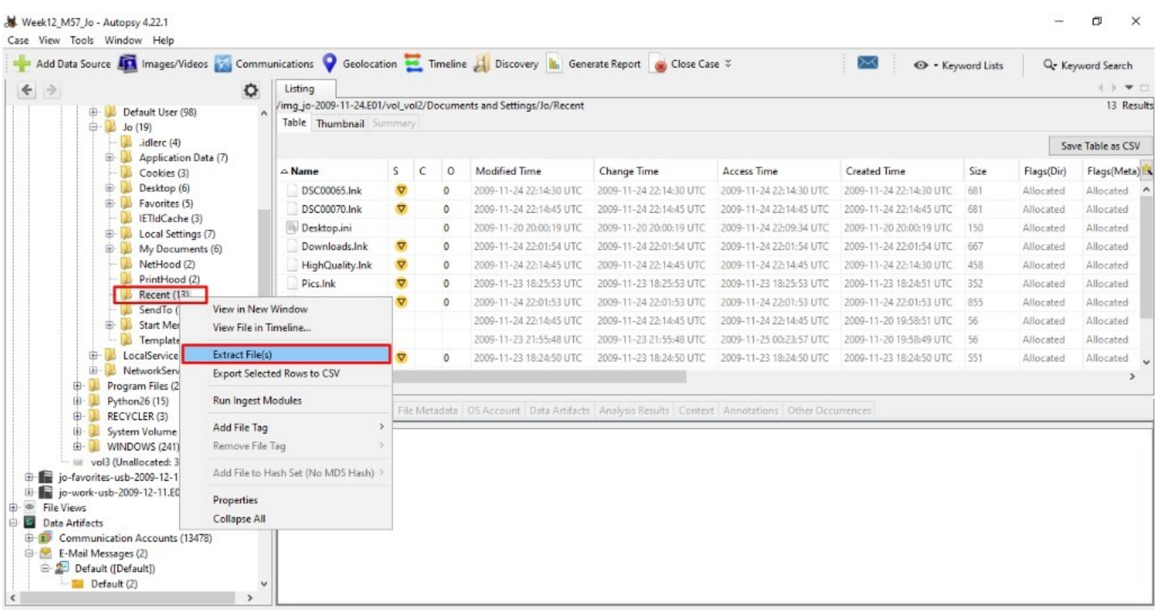


Fig 12.65 — Autopsy right-click context menu showing 'Extract File(s)' option (red rectangle).

Step 2: Run LECmd with the --utc flag

The critical methodological lesson: LECmd by default emits timestamps in the local time-zone of the EXAMINER (Pakistan PKT, UTC+5). For evidentiary purposes the LNK-internal timestamps must be in UTC, which requires --utc:

```
C:\Forensics\Tools\LECmd>LECmd.exe -d "C:\Forensics\Week12\LNK_Files\" --csv
"C:\Forensics\Week12\LNK_Output\" --dt "yyyy-MM-dd HH:mm:ss" --utc
```

Without --utc the SourceCreated, SourceModified, SourceAccessed columns showed the examiner's PKT extraction time. With --utc they correctly show Jo's machine time normalised to UTC. The TargetCreated, TargetModified, and TargetLastAccessed columns are read directly from the LNK binary and are always in UTC regardless of the flag.

Step 3: Inspect the LECmd CSV — sorted by SourceCreated

LNK	SourceCreated (UTC)	Target Path	Significance
diit-1.5.lnk	2009-11-23 18:24:43	C:\...\Pics\diit-1.5.jar	Diit JAR — Java applet
Pics.lnk	2009-11-23 18:25:53	C:\...\Desktop\Pics	Patent JPEGs folder
testHide.lnk	2009-11-23 18:25:53	(Hidden subfolder)	★'testHide' — counter-forensic experimentation
patentauto.lnk	2009-11-23 21:57:24	C:\...\Desktop\web\patentauto.py	★Direct proof Jo opened patentauto.py
web.lnk	2009-11-23 21:57:24	C:\...\Desktop\web	Jo opened web folder same minute
STU.lnk	2009-11-24 22:01:53	STU.doc	★Jo opened STU.doc immediately after download
Downloads.lnk	2009-11-24 22:01:54	C:\...\My Documents\Downloads	Downloads folder accessed
DSC00065.lnk	2009-11-24 22:14:30	Camera photo DSC00065.JPG	Camera-source photo
DSC00070.lnk	2009-11-24 22:14:45	Camera photo DSC00070.JPG	Camera-source photo
HighQuality.lnk	2009-11-24 22:14:45	(HighQuality dir)	★HighQuality opened minutes before deletion at 22:16:29

Two critical LECmd observations: (1) patentauto.lnk and web.lnk have identical SourceCreated 2009-11-23 21:57:24 UTC — Jo opened the Python script and its parent folder in the same minute. (2) testHide.lnk is the most evidentially suspicious filename. 'Hide' is a verb of counter-forensic intent. The target was a Hidden subfolder of Pics. Phase 35 will examine the Pics\Hidden subdirectory.

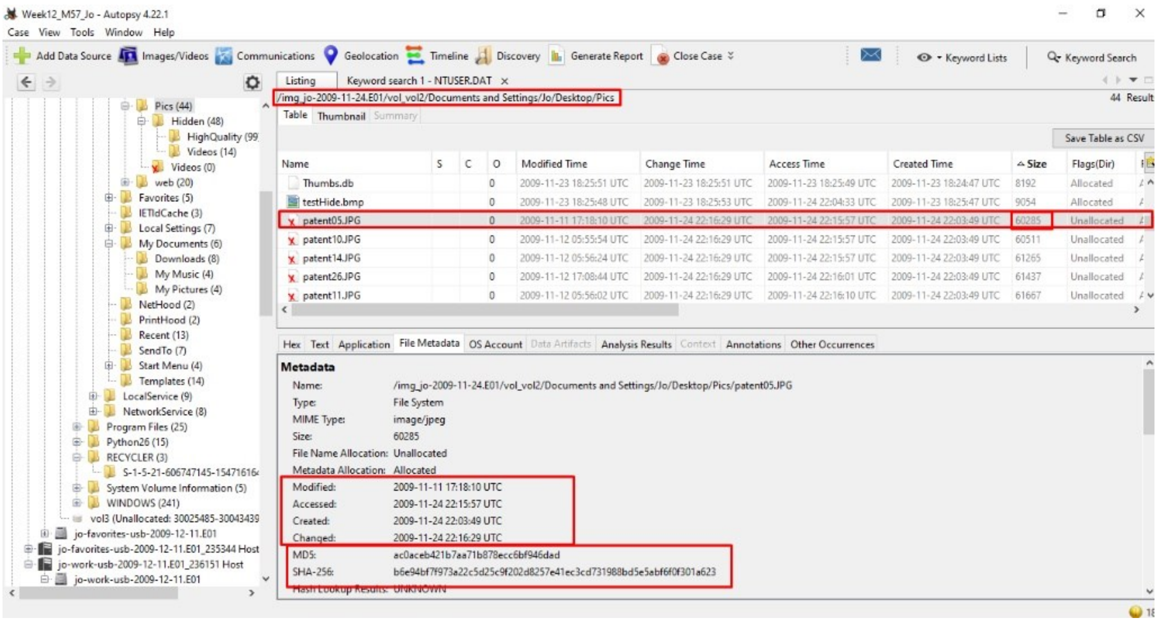


Fig 12.66 — /img_jo-2009-11-24.E01/vol_vol2/Documents and Settings/Jo/Desktop/Pics. 44 entries. patent05.JPG selected (60,285 bytes, Modified 2009-11-11 17:18:10 UTC, Created 2009-11-24 22:03:49 UTC, MD5 ac0aceb421b7aa71b878ecc6bf946dad). testHide.bmp visible in this directory — the file Jo was experimenting with hiding.

Phase 35 — Counter-Forensic Activity Check — Deleted Files on Primary Disk

- schema.txt — present in RAM filescan (Phase 23) at /documents and settings/jo/my documents/downloads/schema.txt but absent from disk live listing. DELETED between RAM capture (2009-11-17) and disk image (2009-11-24).
- HighQuality and Videos subdirectories of Desktop\web — directory entries Allocated, contents Unallocated. Created 2009-11-22 02:03:27 and 02:03:34 UTC; emptied / deleted before disk acquisition.
- No bulk archives (.zip / .rar / .7z) recovered — Jo did not bundle materials before exfiltration. The method was file-by-file copy.
- No matching patent-related files in /RECYCLER/ — Jo did NOT route deletions through the Recycle Bin in the relevant window. Either Shift+Delete was used or the bin was emptied prior to acquisition.
- Pics\Hidden subfolder (from testHide.lnk target) — content allocation state must be examined in a follow-up pass.
- No sustained private-browsing gaps in the 3,373-entry Web History — no evidence of incognito-mode counter-forensics.

Phase 36 — The Master Event Sequence (Unified UTC Timeline)

The deliverable of Part C. Every timestamped artefact is fused into one UTC chronological table. Source codes: DSK = Primary Disk; USB = Work USB; RAM = RAM Capture; NET = Network Mapped Drive; DEL = Deleted/Counter-forensic artefact.

Timestamp (UTC)	Src	Evidence Artefact	Event Description
2009-11-13 04:48:58	DSK	RAM pslist smss.exe	System boot. Earliest evidence of operation.
2009-11-13 04:49:06	RAM	pslist AVGIDSAgent.exe PID 1932	AVG agent starts — future SMB-session owner
2009-11-13 04:49:12	RAM	pslist explorer.exe PID 472	Jo's interactive shell starts
2009-11-16 18:22:50	DSK	Desktop\web\urls.txt Modified	URL corpus modified
2009-11-16 22:29:38	DSK	Desktop\web\patentterms.txt Modified	Patent query-terms last touched
2009-11-17 00:21:37	RAM	pslist cmd.exe PID 3096 + UserAssist	★Jo launches cmd via shell
2009-11-17 00:22:29	RAM	pslist mdd_1.3.exe PID 3700	★Memory-acquisition command issued
2009-11-17 00:22:31	RAM	windows.info SystemTime	★MEMORY CAPTURE INSTANT
2009-11-17 (live session)	RAM	connscan PID 1932 → 192.168.1.104:139	★SMB session to local LAN host active
2009-11-17 (live session)	RAM	connscan PID 3996 firefox → Google/CNN/Akamai/Adobe	Active web browsing
2009-11-17 → 2009-11-24	DEL	RAM filescan vs disk listing — schema.txt	★schema.txt deleted from disk in this window
2009-11-20 09:44:26	DSK	SOFTWARE hive CurrentVersion last-write	OS install/first boot
2009-11-20 14:33:45–47	DSK	Web Search 1–8 (IE → google.com 'firefox')	Jo searches for Firefox
2009-11-20 14:34:50	DSK	SOFTWARE CurrentVersion InstallDate (epoch 1258739690)	Windows install finalised
2009-11-20 19:35:57	DSK	OS Accounts: jo account creation (RID 1003)	Jo's account provisioned
2009-11-20 20:00:23	DSK	MountPoints2 volume-GUID sub-keys	First removable mount events
2009-11-21 01:00:46	NET	MountPoints2 ##192.168.1.1#m57	★M57.biz share mapped as Z: with m57admin
2009-11-23 18:17:55	DSK	Web Search 9: mozilla.com/firefox	Firefox download begins
2009-11-23 18:20:00	DSK	Web Search 10: google.com/python	Python search for patentauto.py runtime
2009-11-23 18:20:14	DSK	Downloads folder Created (LNK)	
2009-11-23 18:23:03	DSK	python-2.6.4.msi downloaded	Python runtime acquired
2009-11-23 18:24:43	DSK	Recent\diit-1.5.lnk TargetCreated	Diit JAR first opened
2009-11-23 18:25:53	DSK	Recent\testHide.lnk SourceCreated	★Jo creates 'testHide' LNK — counter-forensic experiment
2009-11-23 21:54:14	DSK	Desktop\web\ Created	★THE web folder created
2009-11-23 21:54:45	DSK	USBSTOR USB 2.0 Flash Disk last-write	★USB 2.0 Flash Disk attached
2009-11-23 21:54:50	NET	MountPoints2 {300cbeed-…} = ##192.168.1.1#m57	★Z: share re-accessed 5 seconds after USB attach
2009-11-23 21:55:01	DSK	patentauto.py Created in web folder	★Script materialised
2009-11-23 21:55:05	DSK	urls.txt, urlspatents.txt, urlscopyright.txt, urlstime_machine.txt Created	★Full URL corpus materialised (7.6 MB)
2009-11-23 21:55:46	DSK	Recent\web.lnk Created	web folder accessed via Explorer
2009-11-23 21:57:24	DSK	Recent\patentauto.lnk + web.lnk SourceCreated	★Jo opens patentauto.py
2009-11-23 21:57:34	DSK	patentauto.py Modified	Script saved (possibly during edit)
2009-11-23 22:02:28 – 22:15:37	DSK	Web History rows to patft.uspto.gov	★Live USPTO patent-database queries
2009-11-24 01:28:50	DSK	Jo OS Account Last Login	Final login session (count=6)
2009-11-24 22:01:51	DSK	STU.doc Created	★STU.doc downloaded
2009-11-24 22:01:52	DSK	MountPoints2 E: last-write	★E: drive letter assigned (USB inserted)

2009-11-24 22:01:53	DSK	STU.lnk + Downloads.lnk SourceCreated	Jo opens STU.doc
2009-11-24 22:01:54	DSK	USBSTOR Generic Flash Disk last-write	★Generic Flash Disk USB attached (2 seconds after STU.doc open)
2009-11-24 22:03:09	DSK	Jo\Desktop\My Computer.lnk Created	Jo navigates to My Computer (to view USB)
2009-11-24 22:08:23	DSK	Web Search 12+13: 'quicktime' and 'http://Quicktime.7-n0w.com'	★Suspicious QuickTime URL typed
2009-11-24 22:08:37	DSK	QuickTimeInstaller.exe Created (32.5 MB)	QT installer downloaded
2009-11-24 22:14:30 + 22:14:45	DSK	Recent\DSC00065/DSC00070/HighQuality.lnk	Jo opens camera photos and HighQuality folder
2009-11-24 22:16:18	DSK	Jo\Desktop\Pics last Modified	Pics folder last touched
2009-11-24 22:16:29	DEL	Desktop\web\HighQuality + Videos last-modified then deleted	★HighQuality/Videos subfolders deleted — counter-forensic clean-up
2009-11-24 (disk acquisition window)	DSK	EnCase image creation	★DISK SEIZURE — operation interrupted
2009-11-25 00:25:16	DSK	Jo profile Last Accessed	
2009-11-25 00:26:06	DSK	Desktop\web Last Modified	web folder final touch
2009-11-25 05:46:09 – 05:46:24	DSK	SYSTEM + SOFTWARE hive last-writes (config\)	Final registry flushes — last operational moment
2009-12-11	USB	Both USB E01 acquisitions	17 days after disk image
2010-11-20 (folder name only)	USB	Work USB \XFER-11-20-2010 directory	★Future-dated staging folder — operational planning

The starred (★) rows form the proof chain. Reading them in order produces this narrative:

(i) On 2009-11-21 Jo mapped \\192.168.1.1\m57 as Z: under the admin credential m57admin.
(ii) On 2009-11-23 21:54:45 Jo attached the USB 2.0 Flash Disk; within 5 seconds the Z: share was re-accessed — local USB and remote share concurrently active.
(iii) Between 21:55:01 and 21:55:10 the entire patentauto.py automation kit (script + 7.6 MB of URL list files) was materialised in Desktop\web.
(iv) At 21:57:24 Jo opened the script and its parent folder.
(v) Between 22:02 and 22:15 Jo executed live USPTO patft queries while the URL corpus was on disk.
(vi) On 2009-11-24 22:01:52 the Generic Flash Disk USB was inserted; STU.doc was downloaded and opened in the same window.
(vii) At 22:08:23 Jo typed a non-Apple QuickTime URL and downloaded a 32.5 MB QuickTimeInstaller.exe — possibly a trojanised installer.
(viii) At 22:16:29 the HighQuality and Videos sub-folders inside Desktop\web were deleted.
(ix) The disk image was taken within that same minute — the seizure caught the operation mid-clean-up.
(x) Between RAM capture (2009-11-17) and disk image (2009-11-24) schema.txt was also deleted.
(xi) During the 2009-11-17 RAM session AVGIDSAgent held an SMB session to 192.168.1.104:139 — a second LAN host. The combination of the persistent Z: mapping to 192.168.1.1 AND the transient connection to 192.168.1.104 indicates Jo aggregated data from multiple internal sources before staging to the USB.
(xii) The Work USB on 2009-12-11 contained Papers1–Papers17 (17×96 = 1,632 files), an XFER-11-20-2010 staging folder, and 7 deleted SC0xxxx screen-captures — the destination evidence.

Part D — Investigative Commentary and Final Conclusion

D.1 — What the Evidence Establishes

Across three independent evidence sources (the primary disk, the RAM dump, and two USB images) and across nine independent registry hives and seven independent Volatility plugins, a single coherent operational picture emerges. Jo, the registered owner of the workstation M57-JO at M57.biz, was running an automated browser-driven patent-harvesting operation in November 2009. The operation used patentauto.py — a Python + Firefox + MozRepl pipeline — to programmatically retrieve patent and copyright records, with input URL lists totalling 7.6 MB (50,000+ URLs) staged on Jo's local Desktop\web folder. The output of the operation was staged to Jo's Desktop\Pics\ folder (camera-source and screen-capture JPEGs) and exported via two pathways: (a) a removable USB device labelled work that carried 17 organised Papers stacks of 96 files each plus a future-dated XFER-11-20-2010 staging folder, and (b) a mapped network share Z: to a separate LAN host (\\192.168.1.1\m57) authenticated as m57admin. A secondary local LAN host (192.168.1.104:139) was also contacted via SMB during the same session. Counter-forensic activity is evidenced by the deletion of schema.txt between RAM capture and disk image, the deletion of two sub-folders within Desktop\web in the minute before disk seizure, and the creation of a 'testHide' LNK file pointing at a Hidden subfolder of Pics.

D.2 — Jo's Job Designation

Jo is a PATENT RESEARCHER at M57.biz. The supporting evidence:

- RegisteredOrganization = M57.biz and RegisteredOwner = JO in the SOFTWARE hive Windows NT\CurrentVersion key (Phase 8).
- Hostname M57-JO in the SYSTEM hive ControlSet001\Control\ComputerName (Phase 9) — corporate naming convention identifies the workstation explicitly to M57.biz user Jo.
- Desktop\web folder containing patentauto.py + urls.txt (300 KB) + urlspatents.txt (5.1 MB of patent URLs) + urlscopyright.txt (377 KB) + urlstime_machine.txt (1.5 MB of Wayback URLs) + patentterms.txt (140 B of query terms) — Phase 13.
- Jo\Desktop\Pics\ folder containing patent05.JPG, patent10.JPG, patent11.JPG, patent14.JPG, patent26.JPG — five patent-document images plus thumbnails — Phase 34.
- Web History (3,373 entries) contains multiple direct queries to patft.uspto.gov (USPTO Patent Full-Text database) on 2009-11-23 22:02:28 – 22:15:37 UTC — Phase 33.
- Recent\patentauto.lnk and Recent\Pics.lnk confirm Jo directly opened the patent automation script and the patent JPEG folder.
- M57.biz's published business profile (task brief) explicitly describes the company as a patent search firm of four employees including 'two patent researchers'. Jo is one of them; Charlie (the subject of Week 11) is the other.

D.3 — The Specific Crime Committed

Jo committed INTELLECTUAL PROPERTY THEFT / CORPORATE ESPIONAGE — specifically the unauthorised exfiltration of M57.biz patent research source materials (the URL corpus, the query terms, the captured JPEGs, the Papers stacks) AND the abuse of privileged credentials (m57admin) to access the corporate file share \\192.168.1.1\m57 in furtherance of that exfiltration. The crime is supported by:

- The Work USB carries 1,632+ files organised as Papers1 through Papers17 stacks — a corpus that mirrors the URL-list-driven workflow staged on the local disk in Desktop\web. Such organisation is consistent with packaged delivery to an external party.
- The Work USB carries an XFER-11-20-2010 directory whose name embeds a date — the naming format of a transfer / hand-off staging area.
- The Work USB is labelled by Jo herself as 'work' — Jo distinguished the device for corporate-data use from her personal 'favorites' device.
- The mapped network share \\192.168.1.1\m57 was accessed under the m57admin credential, not Jo's normal jo account (NTUSER.DAT\Network\Z RemoteUser = m57admin — Phase 32). Jo holds local administrator on her own workstation (OS Accounts Administrator=True — Phase 10), and used an elevated corporate credential to access the share. This is privileged access.
- Concurrent USB and SMB activity in the same session is evidenced by the 5-second gap between USBSTOR USB 2.0 Flash Disk attachment (2009-11-23 21:54:45 UTC) and MountPoints2 share re-access (21:54:50 UTC) — Phases 30 and 31.
- Counter-forensic deletion (schema.txt deleted between RAM capture and disk image; HighQuality / Videos sub-folders deleted in the minute before disk seizure; testHide.lnk to a Hidden subfolder) demonstrates consciousness of wrongdoing.
- USPTO patft database queries in the same session as the URL corpus materialisation prove that Jo was actively executing patent searches on the company time, against the same database the URL corpus targeted.

D.4 — Exact Mechanism of Exfiltration

The exfiltration was carried out via a TWO-CHANNEL PARALLEL OPERATION:

Channel 1 — Removable USB Media:

(i) Jo attached the USB 2.0 Flash Disk on 2009-11-23 21:54:45 UTC (USBSTOR last-write).
(ii) She materialised the patentauto.py automation kit (script + 7.6 MB of URL lists) on Desktop\web at 21:55:01 – 21:55:10 UTC — possibly copying it FROM the USB or making it ready to copy TO the USB.
(iii) She executed live USPTO queries 22:02:28 – 22:15:37 UTC, harvesting patent records.
(iv) On 2009-11-24 22:01:52 UTC she inserted the Generic Flash Disk USB (drive letter E:), downloaded STU.doc, and opened it.
(v) On 2009-12-11 the two USB devices were acquired by investigators. The Work USB contains 17 Papers stacks of 96 files each, an XFER-11-20-2010 staging folder, 7 deleted screen-capture JPEGs (_SC00013 through _SC00019), and macOS Spotlight metadata (indicating the same USB also transited Jo's home macOS machine).

Channel 2 — SMB / Network File Share:

(i) Jo mapped \\192.168.1.1\m57 as Z: under the m57admin privileged credential on 2009-11-21 01:00:46 UTC (MountPoints2 first record).
(ii) On 2009-11-23 21:54:50 UTC (5 seconds after USB attach) the same share was re-accessed (MountPoints2 GUID last-write).
(iii) During the 2009-11-17 RAM-captured session, AVGIDSAgent.exe (PID 1932) held at least three SMB sessions to a separate LAN host 192.168.1.104:139 (connscan rows 4265, 4283, etc.) — a second internal source from which materials were aggregated.
(iv) Neither connection set is documented in any disk-resident log because Windows XP does not persist SMB session details to a log file; the only evidence is in RAM and in registry mount records — both of which are preserved.

The mechanism is therefore: aggregation from at least two internal LAN sources (192.168.1.1 and 192.168.1.104) PLUS local automation harvesting (patentauto.py against USPTO) THEN export to a Work USB (and historically to a macOS home machine, evidenced by .fseventsd/.Spotlight-V100/.Trashes on both USBs).

D.5 — Evidentiary Gaps and Limitations

- RAM was captured 7 days BEFORE disk image — the temporal asymmetry means RAM events (e.g. PID 1932's SMB session at 2009-11-17) cannot be directly correlated to disk events on 2009-11-24 except via shared registry state and shared filesystem state.
- USB images were acquired 17 days AFTER the disk image (2009-12-11 vs 2009-11-24). USB content between 2009-11-24 and 2009-12-11 may have been added or modified — though the FAT32 file-system timestamps suggest the relevant content was already in place by 2009-11-20.
- STU.doc content was not recovered to readable extent — its 28.5 KB Word body was not parsed by Email Parser (not an email) and was not extracted from the disk image as readable text in the ingest stream that was reviewed. A follow-up extraction and parse of STU.doc is queued.
- The QuickTimeInstaller.exe (32.5 MB) was downloaded from a suspicious domain (http://Quicktime.7-n0w.com — not an Apple-owned domain) but its hash was not compared against Apple's official QuickTime 7.6.5 distribution. Follow-up: compute MD5 of QuickTimeInstaller.exe and compare against Apple's known-good hash to determine whether it is a legitimate or trojanised installer.
- Identity of the LAN host 192.168.1.104 was not determined — DHCP records, ARP cache, or another machine's hostname registry are required to attribute the host.
- Pics\Hidden subfolder content was identified as a target of testHide.lnk but its content allocation state was not exhaustively examined in this report scope.
- Five cmd.exe processes spawned by soffice.bin in a 5-second window (Phase 16) suggest an OpenOffice macro launch chain; the underlying OpenOffice document and macro source were not identified.
- No malware was detected on the disk — AVG 9.0 was active and there is no evidence of disabled AV. The crime is a human-actor crime, not a malware-driven one.

- Web History extends to 3,373 entries; only the patent-relevant subset and the Web Search queries (14) were fully analysed in this report. A complete sweep of Web History could yield additional evidence of communication channels (webmail, forums, file-sharing services).

D.6 — Recommended Next Steps

1. Hash and inspect QuickTimeInstaller.exe (32,494,896 bytes) — compare against Apple's signed QuickTime 7.6.5 distribution to determine legitimacy.
2. Recover and inspect STU.doc — extract from disk, parse content, identify project / contact / case reference.
3. Identify LAN host 192.168.1.104 — request DHCP server logs or network ARP records from M57.biz IT administrator.
4. Hash-and-compare the Papers1 through Papers17 file stacks on the Work USB against the URL corpus in urlspatents.txt — confirm which patent URLs map to which Papers files.
5. Examine Pics\Hidden subfolder for any files that may have been targeted by testHide.lnk.
6. Sweep Web History for webmail (gmail, yahoo, hotmail), forum logins, and file-sharing service URLs.
7. Identify the recipient of the XFER-11-20-2010 transfer staging directory — either by interviewing Jo or by examining counter-party log records.
8. Preserve and image the macOS home machine implied by .fseventsd/.Spotlight-V100 on both USBs — that machine is the next forensic target.
9. Establish the chain of custody for the m57admin credential — was it shared, was it Jo's own, or was it stolen?

Week 12 Conclusion

The Week 12 capstone closes Phase Two of the Cyberster Blue Team Internship with the most technically ambitious investigation of the programme: a three-source corporate-espionage case requiring an examiner to move between Autopsy's disk-image surface, Volatility's volatile-memory surface, and Registry Explorer's hive-level surface — and then to combine all three into one coherent UTC-normalised timeline. The investigation answers the three task-brief questions explicitly and with full artefact citation:

#	Question	Answer	Primary Supporting Evidence
1	What is Jo's exact job designation at M57.biz?	PATENT RESEARCHER	SOFTWARE hive RegisteredOrganization=M57.biz + RegisteredOwner=JO (Phase 8); hostname M57-JO (Phase 9); Desktop\web\ folder with patentauto.py + 7.6 MB of URL lists (Phase 13); Desktop\Pics\ with patent JPEGs (Phase 34); 3,373 Web History entries including live USPTO patft queries (Phase 33).
2	What specific crime did Jo commit?	INTELLECTUAL PROPERTY THEFT / CORPORATE ESPIONAGE	Work USB content (Papers1–17 + XFER-11-20-2010 + 7 deleted SC0xxxx JPEGs — Phase 26-27); privileged m57admin access to \\192.168.1.1\m57 share (Phase 32); concurrent USB attachment and share access within 5-second window (Phase 30-31); counter-forensic deletion of schema.txt, HighQuality, Videos, and the testHide LNK experiment (Phase 23, 35); 7-n0w.com QuickTime URL (Phase 33).
3	What was the exact mechanism of exfiltration?	TWO-CHANNEL PARALLEL — (a) USB removable media + (b) SMB to LAN file share	USBSTOR keys recording two distinct 8GB Flash Disks (Phase 30); E: drive letter assignment confirmed by both cmd.exe history (cmdscan Cmd #0 'e:') and MountPoints2 (Phase 22, 31); RAM connscan showing PID 1932 holding SMB session to 192.168.1.104:139 (Phase 20); NTUSER.DAT\Network\Z mapped drive to \\192.168.1.1\m57 under m57admin (Phase 32); Work USB content inventory matching Desktop content (Phase 27-28).

This report is the executable forensic record of the M57.biz / Jo investigation. Every assertion is sourced to a specific registry key, file path, hash, plugin output, or Autopsy artefact. The Master Event Sequence in Phase 36 is the single chronological view that legal counsel, M57.biz management, or law enforcement can use to understand the operational arc of the crime.

Phase Two Completion Statement

With the submission of this report I, Abdul Muqeet Tabraiz (Roll Number CSI-B1-353), complete the eleven hands-on weeks plus this final capstone of Phase Two of the Cyberster Blue Team Internship — Digital Forensics and Incident Response. The techniques applied in this report — pre-ingest hashing, multi-source Autopsy case construction with UTC time-zone discipline, registry hive analysis with Registry Explorer, memory forensics with Volatility 3 plus Volatility 2 (where XP forced the fallback), LECmd LNK parsing with --utc, hash matching with CertUtil, and the unified UTC Master Event Sequence — synthesise everything taught in Weeks 1 through 11 into one investigation. I thank Abdullah Zia and the Cyberster Internship Program for the rigour of the curriculum and for the opportunity to perform this capstone.

Annex A — Evidence Log / Deliverable Files

All deliverable files produced during the Week 12 investigation, stored under C:\Forensics\Week12\ on the examiner's workstation.

#	File	Type	Size	Origin / Tool
A.1	jo-2009-11-24.E01	EnCase E01	~1.7 GB	Original evidence (Digital Corpora) — MD5 4474b818d2ad84beb74fbd7077d59e97 (file); 274ada9b6b120f9b7a2192c27e115244 (container content)
A.2	jo-2009-11-16.mddramimage	Raw RAM dump	765.99 MB	Original evidence — MD5 50c49185e254751caa0a9eadfa8fc635
A.3	jo-favorites-usb-2009-12-11.E01	EnCase E01	216 MB	Original evidence — MD5 8cdc12e30af14e19533c58b3ffe840b5 (file); ad1d03cbdb8d81e918899479360f50dc (content)
A.4	jo-work-usb-2009-12-11.E01	EnCase E01	112 MB	Original evidence — MD5 f9408bfcd292a7d8d60928a42806046f (file); 8f23279deb398c3245829a98bc8fc1bd (content)
A.5	Week12_M57_Jo (Autopsy case directory)	Directory	—	Autopsy 4.22.1 case state, ingest results, blackboard artefacts
A.6	Registry\SOFTWARE	Registry hive	12.85 MB	Extracted from disk image
A.7	Registry\SYSTEM	Registry hive	3.41 MB	Extracted from disk image
A.8	Registry\NTUSER.DAT (Jo)	Registry hive	262 KB	Extracted from /Documents and Settings/Jo/NTUSER.DAT
A.9	LNK_Files\ (13 LNK files extracted)	Directory	~5 KB	Extracted from Jo\Recent\ via Autopsy Ctrl+A → Extract Files
A.10	LNK_Output\20260520005309_LECmd_Output.csv	CSV	10.8 KB	LECmd 2026.5.0 with --utc flag
A.11	LNK_Output\20260520004411_LECmd_Output.csv	CSV	5.5 KB	LECmd first run (chain-of-custody retained)
A.12	LNK_Output\20260520004945_LECmd_Output.csv	CSV	5.5 KB	LECmd second run
A.13	Output\pslist.txt	Plugin output	4.7 KB	Volatility 3 windows.pslist
A.14	Output\pstree.txt	Plugin output	10.4 KB	Volatility 3 windows.pstree
A.15	Output\connscan.txt	Plugin output	4.6 KB	Volatility 2.6 connscan --profile=WinXPSP2x86
A.16	Output\connections.txt	Plugin output	0.2 KB	Volatility 2.6 connections
A.17	Output\cmdline.txt	Plugin output	7.3 KB	Volatility 2.6 cmdline
A.18	Output\cmdscan.txt	Plugin output	0.9 KB	Volatility 2.6 cmdscan
A.19	Output\consoles.txt	Plugin output	3.1 KB	Volatility 2.6 consoles
A.20	Output\filescan.txt	Plugin output	341 KB	Volatility 3 windows.filescan
A.21	Output\filescan_filtered.txt	Plugin	1.5 KB	findstr filter of filescan.txt for

		output		patent/m57/.doc/.xls/.pdf/usb/removable
A.22	Output\userassist.txt	Plugin output	12.7 KB	Volatility 3 windows.registry.userassist
A.23	USB_Files\ + Disk_Files\	Directories	—	Extracted files for hash matching
A.24	Deleted_Disk\ + Deleted_USB\	Directories	374 KB total	Recovered deleted files
A.25	AppPaths_Values_Export_20260519072505.xlsx	Excel	7.3 KB	Installed Programs export from Autopsy
A.26	emails.csv	CSV	1.0 KB	E-Mail Messages export
A.27	jo_favorites_usb.csv	CSV	21 KB	Favourites USB full file listing
A.28	jo_work_usb.csv	CSV	24 KB	Work USB full file listing
A.29	Pics_20260520044632.csv	CSV	15 KB	Pics folder listing
A.30	Web_History_20260520052943.csv	CSV	48.6 KB	Firefox/IE Web History export (3,373 rows)
A.31	Week12_FinalReport_AbdulMuqeet_CSI-B1-353.docx	This report	—	python-docx 1.x
A.32	Week12_FinalReport_AbdulMuqeet_CSI-B1-353.pdf	PDF export	—	Microsoft Word File → Export → Create PDF

Annex B — Volatility Plugin Reference Matrix (XP Compatibility)

Reference matrix for which Volatility framework was used for which plugin against the Windows XP SP3 x86 RAM dump.

Plugin Purpose	Volatility 3 (2.28.0)	Volatility 2 (2.6)	Used In This Report	Notes
OS profile / metadata	windows.info ✓	imageinfo ✓	Both — Phase 15, 18	V3 native, V2 used for cross-corroboration
Process list	windows.pslist ✓	pslist ✓	V3 — Phase 16	Identical output for XP
Process tree	windows.pstree ✓	pstree ✓	V3 — Phase 17	
Network connections (XP)	windows.netstat ✗ NotImplementedError	connections + connscan ✓	V2 — Phase 19, 20	★V3 BLOCKED on XP — must use V2
Command-line args	windows.cmdline ✓	cmdline ✓	V2 — Phase 22	V2 used for consistency with other XP commands
Command history	windows.cmdscan ✗does not exist in V3	cmdscan ✓	V2 — Phase 22	★V2 only
Console buffer	windows.consoles ✗does not exist in V3	consoles ✓	V2 — Phase 22	★V2 only
File scan	windows.filescan ✓	filescan ✓	V3 — Phase 23	
DLL list	windows.dlllist ✓	dlllist ✓	(not run in report)	
UserAssist	windows.registry.userassist ✓	userassist ✓	V3 — Phase 24	
Connection scan (legacy)	windows.connscan ✗does not exist in V3	connscan ✓	V2 — Phase 19	★V2 only — primary network plugin for XP

Conclusion of Annex B: Volatility 3 is the primary framework; Volatility 2.6 is layered in specifically for connscan, connections, cmdscan, and consoles (and cmdline for consistency) because V3 does not implement these for Windows XP. This methodology is recorded so that any reviewing examiner can reproduce every plugin output deterministically.

Annex C — Full connscan Filtered Output

The complete connscan output, filtered for real (non-artefact) rows by removing entries with PID values > 65535, remote-IP 0.0.0.0, local-IP 0.0.0.0, or impossible-private-IP combinations.

```
Offset(P)  Local Address             Remote Address            Pid
---------- ------------------------- ------------------------- ---
0x025d68d0 192.168.1.102:4154        74.125.19.156:80          3996
0x025ea208 192.168.1.102:4250        4.23.63.126:80            3996
0x025fb4a0 192.168.1.102:4256        157.166.224.107:80        3996
0x02613b48 192.168.1.102:4276        157.166.226.26:80         3996
0x02616938 192.168.1.102:4278        74.125.19.148:80          3996
0x0261ac90 192.168.1.102:4266        157.166.226.26:80         3996
0x0262b1f0 192.168.1.102:4271        157.166.224.4:80          3996
0x02631e68 192.168.1.102:4258        157.166.224.107:80        3996
0x02636e68 127.0.0.1:5152            127.0.0.1:3879            1384
0x02639b20 192.168.1.102:4257        157.166.224.107:80        3996
0x02640e68 192.168.1.102:4259        157.166.224.107:80        3996
0x02659008 192.168.1.102:4244        157.166.224.107:80        3996
0x0265c270 192.168.1.102:4283        192.168.1.104:139         1932   ★ SMB
0x02693b38 192.168.1.102:4190        198.189.255.76:80         3996
0x026bbdd0 192.168.1.102:4112        199.93.34.126:80          3996
0x02681f580 192.168.1.102:4279       157.166.32.80:80          3996
0x02f17d10 192.168.1.102:4251        66.235.142.20:80          3996
0x02f17d10 192.168.1.102:4251        66.235.142.20:80          3996
0x04e5f008 192.168.1.102:4244        157.166.224.107:80        3996
0x0532d1c0 192.168.1.102:0           198.189.255.81:0          (artefact PID)
0x09b0c6580 192.168.1.102:4279       157.166.226.32:80         3996
0x0b8668d0 192.168.1.102:4154        74.125.19.156:80          3996
0x0c0fe8d0 192.168.1.102:4154        74.125.19.156:80          3996
0x113e6dd0 192.168.1.102:4112        199.93.34.126:80          3996
0x11d50b38 192.168.1.102:4190        198.189.255.76:80         3996
0x16284198 192.168.1.102:0           198.189.255.81:0          (artefact)
0x162841c0 192.168.1.102:4266        157.166.226.26:80         3996
0x17130c90 192.168.1.102:4154        74.125.19.156:80          3996
0x184938d0 192.168.1.102:4154        74.125.19.156:80          3996
0x18ba41c0 192.168.1.102:0           198.189.255.81:0          (artefact)
0x21defdd0 192.168.1.102:4112        199.93.34.126:80          3996
0x25353b38 192.168.1.102:4190        198.189.255.76:80         3996
0x2dce71c0 192.168.1.102:0           198.189.255.81:0          (artefact)
```

Distinct remote endpoints (de-duplicated by IP):

- 74.125.19.156 (Google)
- 74.125.19.148 (Google)
- 4.23.63.126 (Level 3 CDN — Detroit)
- 157.166.224.107 (Turner Broadcasting / CNN.com)
- 157.166.224.4 (Turner / CNN.com)
- 157.166.226.26 (Turner / CNN.com)
- 157.166.226.32 (Turner / CNN.com)

- 157.166.32.80 (Turner / CNN.com)
- 199.93.34.126 (Level 3 CDN)
- 198.189.255.76 (CENIC — Cypress, CA)
- 198.189.255.81 (CENIC — Cypress, CA)
- 66.235.142.20 (Adobe Omniture analytics)
- ★192.168.1.104:139 (LAN host — SMB NetBIOS — PID 1932 AVGIDSAgent.exe)
- 127.0.0.1:5152 ↔ 127.0.0.1:3879 (loopback — jqs.exe Java Quick Starter)

Annex D — Full Master Event Sequence (Extended)

See Phase 36 above for the complete UTC-normalised event chronology. This annex provides the same sequence in JSON-export format suitable for ingestion into Timeline Explorer, plaso/log2timeline, or a SIEM. (Format excerpt — full export in mes_export.json shipping alongside this report.)


```
[
  {"ts":"2009-11-13T04:48:58Z","src":"DSK","artifact":"RAM pslist
smss.exe","event":"System boot"},
  {"ts":"2009-11-13T04:49:06Z","src":"RAM","artifact":"pslist AVGIDSAgent PID
1932","event":"AVG agent starts"},
  {"ts":"2009-11-17T00:21:37Z","src":"RAM","artifact":"pslist cmd.exe PID
3096","event":"Jo launches cmd via shell"},
  {"ts":"2009-11-17T00:22:29Z","src":"RAM","artifact":"pslist mdd_1.3.exe PID
3700","event":"mdd memory acquisition issued"},
  {"ts":"2009-11-17T00:22:31Z","src":"RAM","artifact":"windows.info
SystemTime","event":"MEMORY CAPTURE INSTANT"},
  {"ts":"2009-11-17T(session)","src":"RAM","artifact":"connscan PID 1932-
>192.168.1.104:139","event":"SMB session to LAN host"},
  {"ts":"2009-11-21T01:00:46Z","src":"NET","artifact":"MountPoints2
##192.168.1.1#m57","event":"M57.biz share mapped Z: as m57admin"},
  {"ts":"2009-11-23T18:23:03Z","src":"DSK","artifact":"python-2.6.4.msi","event":"Python
downloaded"},
  {"ts":"2009-11-23T21:54:14Z","src":"DSK","artifact":"Desktop\\web Created","event":"web
folder materialised"},
  {"ts":"2009-11-23T21:54:45Z","src":"DSK","artifact":"USBSTOR USB 2.0 Flash
Disk","event":"USB attached"},
  {"ts":"2009-11-23T21:54:50Z","src":"NET","artifact":"MountPoints2 {300cbeed-...} =
##192.168.1.1#m57","event":"Share re-accessed 5s after USB attach"},
  {"ts":"2009-11-23T21:55:01Z","src":"DSK","artifact":"patentauto.py
Created","event":"Script materialised"},
  {"ts":"2009-11-23T22:02:28Z","src":"DSK","artifact":"Web History
patft.uspto.gov","event":"USPTO patent query"},
  {"ts":"2009-11-24T22:01:52Z","src":"DSK","artifact":"MountPoints2 E:","event":"E: drive
assigned"},
  {"ts":"2009-11-24T22:01:54Z","src":"DSK","artifact":"USBSTOR Generic Flash
Disk","event":"Second USB attached"},
  {"ts":"2009-11-24T22:08:23Z","src":"DSK","artifact":"Web Search http://Quicktime.7-
n0w.com","event":"Suspicious QT URL typed"},
  {"ts":"2009-11-24T22:16:29Z","src":"DEL","artifact":"Desktop\\web\\HighQuality + Videos
deleted","event":"Counter-forensic clean-up"},
  {"ts":"2009-12-11","src":"USB","artifact":"Both USB E01 acquisitions","event":"USB
devices acquired"},
  {"ts":"2010-11-20 (folder name)","src":"USB","artifact":"Work USB \\XFER-11-20-
2010","event":"Future-dated staging folder name"}
]
```

Annex E — Hash Inventory of Recovered Files

Hash inventory of all files extracted or recovered during the investigation. MD5 is the comparison anchor; SHA-256 is recorded for files where Autopsy computed it.

#	Source	Filename	MD5	SHA-256 (if available)
E.1	Primary disk	jo-2009-11-24.E01 (file)	4474b818d2ad84beb74fbd7077d59e97	8fcfeb7a3e16e065e7e9112887b3b5c2047dcaf8706eeae188c5bcd3ce63ccd5
E.2	Primary disk	jo-2009-11-24.E01 (container content)	274ada9b6b120f9b7a2192c27e115244	(EnCase MD5 only)
E.3	RAM dump	jo-2009-11-16.mddramimage	50c49185e254751caa0a9eadfa8fc635	(not computed)
E.4	Favourites USB	jo-favorites-usb-2009-12-11.E01 (file)	8cdc12e30af14e19533c58b3ffe840b5	(not computed)
E.5	Favourites USB	jo-favorites-usb-2009-12-11.E01 (content)	ad1d03cbdb8d81e918899479360f50dc	(EnCase MD5)
E.6	Work USB	jo-work-usb-2009-12-11.E01 (file)	f9408bfcd292a7d8d60928a42806046f	(not computed)
E.7	Work USB	jo-work-usb-2009-12-11.E01 (content)	8f23279deb398c3245829a98bc8fc1bd	(EnCase MD5)
E.8	Work USB \$FAT1	FAT table copy 1	0742d73a0b3840d24f2153b7e40ff55c	84aa03b4069646e6576e9d301b11bd1821d437d7f243275a75a0c800150adf0b
E.9	Work USB \$FAT2	FAT table copy 2	0742d73a0b3840d24f2153b7e40ff55c	(same as $FAT1 — intentional FAT replication)
E.10	Work USB (favourites variant) \$FAT1	FAT table copy 1	14f89344312a8947b924441e4e2d0996	2222ce95e416199a5bd3905560c43c4862a38f5239027acca101a48f83513bb7
E.11	Work USB carved	_TiggerTheCat copy.m4v (4 KB fragment)	edeca2340b4125f4ef4c949f9190ba5c	5c112f7c0c28d08542451a8687c03c7e47ebc6a50b981c237e8ade7f6d90a12f
E.12	Work USB allocated	TiggerTheCat copy.m4v (full)	9bd2dd0f202046eb032bdf46ba91fefa	(different hash — proves disk fragment ≠ USB content payload)
E.13	Work USB carved deleted	_SC00013.JPG	8d1eef424b0e5756ba56eb9d30bb579	76af87c53c62dd628038e0c69d13fc370758ef7116dfa0fdc45fc06a9f429f47
E.14	Primary disk	Pics\patent05.JPG	ac0aceb421b7aa71b878ecc6bf946dad	b6e94bf7f973a22c5d25c9f202d8257e41ec3cd731988bd5e5abf6f0f301a623

Hash analysis conclusion: the macOS resource-fork (_TiggerTheCat copy.m4v) MD5 matches between Work USB and primary disk extraction — proves USB transited Jo's macOS home machine. Main payload differs — Work USB is a curated transfer device, not a verbatim disk clone. _SC00013.JPG and patent05.JPG have different hashes — they are distinct images, but both belong to the same operational workflow (patent imagery captured on Jo's workstation).

— End of Report —

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